



Phase-field Simulation of Dendritic Solidification

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Abstract

We consider the solidification of a pure material from its undercooled melt, taking into account effects of surface tension, kinetic undercooling and crystalline anisotropy. The classical mathematical formulation consists of two heat equations coupled with boundary conditions prescribed at the moving solid-liquid interface. Analytic solutions are known only for a few special cases. Since the problem has important industrial applications one is interested in numerical methods for accurate simulation.

The classical formulation is not well suited for computation since it requires explicit tracking of geometrically complex free boundaries. Phase-field methods have emerged as a popular alternative. They are based on the notion that there exists a smooth approximation of an indicator function, or phase-field, uniquely identifying the phase at a given location and time. Phase-field models are derived from thermodynamics and are formulated as reaction-diffusion equations valid in the entire domain. Free boundaries are replaced by diffuse interfaces of width ϵ corresponding to transitions in the phase-field variable. There is no need to explicitly track interfaces since they are implicitly defined by the computed solution.

The main purpose of this work is to find computationally efficient methods for phase-field simulation of dendritic solidification. To this end finite difference approximations and their implementations are considered. A comparison between a first order accurate semi-explicit time-stepping scheme and an implicit scheme based on BDF-discretization is presented. Numerical experiments in two space dimensions show that for sufficiently low error tolerances the BDF-scheme gives a solution in less simulation time.

To further reduce simulation times for the implicit scheme an implementation has been written for parallel distributed memory architectures. Performance measurements on an IBM SP2 show that a parallel efficiency exceeding 70% can be obtained for grids as small as 200×200 points on 16 processors.

Agreement between phase-field model and classical formulation is investigated in the limit of vanishing diffuse interface thickness. It is shown how the accuracy in the interface conditions can be increased from first to second order in ϵ by modifying a single scalar parameter. Simulation can be performed with larger ϵ, and hence smaller computational grids, without loss of accuracy.

An alternative limit where the undercooling tends to zero much faster than the diffuse interface thickness is considered. The analysis shows that the interfacial temperature becomes independent of ϵ to leading order only when the modification is applied. At low undercoolings the smallest geometric scale in the dendrite increases, and solidification proceeds slower. The correction predicted by analysis allows simulation for smaller undercoolings with the same computational resources.

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List of Papers

Paper 1 C. Andersson, *Time-stepping schemes for phase-field simulation of dendritic solidification*, Tech. Rep. TRITA-NA-0216, Department of Numerical Analysis and Computer Science, Royal Institute of Technology, 2002.

Paper 2 C. Andersson, *Computation of dendrites on parallel distributed memory architectures*, in Simulation and Visualization on the Grid, B. Engquist, L. Johnsson, M. Hammill, and F. Short, eds., vol. 13 of Lecture Notes in Computational Science and Engineering, Springer Verlag, June 2000, pp. 195–208.

Paper 3 C. Andersson, *Third order asymptotics of a phase-field model*, Tech. Rep. TRITA-NA-0217, Department of Numerical Analysis and Computer Science, Royal Institute of Technology, 2002.

Paper 4 C. Andersson, *Analysis of a phase-field model for solidification in the limit of vanishing undercooling*, Tech. Rep. TRITA-NA-0218, Department of Numerical Analysis and Computer Science, Royal Institute of Technology, 2002.

Chapter 1

Introduction

Solidification and dendritic growth is very important from a practical point of view. Some properties of solids, e.g. ductility, electrical conductivity and mechanical strength, are determined by the microscopic structures produced upon solidification. One would like to gain control of the structure formation to obtain the desired properties in manufacture. Applications can be found in for example casting and semiconductor production.

The pattern formation process is not yet fully understood. It is for example known that the dendrite tip radius multiplied by the tip velocity is constant, but it is not clear how this constant is selected for a given undercooling. From a scientific point of view one would like to understand the underlying mechanisms in order to create better models.

Here we especially consider the solidification of a pure material from its undercooled melt, taking into account effects of surface tension, kinetic undercooling and crystalline anisotropy. The classical mathematical formulation has the form of a Stefan problem. Two diffusion equations govern the transport of heat in the solid and liquid phases, and are coupled by boundary conditions prescribed at the solid-liquid interface. During solidification the interfaces will move, resulting in a free boundary problem.

In most situtations it is impossible to find analytic solutions to the Stefan problem of practical use. For this reason a considerable effort has been put into developing numerical algorithms over the last twenty years. In particular, phase-field methods have become increasingly popular since they allow simulation without explicit treatment of free boundaries. Chapter 2 gives a brief overview of the classical mathematical description and discusses some of the difficulties associated with direct numerical simulation. The derivation of a phase-field model from thermodynamics is outlined in Chapter 3. The derivation is to large extent based on the work of Wang et al. [40].

The main purpose of this work is to find computationally efficient methods for numerical simulation of dendritic solidification. It can be roughly divided into

two parts. The first part treats the numerical algorithms and their implementation on parallel distributed memory computers. Section 4.1 gives some brief preliminaries on approximation by finite difference methods. Paper 1 compares two different approaches to time-stepping with respect to achieved accuracy and total execution time. All aspects of the numerical schemes are covered in some detail, with the emphasis on linear and non-linear solvers.

The more accurate of the two time-stepping scheme is expensive in terms of floating point operations and also requires more storage. Paper 2 describes a parallel implementation aimed at reducing the simulation time and/or allowing larger problems to be solved. The efficiency of the parallel code is measured by the metrics introduced in Section 4.3.

The second part of the thesis deals with the analysis of phase-field models. Phase-field models are derived from thermodynamics, and it is necessary to show that the resulting equations yield solutions satisfying the free boundary problem one wishes to solve. Karma and Rappel noted that a more accurate analysis can be exploited to reduce the sizes of computational grids without losing agreement with the free boundary formulation [21]. The modification requires only a small correction to a single physical parameter, as outlined in Section 4.2.

In paper 3 asymptotic approximations to solutions of a phase-field model are constructed. It is shown that the correction suggested by Karma and Rappel [21] increases accuracy in the interface conditions from first to second order. This property does not follow automatically from more accurate analysis. The analysis shows that the fifth degree polynomial modeling the release of latent heat originally proposed in [40] can never lead to third order accuracy. Furthermore, if there are polynomials in the phase-field model leading to third order accuracy they are of degree ten or higher. Polynomials of such high degree are not desirable for computation beacuse of sensitivity to round-off errors.

Paper 4 presents an alternative asymptotic analysis where the limits are chosen to be more appropriate for numerical simulation at low undercoolings. It is shown that the correction determined in paper 3 increases the order of accuracy by almost one in this limit. First order convergence is obtained instead of "almost zero". The limit itself is not of practical interest but allows investigation of the phase-field model at low undercoolings.

Chapter 2

Solidification of a pure material

2.1 The Stefan problem

Let us consider the simple example of a single growing crystal. We can think of the computational domain Ω as consisting of the regions $\Omega_-(t)$ and $\Omega_+(t)$ corresponding to the solid and liquid phases at a given time t. The two phases are separated by a free boundary $\Gamma(t)$ representing the solid-liquid interface, see Figure 2.1. θ is here defined as the angle between the outward normal and some reference direction. It is needed to account for preferred growth directions in the crystal. In three dimensions θ should be a vector of two components to properly account for anisotropy.

Given an initial interface location, temperature distribution, and outer boundary conditions, we wish to determine the interface location $\Gamma(t)$ and temperature $T(\mathbf{x}, t)$. Taking into account effects of surface tension, kinetic undercooling and anisotropy one can formulate a Stefan problem in physical variables,

$$\begin{cases} \partial_t T = & D\nabla^2 T, & \mathbf{x} \in \Omega_\pm(t), \\ cD\partial_n T\big|_-^+ = & -Lv_n, & \mathbf{x} \in \Gamma(t), \\ T = & T_M - \widetilde{\sigma}(\theta) T_M \mathcal{K}/L - v_n/\mu(\theta), & \mathbf{x} \in \Gamma(t) \end{cases} \tag{2.1}$$

where T_M is the melting temperature, D the thermal diffusivity, c the heat capacity, L the latent heat of fusion, v_n the normal velocity of the interface, $\mathcal{K}$ the interfacial curvature taken to be positive if the center of curvature is inside the solid, $\mu(\theta)$ the interfacial mobility and $\widetilde{\sigma}(\theta)$ a function depending on the surface tension $\sigma(\theta)$.

In (2.1) it has been assumed that all physical parameters are constant and equal in the bulk solid and liquid phases. This is a valid assumption for certain materials, such as succinonitrile (SCN) and will reduce the amount of notation needed. The assumption is not vital for the phase field modeling [3].

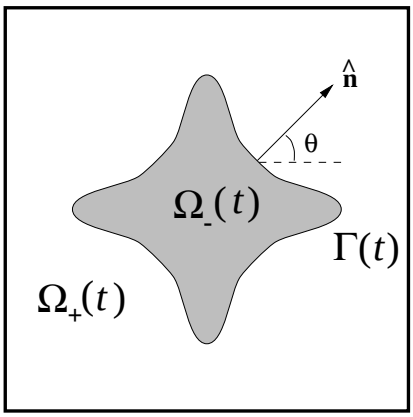


Figure 2.1. Schematic overview of the considered geometry

The first equation of (2.1) describes the diffusion of heat inside the bulk solid and liquid phases. The second equation gives an expression for the interfacial velocity, proportional to the discontinuity in the normal derivative of the temperature across the interface. It is needed for conservation of energy and corresponds to release and/or absorption of latent heat. The third equation, or the Gibbs-Thomson condition, models the change of melting temperature according to kinetic and capillary effects.

The Gibbs-Thomson condition of (2.1) is not present in the classical Stefan problem often seen in mathematical textbooks. Instead the third equation is replaced by $T = T_M$, i.e. it is assumed that the phase transition takes place at equilibrium. The more complicated boundary condition is needed in order to obtain dendritic structures. In an attempt to motivate why the classical Stefan problem is not sufficient in the current context, and explain why the dendritic structures occur, the underlying physics of the "pattern selection process" will be briefly overviewed. More detailed descriptions of the underlying physics can be found in [27, 28].

A pure liquid is held at a temperature below melting temperature. This is a metastable state and solidification can be initiated by disturbing the system, e.g. by adding a crystal seed. As solidification proceeds latent heat will be released at the solid-liquid interface, and the growth rate will depend on how fast this heat can be conducted away. From this point of view it is favorable to have a large interfacial area since it implies faster dissipation. On the other hand, the interface is associated with a surface energy which would in turn imply that a small interfacial area is to prefer. The actual shape of the crystal is determined by a compromise between these two competing effects. Surface energy has a stabilizing effect on the growth since it determines a smallest scale in the geometry. The process is unstable in the sense that small perturbations in initial data can significantly alter the time-dependent shape of the crystal.

2.2 Related physical problems

The Stefan problem is significant for applications in semiconductor production where one is interested in the growth of pure crystals. In order to simulate realistic casting (2.1) needs to be complemented with equations to account for the diffusion of alloying elements. For each additional component in the alloy a diffusion equation for a concentration is added to the Stefan problem, and also an interface condition prescribing a discontinuity in the concentration across the interface. The phase field approach has been successfully used to model solidification of binary alloys, both in isothermal [42, 43] and non-isothermal settings [10, 30].

For casting applications it may be unrealistic to assume that the melt is stationary during the solidification process. Phase-field models have been formulated which take into account fluid flow by accounting for convective effects in the derivation of the model [4, 37, 38].

2.3 Direct numerical simulation

Despite the apparent simplicity of equation (2.1) analytic solutions are known only for a few very special cases. For example a planar interface moving with constant velocity for undercooling $\Delta = c(T_M - T_\infty)/L > 1$, where T_∞ is the uniform temperature of the liquid far away from the interface. The Ivantsov parabolas are a more useful family of analytic solutions in two and three space dimensions. In the absence of surface tension they give a paraboloidal dendrite tip and the associated temperature for a given undercooling. The Ivantsov solutions are formulated in terms of the product between tip velocity and tip radius (the Peclet number) [28]. They are however not capable of determining the velocity or curvature separately.

Numerical computations is therefore a necessary tool for modeling solidification in realistic settings. The major difficulties of direct simulation are associated with keeping track of the moving solid-liquid interfaces so that the correct boundary conditions are prescribed.

Front-tracking methods fit naturally in the framework of free boundary problems, see e.g. [2, 20]. The interface is represented by a discrete set of marker points which are connected by interpolants to obtain the location of the free boundary. The time-dependent problem is then solved by advecting the marker points according to the velocity given by (2.1), and solving the heat equations in solid and liquid given the location of the boundary.

In [20] an approach based on the immersed boundary method [35] is used to transfer data between a fixed temperature grid and interfaces. Time-stepping essentially reduces to solving a system of non-linear equations for the normal velocities of the marker points guaranteeing that the Gibbs-Thomson condition

holds. Almgren considers a different algorithm in [2] where marker points are advected by minimizing an energy functional chosen such that the Gibbs-Thomson condition is satisfied.

Due to the complicated interfacial shapes which arise it will be necessary to add and delete points during simulation. Merging interfaces present a problem since it becomes necessary to redefine the connectivity of marker points to represent the new geometry. This becomes increasingly difficult with the number of space dimensions.

Over the past decade phase-field methods have emerged as a popular alternative for computation. Using ideas from continuum models of phase transitions a set of reaction-diffusion equations valid in the entire computational domain is derived from thermodynamics. The sharp interfaces of (2.1) are replaced by thin diffuse interfaces of width ϵ implicitly given by the solution. Thus the need to explicitly represent moving boundaries is eliminated.

Level set methods can be used as an alternative to phase-field modeling [11, 33]. Front-tracking is avoided by introducing a signed distance function, or level set function, which for every point in the domain gives the distance to the closest solid-liquid interface. In computation the level set is advected according to an extension of the interface conditions. The temperature is updated so that the Stefan problem is satisfied by requiring that the temperature interpolant satisfies the Gibbs-Thomson condition at solid-liquid interfaces [11].

Compared to phase-field methods the level set approach has the advantage of being independent of the artificial diffuse interface width ϵ. Simulation is performed using the sharp interface formulation so there is no need to derive a new set of equations. There are however some additional technicalities in the formulation of numerical schemes. Interpolation is needed to ensure that the interface conditions hold. It is also necessary to "reinitialize" the level set function so that it remains a signed distance function [11].

To date phase-field methods have been used much more extensively than level set methods. The phase-field model considered in this work is briefly overviewed in Chapter 3.

Chapter 3

The phase-field approach

3.1 Derivation of a phase-field model

There are several different phase-field models describing the solidification of a pure material from its undercooled melt. They are all based on the key assumption that there is a smooth approximation of an indicator function taking on distinct values in the bulk solid and liquid phases, the so called *phase-field variable*.

The derivation proceeds by formulating the free energy/entropy of the system and requiring that the temperature and phase-field evolve such that the free energy decreases [29] or the entropy increases [34, 40]. The latter approach results in thermodynamically consistent phase-field models a priori guaranteeing that the second law of thermodynamics holds. It is not clear from simulation whether the thermodynamically consistent models give better results than the ones based on derivation from free energy [24]. Since non-isothermal situations are considered the entropy is however more appropriate from a physical point of view [34].

For both approaches a set of reaction-diffusion equations is obtained but with different non-linear reaction terms. The difference is also affected by the choice of constant values taken by the phase-field, which has consequences for the interpolants $G(\phi)$ and $p(\phi)$ discussed below.

A brief outline of the derivation of a thermodynamically consistent phase-field model is now given. The presentation is based on the work of Wang et al. [40] where the derivation is carried out in full detail, and to lesser extent on the more general work of Penrose and Fife [34]. To avoid some technicalities we consider the isotropic case where the angular dependence of $\widetilde{\sigma}(\theta)$ and $\mu(\theta)$ is not taken into account. The same approach works also for the anisotropic case [32], as discussed in Section 3.4.

We begin by postulating the existence of a phase-field variable, which is an approximation of a non-conserved indicator function such that

$$\phi(\mathbf{x},t) = \begin{cases} 0 & \mathbf{x} \in \Omega_-(t) \\ 1 & \mathbf{x} \in \Omega_+(t) \end{cases}, \tag{3.1}$$

c.f. Figure 2.1. Furthermore, it is assumed that $\phi(\mathbf{x},t)$ varies smoothly between 0 and 1 only in thin regions $\Gamma_\epsilon(t)$ of width $\mathcal{O}(\epsilon)$ corresponding to diffuse solid-liquid interfaces. The small positive parameter ϵ is related to physical quantities below.

To derive evolution equations for phase-field and temperature, we assume that the entropy and internal energy of any given subvolume $V \subset \Omega$ can be written as

$$\mathcal{S} = \int_V \left[s(e,\phi) - \frac{\delta^2}{2}\big(\nabla\phi\big)^2 \right] dV, \tag{3.2}$$

$$\mathcal{E} = \int_V e(T,\phi) dV, \tag{3.3}$$

respectively. Here s is an entropy density and e an internal energy density. The second term in the integrand of (3.2) is a gradient entropy term needed to account for contributions from the solid-liquid interface, analogous to the Landau-Ginzburg energy term in the free energy [34]. According to the first and second laws of thermodynamics, we assume that T and ϕ evolve in such a way that the energy is conserved and the local entropy production in V is non-negative. The second requirement is met if

$$\tau\partial_t\phi = \frac{\delta\mathcal{S}}{\delta\phi}, \tag{3.4}$$

where $\tau > 0$ can be interpreted as the relaxation time of the phase-field [34, 40].

To formulate a set of partial differential equations the internal energy density is written

$$e(T) = e_-(T) + p(\phi)L(T) = e_+(T) + \big(p(\phi)-1\big)L(T), \tag{3.5}$$

where $e_\pm(T)$ are the classical internal energy densities in liquid and solid, and the latent heat $L(T) = e_-(T) - e_+(T)$ is their difference. The function $p(\phi)$ is at this point unknown but will be specified below. It is required to satisfy the normalization $p(0) = 0$ and $p(1) = 1$. It will be assumed that the internal energy density in the liquid phase is linearly dependent on the temperature.

To identify the functional derivative in (3.4) the internal energy is related to the Helmholtz free energy through thermodynamic relations [34],

$$f(T,\phi) = T\left(-\int_{T_M}^{T} \frac{e_+(\zeta)}{\zeta^2} d\zeta - \big(p(\phi)-1\big)Q(T) + G(\phi) \right), \tag{3.6}$$

where

$$Q(T) = \int_{T_M}^{T} \frac{L(\zeta)}{\zeta^2} d\zeta, \tag{3.7}$$

and $G(\phi)$ is another function to be specified. The free energy is required to be continuous with respect to temperature at the melting point T_M, and to have local minima in the solid and liquid phases. We obtain the additional constraints

$$\begin{aligned} &G(0) = G(1), \\ &\big[G'(\phi) - p'(\phi)Q(T)\big]_{\phi=0,1} = 0, \\ &\big[G''(\phi) - p''(\phi)Q(T)\big]_{\phi=0,1} > 0. \end{aligned} \tag{3.8}$$

From (3.2)–(3.6), and the first and second law of thermodynamics the dynamic equations can be written

$$\begin{aligned} &\tau\partial_t\phi = Q(T)p'(\phi) - G'(\phi) + \delta^2\nabla^2\phi, \\ &\partial_t\big(T + \tfrac{1}{c}(p(\phi) - 1)L(T)\big) = D\nabla^2 T. \end{aligned} \tag{3.9}$$

Expressions for the functions $p(\phi)$ and $G(\phi)$ are needed to fully specify the model. Following [40, Model I] we choose $G(\phi)$ to be a symmetric double well potential,

$$G(\phi) = \frac{1}{4a}\phi^2(1-\phi)^2 = \frac{1}{4a}g(\phi), \tag{3.10}$$

where a is the inverse depth of the potential, and

$$p(\phi) = \frac{\int_0^\phi g(\zeta)d\zeta}{\int_0^1 g(\zeta)d\zeta} = \phi^3(6\phi^2 - 15\phi + 10). \tag{3.11}$$

These functions satisfy the necessary normalizations and (3.8) for all temperatures. Hence the free energy (3.6) has local minima at $\phi = 0, 1$ for all values of T. This corresponds to one of the solid or liquid phases being stable, and the other phase metastable, regardless of the supercooling of the liquid or superheating of the solid. The choice of $p(\phi)$ has the advantage that the constant values taken in the bulk phases are given by (3.1) regardless of T [34], but is not physically realistic for T far from melting temperature.

We note that by letting the latent heat be a function of T in (3.5) the heat capacities in the solid and liquid phases are allowed to depend differently on temperature. For simplicity it will now be assumed that the latent heat is constant, and that $L(T) = L(T_M) = L_0$, which corresponds to equal constant heat capacities $c_- = c_+ = c$. Equation (3.7) becomes

$$Q(T) = \frac{L_0}{T_M}\left[\frac{T - T_M}{T_M} + \mathcal{O}\left(\left(\frac{T - T_M}{T_M}\right)^2\right)\right], \tag{3.12}$$

after Taylor expansion around $T = T_M$. If $|T - T_M| \ll T_M$ equations (3.9) and (3.12) result in the linear approximation

$$\begin{aligned} &\tau\partial_t\phi = \tfrac{L_0}{T_M^2}(T - T_M)p'(\phi) - \tfrac{1}{4a}g'(\phi) + \delta^2\nabla^2\phi, \\ &\partial_t\big(T + \tfrac{L_0}{c}p(\phi)\big) = D\nabla^2 T. \end{aligned} \tag{3.13}$$

Since the linearization is relevant only for T close to T_M it is not a limitation to have local minima of the free energy (3.6) for all temperatures T.

The key assumptions needed for deriving the phase-field model are the existence of a phase-field variable defined by (3.1) evolving according to (3.4), and a system entropy on the form (3.2). All other assumptions can either be generalized or follow from standard physical modeling [34].

One should note that a, δ and τ introduced during the phase-field modeling are not physically measurable quantities. By considering the phase-field model under equilibrium conditions $T = T_M$ it is possible to relate a and δ to interface thickness and surface tension σ. An asymptotic analysis of the phase-field model in the limit of vanishing interface thickness then gives an expression for τ. Furthermore, the asymptotics show that (2.1) and (3.13) are indeed equivalent in the limit of vanishing interface thickness.

Details on the asymptotic analysis of this phase-field model can be found in [40]. In [32] an asymptotic analysis of the model is given for anisotropic surface energy and mobility. Caginalp has shown how phase-field models can give rise to other free boundary problems by considering different asymptotic limits [8]. In [9] convergence to (2.1) is proved rigorously in the one-dimensional and radially symmetric cases.

Asymptotics is a necessary tool in the formulation of phase-field models also for related problems, such as the solidification of binary alloys [10, 43]. It has proved useful also for improving the efficiency in numerical computation, as discussed in Section 4.2.

3.2 Non-dimensionalization

Based on results from asymptotic analysis the phase-field model can be written on dimensionless form to reduce the number of parameters. We introduce the dimensionless temperature

$$u = \frac{T - T_M}{T_M - T_\infty},$$

where T_∞ is some reference temperature such as the initial temperature on the boundaries. Space and time are scaled by

$$x = \frac{x_{dim}}{w},$$
$$t = \frac{t_{dim}}{w^2/D},$$

where the subscript dim denotes a dimensional quantity, w is a reference length scale, and w^2/D is the thermal diffusion time. Defining dimensionless variables

$$\epsilon = \frac{\sqrt{a}\delta}{w} = \frac{\epsilon_{dim}}{w}, \tag{3.15}$$

$$m = \frac{\mu\sigma T_m}{DL_0}, \tag{3.16}$$

$$\alpha = \frac{wL_0^2}{6\sqrt{2}c\sigma T_M},$$

$$\Delta = \frac{c(T_m - T_0)}{L_0},$$

(3.13) can be rewritten as

$$\frac{\epsilon^2}{m}\partial_t\phi = \epsilon\alpha\Delta u p'(\phi) - \frac{1}{4}g'(\phi) + \epsilon^2\nabla^2\phi, \tag{3.17}$$

$$\partial_t\Big(u + \frac{1}{\Delta}p(\phi)\Big) = \nabla^2 u. \tag{3.18}$$

Δ is here the dimensionless undercooling (or inverse Stefan number) which typically lies between 0 and 1. $\alpha\Delta$ can be interpreted as a scaled inverse capillary length, and m is the interfacial mobility. The diffuse interface thickness is of order ϵ as given by (3.15).

We note that (2.1) can also be expressed using the dimensionless scalings introduced above. The Stefan problem takes the form

$$\begin{cases} \partial_t u = & \nabla^2 u & \mathbf{x} \in \Omega_\pm(t) \\ \partial_n u\big|_-^+ = & -v_n/\Delta, & \mathbf{x} \in \Gamma(t) \\ u = & -\frac{1}{6\sqrt{2}\alpha\Delta}\Big(\mathcal{K} + \frac{v_n}{m}\Big) & \mathbf{x} \in \Gamma(t) \end{cases}, \tag{3.19}$$

where v_n and $\mathcal{K}$ should be interpreted as dimensionless quantities.

3.3 Model properties

The phase-field model (3.17)–(3.18) and the equivalent dimensional formulation (3.13) are valid in the entire computational domain. Equation (3.18) corresponds to conservation of energy. The non-linear terms are needed to account for the release/absorption of latent heat at the solid-liquid interface. Equation (3.17) is the evolution equation for the phase-field, and does not have a direct physical interpretation.

Since the phase-field model is valid everywhere there is no need to explicitly track geometrically complex solid-liquid interfaces. The location is implicitly given by the definition of the phase-field variable. One could for example let the location of the free boundary be defined by $\Gamma(t) = \{y(t) : \phi(y(t), t) = 1/2\}$, and the diffuse interface be given by $\Gamma_\epsilon(t) = \{y(t) : 0.1 < \phi(y(t), t) < 0.9\}$. Although

neither of the representations are needed during computation they are most likely of interest during post-processing to determine the shape of the crystal.

Equivalence between the phase-field equations and the sharp interface model is formally shown in the limit $\epsilon \to 0$, thus resulting in a singularly perturbed set of reaction-diffusion equations. For purposes of computation it is impossible to let $\epsilon \to 0$ since the grid resolution must be at least of order ϵ in the vicinity of interfaces to resolve the transition of ϕ from zero to one. Even though the Stefan problem is only an approximation of a physical process it is impossible to choose a physically realistic value of ϵ in simulation. Interfaces have a finite width on the scale of Angstroms which can for example be compared to the radius of a SCN dendrite tip which is on the order of 0.1 mm [28].

There is also an upper bound on the value of ϵ determined by the dynamical properties of the phase-field model. If a too large value is used in simulation unphysical oscillatory intermediate states appear in the transition region of the phase-field variable, and the model loses its meaning [1, 12, 14]. These restrictions on ϵ can be relaxed by more accurate asymptotic analysis, as discussed in Section 4.2, and papers 3 and 4.

The main drawback of the phase-field methods is that the accuracy depends on the parameter ϵ. In practice the value of ϵ is determined as a compromise between desired accuracy and available computational resources. We shall discuss this issue further in Section 4.2 where implications of asymptotic analysis on computational efficiency is considered.

There are a number of other differing length and time scales in the problem which make the formulation of efficient numerical schemes challenging. The spatial discretization must be chosen such that the small-scale local geometry is resolved while the grid should cover a sufficiently large domain to capture the entire crystal. At low undercoolings the temperature field extends much faster than the solidification front propagates [6]. To ensure that computed dendrites are unaffected by boundary conditions imposed on temperature it is necessary to discretize large domains. Since the crystal growth is slow long-time integration becomes necessary.

The time step restrictions are governed by conditions in the vicinity of interfaces. Changes away from interfaces, where the phase-field model reduces to a standard heat equation, occur on a much slower time scale. Choosing time steps becomes even more difficult for alloy solidification where the diffusion times of heat and solute can vary greatly.

3.4 Inclusion of anisotropy

For many materials, including metals, the surface energy of the solid-liquid interface and the kinetic coefficient depend on orientation. These effects were not accounted for in the derivation outlined in Section 3.1. Since anisotropy has a crucial impact on the shape of the dendrites it is necessary to modify the phase-field model for simulation in realistic physical settings. In [32] McFadden et. al.

derive a phase-field model with anisotropy in two space dimensions using the same approach as in Section 3.1.

Letting the parameters τ and δ from (3.9) depend on θ (c.f. Figure 2.1) a model can be formulated which is equivalent to the Stefan problem (2.1) in the limit $\epsilon \to 0$. The modified version of (3.9) becomes

$$\begin{aligned} \tau(\theta)\partial_t\phi &= Q(T)p'(\phi) - G'(\phi) + \bar{\delta}^2\nabla_a^2(\theta)\phi, \\ \partial_t\big(T + \tfrac{1}{c}(p(\phi)-1)L(T)\big) &= D\nabla^2 T, \end{aligned}$$

where the ordinary Laplacian of the first equation has been replaced by

$$\nabla_a^2(\theta) = -\frac{\partial}{\partial x}\Big(\eta(\theta)\eta'(\theta)\frac{\partial}{\partial y}\Big) + \frac{\partial}{\partial y}\Big(\eta(\theta)\eta'(\theta)\frac{\partial}{\partial x}\Big) + \nabla\big(\eta^2(\theta)\nabla\big). \tag{3.21}$$

Here $\eta(\theta)$ is a function chosen to model the angular dependence of the coefficient in front of the Landau-Ginzburg term in the entropy functional (3.2). It is typically chosen as

$$\delta(\theta) = \bar{\delta}\eta(\theta) = \bar{\delta}[1 + \gamma\cos(k\theta)], \tag{3.22}$$

in the notation of [44]. γ is the relative strength of the anisotropy, the integer k a mode number determining the number of symmetry axes, and $\bar{\delta}$ a constant governing the size of the gradient contributions.

One of the main reasons for using the phase-field approach is to avoid explicit representation of interfaces. Therefore the angle θ must be formulated in terms of the phase-field variable. We do this by identifying the interface with level sets of the phase-field. The unit normal can then be written

$$\widehat{\mathbf{n}} = \frac{\nabla\phi}{\|\nabla\phi\|_2} = \cos(\bar{\theta})\,\widehat{\mathbf{x}} + \sin(\bar{\theta})\,\widehat{\mathbf{y}},$$

where $\bar{\theta}$ is the angle between $\widehat{\mathbf{n}}$ and the x-axis. If the angle between the reference direction and the x-axis is θ_0, it follows from simple trigonometry that

$$\theta = \bar{\theta} - \theta_0 = \tan^{-1}\left(\frac{\partial_y\phi}{\partial_x\phi}\right) - \theta_0. \tag{3.24}$$

The modified Laplacian (3.21) is clearly non-linear in ϕ, which is significant in the formulation of numerical schemes.

In the dimensionless phase-field model the choice of $\eta(\theta)$ introduces an angular dependence also in the kinetic undercooling term of the Gibbs-Thomson condition [6, 32]. By letting $\tau(\theta) := \bar{\tau}\eta(\theta)$, where $\eta(\theta)$ is given by (3.22), the contribution becomes independent of the angle between the solid-liquid interface and the reference direction [6]. If we instead choose $\tau(\theta) = \bar{\tau}$ the only modification of (3.17) is that the Laplacian should be replaced by (3.21). Equation (3.18) does not need to be changed to account for anisotropy.

Chapter 4

Numerical simulation

A few different approaches to direct simulation of the Stefan problem were briefly overviewed in Section 2.3. The phase-field approach has been gaining in popularity and is emerging as the method of choice for simulation of dendritic solidification. There is an extensive amount of literature on the use of phase-field models in computation. They have been used to model a number of different physical phenomena, including solidification of pure materials [22, 25, 44], solidification of alloys [5, 10, 30, 42], solid to solid phase transformations [41] and solidification under external flow [39, 38] just to name a few. Several different numerical approximation schemes have been used. Although uniform finite difference discretizations seem to be the most common there are examples of adaptive algorithms based on both finite differences [6] and finite elements [36, 39].

In this chapter some aspects of numerical approximation will be covered. The aim is not to fully cover finite difference techniques but to introduce some concepts which are discussed in greater detail in the papers below.

4.1 Finite difference methods

There are a number of general techniques for numerical approximation of partial differential equations (PDE), including finite difference, finite element and spectral methods. They all result in a set of algebraic equations, whose solution constitutes an approximation to the solution of the continuous problem. Here we have chosen to work with finite difference methods, which lead to structured systems approximating the PDE. Thus hierarchical computer memory systems can be efficiently used. Identification of data dependencies becomes trivial which is useful when formulating parallel algorithms, as discussed in Section 4.3.

Finite difference techniques are conceptually simple. Algebraic equations are obtained by replacing partial derivatives by difference quotas. By solving the system an approximate value of the solution is obtained in every point on the computational grid. For time-dependent PDE we think of the computational

grid as a discrete representation of space only. Starting from some initial state, the solution is advanced in time by solving a set of algebraic equations per time step.

Numerical schemes can be constructed via the method of lines. By replacing spatial derivatives with difference quotas a set of ordinary differential equations (ODE) for the gridvalues is obtained. One can then proceed with the temporal discretization using standard numerical methods developed for ODE. We distinguish between implicit and explicit time-stepping schemes. For the latter the algebraic systems are trivial and the solution at the next time level can be explicitly formulated using the current approximation. For implicit schemes it is necessary to actually solve a system to update the solution. In the present application the systems are non-linear.

For analysis of finite difference methods we are primarily interested in stability and accuracy. Stability imposes conditions on the relation between time step and grid spacing which must hold to prevent uncontrolled growth of errors. The accuracy is a measure of how well the discretization approximates the PDE and hence also at which rate the approximation converges to the exact solution as grid spacing and time step decrease. Please refer to [19] for proper definitions and a detailed exposition on the analysis of finite difference methods.

In computation one would like to determine an approximate solution to a given tolerance as fast as possible. The stability and accuracy of numerical schemes have a crucial impact since they govern the range of time steps and grid spacings resulting in the desired accuracy. For explicit schemes the simulation time increases with the number of time steps taken, and the execution time per time step grows linearly with the number of grid points. This is not necessarily true for implicit schemes, when a system of equations must be solved each time step. The stability conditions for implicit schemes are however often much less restrictive.

In paper 1 two different time stepping schemes are compared with respect to achieved accuracy and total execution time.

4.2 Asymptotics and implications on computational efficiency

Asymptotic analysis of phase-field models is necessary to show equivalence with corresponding sharp interface problems but also has significant implications on computational efficiency. This was first observed by Karma and Rappel [21] and has since attracted considerable attention, see e.g. [3, 13, 22, 31].

The early analysis verified that solutions to the phase-field equations satisfy the Stefan problem with errors of size $\mathcal{O}(\epsilon)$ [7, 8, 9, 40]. Thus convergence is guaranteed in the limit $\epsilon \to 0$. More accurate asymptotics have subsequently determined the interface conditions to $\mathcal{O}(\epsilon^2)$, and requirements on the phase-field model for second order agreement with the Stefan problem [3, 13, 21]. Since the

grid resolution is determined by ϵ this allows simulation on smaller grids without increasing errors in the interface conditions.

Karma and Rappel formulated the observations in terms of the sharp-interface and thin-interface limits which have more direct physical interpretation. In the first order asymptotics, or the sharp-interface limit, the temperature is constant in the diffuse interface. In the second order asymptotics the $\mathcal{O}(\epsilon)$-contribution to the temperature varies across the interface. Taking the contribution into account can therefore be seen as allowing a small variation of u, or equivalently, having a thin interface of width different from zero.

Their analysis showed that the phase-field model gives an error of size $\mathcal{O}(\epsilon)$ in the Gibbs-Thomson condition (3.19) because the mobility m is determined by first order asymptotics. To obtain second order agreement it is therefore necessary to add a correction prior to simulation. For the model considered here the correction takes the form

$$\frac{1}{m} = \frac{1}{m_0} + \frac{209}{35}\epsilon\alpha, \tag{4.1}$$

where m_0 is the value predicted by (3.16).

It can be shown that simulation without applying (4.1) is equivalent to assuming a diffuse interface thickness much smaller than the capillary length [21, 22]. Since the capillary length is a measure of the smallest geometric scale, the effects of the thin interface correction (4.1) are significant. Computation on smaller grids is made possible by modifying a single scalar parameter prior to simulation. A larger value of ϵ can also be used without introducing unphysical intermediate states in the phase-field variable. Furthermore, the correction allows simulation in the absence of kinetic undercooling ($m_0 = \infty$), which is a reasonable assumption at low undercoolings [3].

The exact form of (4.1) depends on properties of the functions $p(\phi)$ and $g(\phi)$ in the phase-field model. Almgren [3] has shown that the functions must satisfy certain integral constraints for a thin interface correction to exist. His analysis was performed for the more general problem of binary alloy solidification.

4.3 Parallel computing

Parallel computing is a cost efficient way of overcoming limitations of current hardware technology such as processor clock frequency and memory bandwidth. By letting several computers cooperate to solve a single computational task a problem can be solved faster than a single conventional computer would allow. It also enables larger problem sizes, which can be exploited to increase the accuracy or to simulate more complex systems.

There are several types of parallel architectures. The two most common ones are the shared and distributed memory machines which differ in the way computer memory is handled. On a shared memory computer all processors share the same address space and have access to a large global memory. On a

distributed memory computer each processor has access to its own local memory only.

Here we consider distributed memory architectures. Since each processor has its own local memory the bandwidth scales with the number of computers used. Non-local data can however not be accessed directly. There is a need for communication - or message passing - between processors. One of the critical issues when designing a program is to keep the ratio of communication to computation small since communication is essentially an undesirable overhead. For this reason it may be necessary to choose different algorithms than one would have on conventional computer architectures.

We distinguish between local communication involving a few processors, and global communication involving all processors. One would like to avoid the latter if possible since it means that all computers must reach a certain stage in the computation before the execution in any processor can proceed. At these so called synchronization points all processors remain idle while waiting for the slowest one to complete its task. Thus it is important to have a good load balance so that an equal amount of work is performed on all processors. By choosing an appropriate distribution of data onto the parallel machine this can be achieved. The data distribution also affects the amount of communication needed.

4.3.1 Scalability metrics

The ideal parallel program running on p processors should finish p times faster than on a single processor. Due to communication overhead this is hardly ever the case. To measure how efficiently a parallel algorithm exploits the computational resources of a parallel machine one therefore investigates different scalability metrics [26]. The most well known is the speedup defined by

$$\text{speedup} = \frac{T_1(W)}{T(p;W)}.$$

Here $T_1(W)$ is the time needed to solve a problem of size W on a single processor using the fastest known serial algorithm, and $T(p;W)$ the time needed for the parallel algorithm running on p processors to solve the same problem[1].

The parallel efficiency is defined as the speedup over the number of processors. A parallel efficiency of one corresponds to a perfectly scalable algorithm, which runs p times faster on p processors. For an algorithm which is not perfectly scalable there is an asymptotic limit on the speedup which can be obtained. The famous Amdahl's law states that if a fraction f of the code is inherently serial and can not be parallelized the speedup can never exceed $1/f$, even on an infinite number of processors [18].

[1] The problem size is measured as the number of floating point operations required by the fastest known serial algorithm running on a single processor.

The definition of the speedup is based on the assumption that a problem of a given size W is to be solved. It does not account for the ability to solve larger problems on a parallel machine. The scaled speedup is an alternative metric, which compares how long a given parallel program would have taken to run on a hypothetical serial processor with the same amount of RAM memory as the parallel machine [18]. The problem size is increased linearly with the number of processors, and the scaled speedup is defined by

$$\text{scaled speedup} = \frac{T_1(W \cdot p)}{T(p; W \cdot p)}$$

By increasing the problem size memory and surface-to-volume effects are kept constant. The limitations of Amdahl's law no longer apply and the parallel efficiency approaches $1 - f$ as the number of processors tends to infinity [18].

What kind of scalability metric to use depends on what one wishes to use the increased computational resources for. A more thorough coverage of scalability analysis and other metrics can be found in e.g. [16, 26].

Chapter 5

Computed dendrites

In this chapter three examples of computed dendrites are given. They are intended to exemplify the type of dendritic structures observed in simulation. Hence only computational parameters and plots are given and no conclusions are drawn. Additional simulations can be found in papers 1, 2 and 4 where different aspects of the phase-field model are investigated numerically.

All simulations in this chapter use dimensionless parameters $\alpha = 400$, $\Delta = 0.5$ and $m = 0.05$, corresponding to pure nickel at a certain undercooling [44]. The diffuse interface thickness is taken to be $\epsilon = 0.005$. Anisotropy is modeled by (3.22) with $\gamma = 0.015$, and $k = 4$ or 6 depending on symmetry. From physics one would expect the fourfold symmetry to be more appropriate since that better approximates the BCC-lattice observed in solid nickel.

The plots shown in figures 5.1–5.3 have been computed using the semi-explicit scheme described in paper 1. The uniform discretization satisfies $\Delta x = \Delta y = \epsilon = 0.005$, and a constant time step $\Delta t = 10^{-4}$ has been employed. The grid size was increased during the course of the simulations. When the temperature at an outer boundary differs from the initial value -1 by more than 10^{-3} an additional 50 grid points are added in that direction to ensure that boundary conditions do not affect the growth of the dendrite. This adaptivity is of course ad-hoc, and likely to influence the computed solution. Since the only purpose is to exemplify computed dendrites it has nevertheless been used to reduce execution time. Each simulation took about five days on a single work station with a large amount of RAM memory.

In addition to the phase-field equations described in Chapter 3 a random perturabtion of 1% amplitude was added to the non-linear terms in the right-hand side of equation. This serves to stimualte sidearm growth, and can be physically interpreted as a phenomenological model for material impuritites and thermal noise (3.18) [23, 44].

The simulations have not been compared to actual nickel dendrites but the dendritic structures observed exhibit the qualitative behaviour observed in physical experiments, see e.g. [15].

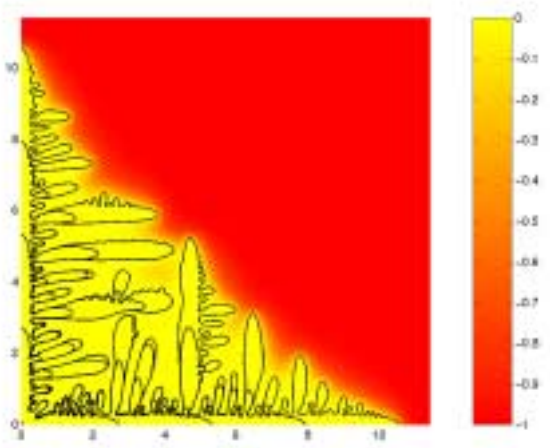


Figure 5.1. Solidification of pure nickel under fourfold anisotropy starting from circular initial seed of radius 0.05. The solid lines show the location of the solid-liquid interface at time levels $t = 0, 0.3, 0.6, 0.9, 1.2$. The background color represents the heat distribution at the final time $t = 1.2$.

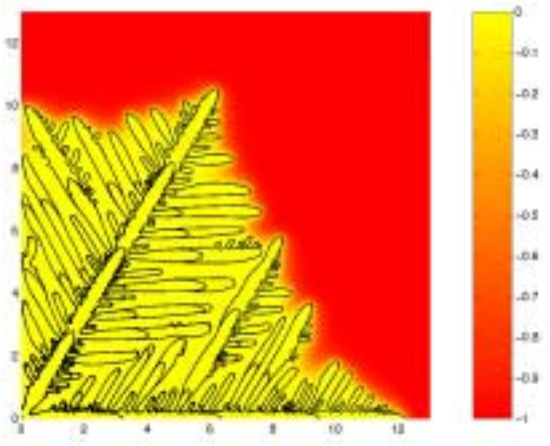


Figure 5.2. Solidification of pure nickel under sixfold anisotropy starting from circular initial seed of radius 0.05. The solid lines show the location of the solid-liquid interface at time levels $t = 0, 0.3, 0.6, 0.9, 1.2$. The background color represents the heat distribution at $t = 1.2$.

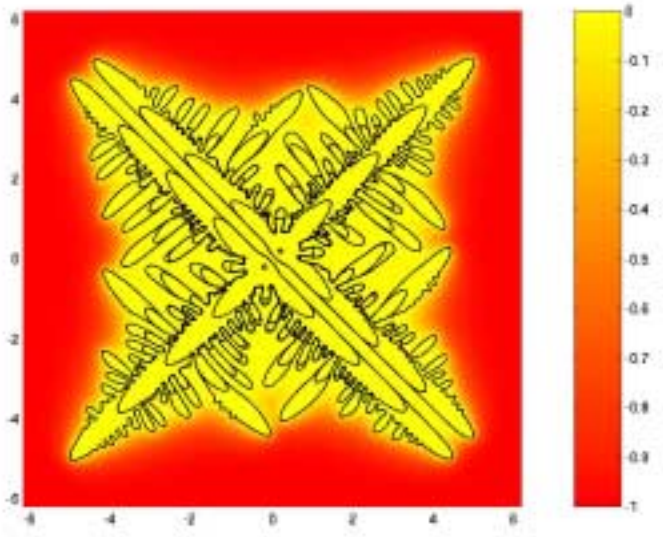


Figure 5.3. Solidification of pure nickel starting from two circular initial seeds of radius 0.05. The solid lines show the location of the solid-liquid interface at time levels $t = 0, 0.25, 0.5, 0.75, 1.0$. The background color represents the heat distribution at $t = 1.0$.

Chapter 6

Summary of papers

6.1 Paper 1: Time-stepping schemes for phase-field simulation of dendritic solidification

In this paper two different approaches to time-stepping are investigated for the phase-field model discretized by finite differences on uniform Cartesian grids. The first scheme is semi-explicit, computationally cheap per time step and has low storage requirements. Since the phase-field variable is updated using an explicit method there are however step size restrictions. The second scheme is based on the implicit BDF-methods. It leads to increased accuracy and relaxed step size restrictions at the expense of increased storage requirements and computational complexity.

For the semi-explicit scheme one needs to solve a linear system of equations each time step, which can be done efficiently using a fast Poisson solver. The implicit scheme leads to non-linear systems solved by a Newton-GMRES algorithm. Solvers are discussed in some detail. Extensions to adaptive grids and parallel computers are considered, although neither are used in this paper.

Numerical experiments showing that the semi-explicit scheme is only first order accurate in time are presented. Convergence rates of first, second and third order BDF-schemes are experimentally validated. The importance of accurate time discretization is shown by a simple one-dimensional example where the semi-explicit scheme fails to capture the correct propagation speed of the interface.

By experiments using realistic parameters it is shown that one benefits from using the BDF-scheme in two dimensions for sufficiently low error tolerances.

6.2 Paper 2: Computation of dendrites on parallel distributed memory architectures

In paper 1 it was shown by computational experiments that one benefits from using the implicit scheme which is more computationally expensive per time step for sufficiently low error tolerances. In this paper the possibility to use parallel computers to obtain simulation times comparable to the semi-explicit scheme also for larger error tolerances is investigated. The program described is implemented in C and uses MPI [17] for message passing in a distributed memory setting. The implementation is verified by reproducing a computation previously published in [6].

The key computational kernels of the Newton-GMRES solver are identified in terms of linear algebra operations. A data distribution is suggested which results in a low communication over computation ratio. The scalability is investigated on a massively parallel IBM SP2, as well as networks of Sun Ultra 5 workstations which do not have access to an optimized communications network. The parallel efficiency exceeds 75% on 16 IBM SP2 processors even for grids as small as 200×200 points. For the same grid size an efficiency of more than 70% is achieved using up to eight Sun computers. Investigations of the scaled speedup demonstrate the possibility of using parallel computers for efficient simulation on larger grids.

6.3 Paper 3: Third order asymptotics of a phase-field model

As first observed by Karma and Rappel in [21] the agreement between the phase-field and sharp interface models can be increased from first to second order in ϵ by modification of the dimensionless parameter m only. The modification allows simulation on smaller grids without loss of accuracy. In this paper we investigate whether it is possible to obtain a third order accurate phase-field model by a similar modification. The focus is therefore not on the numerical algorithms or implementation, as in papers 1 and 2, but on the phase-field model itself.

To identify the interface conditions satisfied by solutions to the phase-field equations a formal asymptotic analysis is performed in the limit of vanishing diffuse interface thickness. The asymptotics is carried out using the dimensionless quantities from Section 3.2 rather than the dimensional phase-field model. For the analysis the function $p(\phi)$ modeling the release of latent heat is unspecified, but required to satisfy certain constraints imposed by assumptions made in deriving the model.

In the process of constructing asymptotic approximations additional constraints on $p(\phi)$ appear which must be satisfied for the temperature to be continuous across the interface. Conditions for the existence of a thin interface

correction are identified. An explicit expression for the correction resulting in second order accuracy is given.

No thin interface correction leading to third order accuracy has been found. Due to difficulties evaluating the interfacial temperature for general $p(\phi)$ we have restricted the investigation to all possible polynomials of degree nine or less. That allows us to determine the interfacial conditions by using a C++ program for semi-symbolic computation.

From the analysis and the semi-symbolic code we conclude that if there are polynomials $p(\phi)$ leading to third order agreement they must be of degree ten or higher. For computation one would prefer not to use polynomials of such high degree due to round-off errors.

6.4 Paper 4: Analysis of a phase-field model for solidification in the limit of vanishing undercooling

In the last paper an alternative asymptotic analysis is presented in the limit of vanishing undercooling Δ. The limit can be motivated by an interest to perform numerical simulation in the low undercooling regime. As the undercooling decreases the smallest geometric scale increases, and the solidification proceeds slower. Thus larger domains must be discretized before dendritic structures are observed, and long time integration is needed for them to arise. It is desirable to have a diffuse interface thickness $\epsilon \gtrsim \Delta$ to reduce computational complexity. In simulation it is often necessary to use a smaller ϵ to avoid the appearance of unphysical oscillations in the phase-field variable.

For the analysis we assume that Δ approaches zero much faster than ϵ. The construction of asymptotic approximations is based on the same techniques used in paper 3. For technical reasons stronger assumptions on the interfacial velocity and curvature are required. As a consequence the limit $\Delta \to 0$ corresponds to a stationary planar interface. The limit itself is not interesting from a computational point of view. The asymptotics is merely used to investigate the agreement between the sharp interface and phase-field models.

It is shown that the Gibbs-Thomson condition has an error of size ϵ. By applying the same thin interface correction derived in paper 3 this leading order error can be eliminated, resulting in an error of size Δ. Since it is assumed that ϵ approaches zero much slower than Δ the order of accuracy is increased from "almost zero" to one. The analysis presents a motivation for the usefulness of the thin interface correction at low undercoolings which does not directly rely on scaling arguments or the approximation $m_0 = \infty$. Furthermore, the correction allows larger ϵ to be used before unphysical intermediate states appear.

Numerical experiments supporting the conclusions of the analysis are presented.

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Paper 1

Time-stepping schemes for phase-field simulation of dendritic solidification

Christer Andersson

Abstract

Two different time-stepping schemes are investigated for a phase-field model of solidification of a pure material. A first order accurate semi-explicit scheme which is computationally cheap per time step is compared to a more accurate fully implicit BDF-discretization. The implementation of fast Poisson and Newton-GMRES solvers are discussed in some detail.

Convergence rates are experimentally validated for decreasing time steps. By computational experiments it is shown that for sufficiently low error tolerances one benefits from using the BDF-scheme in two space dimensions.

1 Introduction

Numerical simulation of dendritic solidification has been a challenging problem. The classical mathematical formulation requires explicit tracking of geometrically complex solid-liquid interfaces where boundary conditions are prescribed. Over the past decade the evolution of phase-field models have allowed simulation of physically realistic cases [20]. By introducing a phase-field variable ϕ approximating a thermodynamic order function, a model where the interfaces are implicitly represented by the solution can be derived. The approach results in a set of reaction-diffusion equations valid in the entire computational domain. There is no need to explicitly track moving boundaries.

Here we have chosen to work with a thermodynamically consistent phase-field model derived by Wang et al. [25, 33]. It is assumed that $\phi(\mathbf{x},t)$ takes the value zero in solid, one in liquid, and varies smoothly between zero and one in thin regions of width $\mathcal{O}(\epsilon)$ corresponding to diffuse solid-liquid interfaces. On dimensionless form the model can be written as

$$\frac{\epsilon^2}{m}\frac{\partial\phi(\mathbf{x},t)}{\partial t} = \epsilon\alpha\Delta u(\mathbf{x},t)p'(\phi(\mathbf{x},t)) - \frac{1}{4}g'(\phi(\mathbf{x},t)) + \epsilon^2\nabla^2\phi(\mathbf{x},t), \tag{1.1}$$

$$\frac{\partial}{\partial t}\Big(u(\mathbf{x},t) + \frac{1}{\Delta}p(\phi(\mathbf{x},t))\Big) = \nabla^2 u(\mathbf{x},t), \tag{1.2}$$

assuming that the heat capacities and thermal conductivities are the same in solid and liquid. Here $u(\mathbf{x},t)$ is the dimensionless temperature. The physical parameters Δ, α and m correspond to the undercooling, inverse capillary length

and interfacial mobility, respectively. The function $g(\phi) = \phi^2(1-\phi)^2$ is a symmetric double well potential introduced in the derivation of the model. The function $p(\phi)$ describing the release of latent heat is taken to be $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ as suggested in [33, Model I].

The phase-field model (1.1)–(1.2) agrees formally with the sharp interface formulation in the limit $\epsilon \to 0$ [3, 33]. For purposes of simulation ϵ is chosen as a compromise between desired accuracy and available computational resources since ϵ is also a measure of the smallest grid resolution. In recent years much interest has been devoted to accurate asymptotic analysis of the phase-field equations since it allows larger values of ϵ, and hence smaller grids, in simulation [3, 12, 21]. Finding numerical methods capable of simulation for sufficiently small ϵ is however still a challenging task.

In this paper we shall consider two different approaches to time stepping based on the method of lines. A widely used semi-explicit scheme is introduced in Section 2.2.1. It is motivated by a desire to avoid non-linear systems during simulation. For discretization on uniform Cartesian grids fast Poisson solvers can be used to update the solution as discussed in Section 2.4. The second approach is based on the implicit BDF-schemes, as outlined in Section 2.2.2, which are more accurate but require the solution of a large sparse non-linear system each time step. The non-linear solver is based on a Newton-GMRES algorithm described in Section 2.3.

The primary purpose is to investigate which of the time-stepping schemes is more appropriate with respect to accuracy and simulation time. One-dimensional computational experiments verifying the order of accuracy are presented in Section 3.1. In Section 3.2 it is shown that in some instances the first order accurate semi-explicit scheme may fail to capture the correct propagation speed of the solid-liquid interface. A comparison of execution time versus achieved accuracy in two space dimensions is given in Section 3.3.

All numerical experiments in this paper have been carried out on uniform Cartesian grids in one and two space-dimensions. For realistic simulations one is interested in three dimensions, which most likely requires adaptive grids to reduce storage requirements. We believe that the conclusions of this paper can be extended also to the three-dimensional case. The Newton-GMRES solver can be used on adaptive grids without modification. For the fast Poisson solver certain types of adaptive grids can be handled after modification to the algorithm.

For simulation in two and three dimensions the use of parallel computers can significantly reduce the execution time. Although we are not directly concerned with parallel implementation here some discussion on how well the different algorithms can be implemented on distributed memory architectures is given. A more detailed discussion of the parallel implementation of the BDF-scheme can be found in [2].

2 Numerical Approximation

The computational experiments in this paper are carried out for the phase-field model (1.1)–(1.2) in one and two space dimensions. No significant differences appear in the numerical methods described below, and the generalization to the three-dimensional case is straight-forward. Some modifications do however need to be made to the phase-field model (1.1)–(1.2) to account for significant effects during solidification.

To include curvature in one space-dimension we let the Laplacian be on the form

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{2}{x} \cdot \frac{\partial}{\partial x}, \tag{2.1}$$

which is equivalent to assuming spherical symmetry. In two dimensions the Laplacian in (1.1) is replaced by

$$-\frac{\partial}{\partial x}\Big(\eta(\theta)\eta'(\theta)\frac{\partial}{\partial y}\Big) + \frac{\partial}{\partial y}\Big(\eta(\theta)\eta'(\theta)\frac{\partial}{\partial x}\Big) + \nabla\big(\eta^2(\theta)\nabla\big) \tag{2.2}$$

to include effects of anisotropy. Here θ is the angle between the normal to the solid-liquid interface and some reference direction, and $\eta(\theta)$ is the function used to describe the dependence of surface energy on orientation. For the simulations it has been chosen as

$$\eta(\theta) = 1 + \gamma\cos(k\theta), \tag{2.3}$$

where γ is the relative strength of the anisotropy and the integer k a mode number determining the number of symmetry axes. We note that by identifying the interface with level sets of the phase-field variable the angle θ is implicitly given by

$$\theta = \arctan\left(\frac{\partial\phi/\partial y}{\partial\phi/\partial x}\right) - \theta_0, \tag{2.4}$$

for some θ_0 determined by the reference direction. Thus there is no need to explicitly determine the location of the solid-liquid interface during simulation.

Equation (2.2) is equivalent to including angular dependence in the gradient terms of the entropy functional when deriving the phase-field model [24]. The modification is necessary for materials where the surface energy depends on orientation, e.g. metals. In three dimensions (2.2) would be replaced by an expression which depends on two angles.

The phase-field model is discretized by the method of lines so that a system of algebraic equations for the grid values is obtained. Section 2.1 discusses approximation of spatial derivatives. In Section 2.2 two different approaches to time stepping are considered. The first of these leads to a system of linear equations to be solved each time step, and an efficient solver under certain simplifying assumptions is discussed in Section 2.4. The second approach results in non-linear systems. A non-linear solver based on a Newton-GMRES algorithm is described in Section 2.3.

2.1 Spatial Discretization

All computations in this paper have been carried out on uniform Cartesian grids,

$$\Omega_1 = \big\{ x_i = x_{min} + i\Delta x,\, 0 \le i < M \big\},$$
$$\Omega_2 = \big\{ \mathbf{x}_{i,j} = (x_{min} + i\Delta x\,,\, y_{min} + j\Delta y),\, 0 \le i \le M,\, 0 \le j < N \big\},$$

where values of Δx and Δy are given by the number of grid points and the size of the domain. The approximation of spatial derivatives in the phase-field equations by finite differences is straight-forward. We note that the construction of the phase-field model with thin diffuse interfaces of width ϵ suggests that adaptive discretizations are more appropriate [6, 22, 26]. Only uniform discretizations are considered here since we are primarily considering implications of time stepping schemes.

By introducing operators D_ϕ and D_u in (1.1) and (1.2) respectively the partial differential equations are approximated by a set of ordinary differential equations for the grid values in two dimensions,

$$\begin{aligned} &\frac{\epsilon^2}{m}\frac{d\phi_{i,j}(t)}{dt} = \epsilon\alpha\Delta u_{i,j}(t)p'(\phi_{i,j}(t)) - \frac{1}{4}g'(\phi_{i,j}(t)) + \epsilon^2 D_\phi(\phi_{i,j}(t))\phi_{i,j}(t), \\ &\frac{d}{dt}\Big(u_{i,j}(t) + \frac{1}{\Delta}p(\phi_{i,j}(t))\Big) = D_u u_{i,j}(t). \end{aligned} \tag{2.5}$$

To conveniently formulate second order represention $D_\phi(\phi)$ and D_u we use the difference operators

$$\begin{aligned} D_{+x}f_{i,j} &= \big(f_{i+1,j} - f_{i,j}\big)/\Delta x, & \qquad D_{+y}f_{i,j} &= \big(f_{i,j+1} - f_{i,j}\big)/\Delta y, \\ D_{-x}f_{i,j} &= \big(f_{i,j} - f_{i-1,j}\big)/\Delta x, & \qquad D_{-y}f_{i,j} &= \big(f_{i,j} - f_{i,j-1}\big)/\Delta y, \\ D_{0x}f_{i,j} &= \big(D_{+x} + D_{-x}\big)f_{i,j}/2, & \qquad D_{0y}f_{i,j} &= \big(D_{+y} + D_{-y}\big)f_{i,j}/2. \end{aligned}$$

The standard five-point stencil can be used to approximate $\nabla^2 u$,

$$D_u u_{i,j} = \big(D_{+x}D_{-x} + D_{+y}D_{-y}\big)u_{i,j}. \tag{2.6}$$

The action of the anisotropic Laplacian (2.2) on ϕ becomes

$$\begin{aligned} D_\phi(\phi_{i,j})\phi_{i,j} = &- D_{-x}\Big((\eta\eta')_{i+1/2,j}D_{0y}\frac{\phi_{i,j} + \phi_{i+1,j}}{2}\Big) \\ &+ D_{-y}\Big((\eta\eta')_{i,j+1/2}D_{0x}\frac{\phi_{i,j} + \phi_{i,j+1}}{2}\Big) \\ &+ D_{-x}\Big((\eta^2)_{i+1/2,j}D_{+x}\phi_{i,j}\Big) + D_{-y}\Big((\eta^2)_{i,j+1/2}D_{+y}\phi_{i,j}\Big), \end{aligned}$$

where $\eta(\theta)$ is given by (2.3), and θ approximated by

$$\theta_{i+1/2,j} = \arctan\left(\frac{D_{0y}(\phi_{i,j} + \phi_{i+1,j})}{2D_{+x}\phi_{i,j}}\right) - \theta_0,$$
$$\theta_{i,j+1/2} = \arctan\left(\frac{2D_{+y}\phi_{i,j}}{D_{0x}(\phi_{i,j} + \phi_{i,j+1})}\right) - \theta_0.$$

between gridpoints. This expression requires all eight neighboring points, which should be compared to just four neighboring points for the standard five-point stencil. For the system of ordinary differential equations to be a valid approximation in all grid points it is necessary to discretize also boundary conditions on (1.1)–(1.2). In what follows we shall assume zero Neumann-conditions $\nabla\phi \cdot \hat{\mathbf{n}} = \nabla u \cdot \hat{\mathbf{n}} = 0$ on all boundaries.

An approximation similar to (2.5) can be constructed for the phase field equations on Ω_1. For the one-dimensional case D_ϕ and D_u are chosen to account for curvature by (2.1), so D_ϕ does not depend on ϕ. The boundary condition on the temperature at the outer boundary is replaced by [34]

$$\frac{\partial u}{\partial x}(x_{max}) + \frac{u(x_{max}) + 1}{x_{max}} = 0.$$

The techniques used to formulate time-stepping schemes do not depend on the number of space dimensions. In the ensuing discussion we shall therefore use notation appropriate for the two-dimensional phase-field model.

It is convenient to formulate a one-to-one mapping from grid values to column vectors. Let such a mapping be given by the natural ordering [28],

$$\phi_{i,j}(t) \to \Phi_{i+jM}(t) \quad , \quad u_{i,j}(t) \to U_{i+jM}(t).$$

The semi-discrete system (2.5) can be written

$$\frac{d}{dt}\begin{bmatrix} \Phi \\ U + p(\Phi)/\Delta \end{bmatrix} = \begin{bmatrix} mD_\phi(\Phi) & 0 \\ 0 & D_u \end{bmatrix}\begin{bmatrix} \Phi \\ U \end{bmatrix} + \begin{bmatrix} F(\Phi, U) \\ 0 \end{bmatrix}, \tag{2.7}$$

where non-linear terms in (2.5) not related to the difference operator D_ϕ have been collected in $F(\Phi,U)$. D_u and D_ϕ should here be interpreted as a sparse matrices, whose structure depend on the mapping from grid values to vectors. The natural ordering results in banded matrices. On non-uniform grids, or with different algorithms for solving algebraic equations, other mappings may be more appropriate.

2.2 Temporal Discretization

Time-stepping schemes are formulated for the large sparse system of ODEs (2.7). Approximate solutions are constructed at equidistant time levels $\{n\Delta t\}_{n=1,2,\ldots}$, starting from initial values $\Phi(0) = \Phi^0$ and $U(0) = U^0$. Extension to adaptive time stepping does not introduce any significant modification to the numerical methods constructed, but would require an algorithm for determining Δt.

2.2.1 Semi-explicit schemes

"Equation splitting" is a common approach to time stepping motivated by a wish to avoid non-linear equations. Given Φ^{n-1} and U^{n-1}, solutions at time level n are computed in two stages:

1. Compute Φ^n from the upper part of (2.7) using an explicit scheme, e.g. forward Euler,

$$\Phi^n = \Phi^{n-1} + m\Delta t D_\phi(\Phi^{n-1})\Phi^{n-1} + \Delta t F(\Phi^{n-1}, U^{n-1}) \tag{2.8}$$

2. Compute U^n from the lower part of (2.7) using an implicit scheme, e.g. Crank-Nicholson or backward Euler,

$$\Big(I - \Delta t D_u\Big)U^n = \left[U^{n-1} - \frac{1}{\Delta}\Big(p(\Phi^n) - p(\Phi^{n-1})\Big)\right], \tag{2.9}$$

   where the time derivative of $p(\phi)$ is approximated using the value of Φ^n computed in stage 1.

Other numerical schemes can of course be used, but the update in stage 1 should be explicit to avoid non-linear equations.

We note that employing the forward Euler scheme imposes step size restrictions on Δt. Ignoring non-linear terms in (2.8), and setting the anisotropy to zero, the well known stability requirement

$$\Delta t\left(\frac{1}{\Delta x^2} + \frac{1}{\Delta y^2}\right) \leq \frac{1}{2m} \tag{2.10}$$

holds in two space dimensions. If the forward Euler scheme were to be applied in stage 2 as well m would be replaced by one in (2.10) (again neglecting non-linear terms). Since m is typically smaller than one more severe restrictions on the time step would be obtained. Because the non-linear contributions are zero away from diffuse interfaces (2.10) is an upper bound on the largest time step allowed.

As will be seen in Section 3.1 the time stepping scheme based on equation splitting is limited to first order accuracy in time. This is a consequence of the splitting, and is true also if more accurate methods are used in stages 1 and 2 above.

2.2.2 Backward difference formulas

We wish to relax the stability restriction (2.10) and increase the order of accuracy. For this purpose the backward difference schemes developed for stiff systems of ordinary differential equations will be considered [18]. For a general ODE $y'(t) = f(t, y)$ the BDF-methods take the form

$$\sum_{i=0}^{k} \alpha_i^{(k)} y\big((n-i)\Delta t\big) = \Delta t f\big(n\Delta t, y(n\Delta t)\big),$$

where $1 \leq k \leq 6$ is the order of accuracy and $\alpha_i^{(k)}$ are known constants. Generalization to adaptive time stepping is relatively straight-forward, and will

introduce a dependence of $\alpha_i^{(k)}$ on the time steps [17]. The BDF-methods have desirable stability properties. In the sense of theory of numerical methods for ODE, the first and second order methods are A-stable, and methods of order up to and including six are $A(\alpha)$-stable [17, 18]. BDF(1) is the well-known backward Euler method.

BDF(k) is a k-step multi-step method and the order of accuracy is determined by the number of previous solutions used to approximate the time derivative. Thus one can obtain a high order of accuracy in the temporal discretization with little extra computational work at the expense of increased storage requirements. As will be seen in Section 3.4 the extra storage required is small compared to that used by the non-linear solver. Solutions must be known at k previous time levels in order to apply the BDF(k)-method. To avoid having to specify initial data on several time levels, simulation is typically initiated with $k = 1$. In each successive time step k is then increased until the desired accuracy is obtained.

Applying BDF-methods to (2.7) results in

$$\sum_{i=0}^{k} \alpha_i^{(k)} \begin{bmatrix} \Phi^{n-i} \\ U^{n-i} + p(\Phi^{n-i})/\Delta \end{bmatrix} - \Delta t \begin{bmatrix} mD_\phi(\Phi^n) & 0 \\ 0 & D_u \end{bmatrix} \begin{bmatrix} \Phi^n \\ U^n \end{bmatrix} - \Delta t \begin{bmatrix} F(\Phi^n, U^n) \\ 0 \end{bmatrix} = 0. \tag{2.12}$$

The BDF schemes are implicit, and Φ^n and U^n appear as unknowns in the non-linear equation (2.12). To advance the solution in time it is necessary to solve a non-linear system each time step.

Other higher order schemes have also been considered. These include explicit Adams and Runge-Kutta methods, as well as implicit Adams methods and singly diagonally implicit Runge-Kutta methods. Numerical experiments for the one-dimensional spherically symmetric phase-field equations indicated that the BDF-methods resulted in solutions to a certain accuracy in the shortest execution time.

2.3 Non-linear solver

2.3.1 Newton iteration

To formulate an algorithm for solving (2.12) we rewrite the equation as $G(\xi) = 0$, where the vector ξ contains the unknown grid values $\phi_{i,j}^n$ and $u_{i,j}^n$ at time $n\Delta t$. A sequence of approximate solutions $\{\xi_i\}$ can be obtained by Newton iteration,

$$\xi_{i+1} = \xi_i - \lambda_i \left(\frac{\partial G}{\partial \xi}(\xi_i) \right)^{-1} G(\xi_i) =: \xi_i + \lambda_i d_i, \tag{2.13}$$

given an initial guess ξ_0. For $\lambda_i = 1$ the classical Newton's method is obtained, and by allowing other λ_i we get the damped Newton iteration. Newton's method is attractive because it guarantees quadratic convergence in the vicinity of the

root for non-singular Jacobians. However, if the initial guess is far from the root so that the linear approximation of $G(\xi)$ is poor, $\{\xi_i\}$ may converge slowly, irregularly or not at all. In the present context this corresponds to a poor initial guess to the solution at the next time level, which will occur more frequently as the time steps are increased.

In the line search framework damping factors λ_i are determined adaptively such that acceptable next iterates are found, thereby increasing the domain of convergence while retaining quadratic convergence in the vicinity of the root [11]. We begin by observing that solving the non-linear system can be transformed into the equivalent problem of finding zeros to

$$g(\xi) = \frac{1}{2}\|G(\xi)\|_2^2,$$

and modifications to Newton's method can be derived by considering methods for determining local minima of g.

Noting that d_i is a descent direction of g at ξ_i, the residual norm is guaranteed to decrease for λ_i sufficiently small. We formulate the condition

$$g\big(\xi_i + \lambda_i d_i\big) \leq g(\xi_i) + \alpha\lambda_i \nabla g(\xi)^T d_i, \qquad \alpha \in (0,1), \tag{2.15}$$

which can be interpreted as requiring the average rate of decrease in g to be at least some fraction α of the initial rate of decrease. Condition (2.15) does not impose any constraints on the λ_i, and may result in $\lambda_i \ll 1$. We formulate the additional constraint that the rate of decrease of g in the direction d_i at the next iterate should be larger than some prescribed fraction of the rate of decrease at the current iterate,

$$\nabla g\big(\xi_{i+1}\big)^T(\xi_{i+1} - \xi_i) \geq \beta \nabla g\big(\xi_i\big)^T(\xi_{i+1} - \xi_i), \qquad \beta \in (\alpha, 1). \tag{2.16}$$

By (2.13) the expression can be formulated in terms of λ_i and d_i.

We refer to (2.15) and (2.16) as the α- and β-conditions. It can be shown that for $\alpha < 1/2$, the α-and β-conditions can be satisfied simultaneously [11]. Furthermore, if the local convergence assumptions underlying Newton's method are met the sequence $\{\xi_i\}$ converges super-linearly to a minima of g for λ_i satisfying (2.15) and (2.16). For i greater than some i_0, the convergence rate will be quadratic [11].

It remains to determine λ_i such that the α- and β-conditions are satisfied which is done by backtracking. In each Newton iteration we begin by investigating whether (2.15) and (2.16) are satisfied for the full Newton step $\lambda_i = 1$. If this is not the case, we note that some information on g is known from the investigation. By fitting an interpolating polynomial in λ_i to the known data we obtain an analytic expression from which to determine the next value of λ_i. After having tried the full Newton step this polynomial is of second degree, and in successive backtrackings a third degree polynomial is used. A simple version of the backtracking line search modification which does not enforce the directional derivative condition (2.16) is given in Algorithm 1.

The actual backtracking is performed in the inner loop of the algorithm, and in its absence the classical Newton iteration is obtained. Some heuristic limits on how much λ_{temp} is allowed to change in a single backtrack are used in the actual implementation to handle the case when the interpolating polynomial is a poor model of $g(\xi_i + \lambda_i d_i)$. To enforce the β-condition an additional iterations is required. It has been omitted to avoid too much technicalities. Full algorithmic descriptions can be found in [11].

It should be noted that Algorithm 1 contains multiple evaluations of the function $g(\xi)$ for the same value of ξ. To reduce the amount of computation temporary storage is used in the actual implementation. Since g is a scalar this is of no importance to the total storage requirements of the code. Also note that ∇g can be rewritten in terms of $G(\xi)$ and $G'(\xi)$ and that no specific function is needed for its computation.

Figure 1 shows the number of iterations required to solve the scalar equation $G(\xi) = \xi - \cos\xi = 0$ as the initial guess approaches the point where $G'(\xi) = 0$. For the worst case Newton's method converges in 962 iterations, whereas algorithm 1 yields convergence in at most 6 iterations. If the β-condition is enforced as well convergence is obtained in 5 iterations for all values of ξ_0.

For purposes of dendritic growth simulation the initial guess ξ_0 is determined by a predictor [17]. The convergence rate of the non-linear solver is to some extent dependent on the initial guess since quadratic convergence is guaranteed only in the vicinity of the root. It is therefore desirable to determine accurate approximations of Φ^n and U^n. Moreover, the predictor should require few floating point operations since it is applied once each time step. Using the solution at time level $n-1$ is a simple predictor which does not require any additional computation.

Regardless of the predictor chosen the initial guess becomes a worse approximation to the solution of (2.12) as the time step increases. With the increased domain of convergence resulting from the backtracking line search modification larger Δt can be chosen while guaranteeing convergence of the Newton iteration.

For practical purposes we are usually not interested in solving the non-linear system to full machine precision. Instead we wish to solve the system to such an accuracy that error contributions from the solver are small compared to discretization errors. For the simulations presented here the maximum norm of the residual is required to be below a certain tolerance before convergence is said to have occurred.

We now proceed by discussing how to compute Newton directions by use of Krylov subspace methods.

2.3.2 Krylov subspace methods

By solving the non-linear system (2.12) using the modified Newton iteration described in the previous section the numerical scheme has been reduced to solving linear systems, $A\mathbf{x} = \mathbf{b}$. This operation is expensive compared to the other operations in the Newton iteration and expected to account for a large fraction of the total run time. It is therefore important to find efficient methods

Choose initial guess ξ_0 and $0 < \alpha < 1/2$
while (not convergence)
  Compute $\mathbf{d}_0 = -\big(G'(\xi_0)\big)^{-1} G(\xi_0)$
  Let $\lambda = 1$, $s = \big(\nabla g\big)^T \mathbf{d}_0$ and $\xi = \xi_0 + \lambda \mathbf{d}_0$
  while $\big(g(\xi) \geq g(\xi_0) + \alpha\lambda s\big)$
    if $\lambda = 1$ /* first iteration, quadratic fit */
      $\lambda_{temp} = -s/\big(2(g(\xi) - g(\xi_0) - s)\big)$
    else /* cubic fit */
      $$\begin{bmatrix} a \\ b \end{bmatrix} = \frac{1}{\lambda - \lambda^{(p)}} \begin{bmatrix} 1/\lambda^2 & -1/(\lambda^{(p)})^2 \cdot \\ -\lambda^{(p)}/\lambda^2 & \lambda/(\lambda^{(p)})^2 \end{bmatrix} \cdot \begin{bmatrix} g(\xi) - g(\xi_0) - \lambda s \\ g(\xi^{(p)}) - g(\xi_0) - \lambda^{(p)} s \end{bmatrix}$$
      if $(a = 0)$ /* no cubic term */
        $\lambda_{temp} = s/(2b)$
      else
        $\lambda_{temp} = (\sqrt{b^2 - 3a} - b)/(3a)$
      end
    end
    Let $\lambda^{(p)} = \lambda$ and $\xi^{(p)} = \xi$
    Let $\lambda = \lambda_{temp}$ and $\xi = \xi_0 + \lambda \mathbf{d}_0$
  end
  /* α-condition satisfied */
  Let $\xi_0 = \xi$
end

Algorithm 1: Backtracking line search modification of Newton's method.

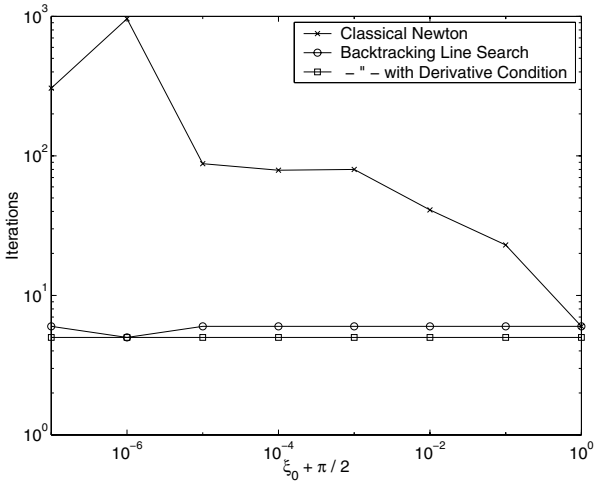


Figure 1: Number of iterations needed to solve $\xi - \cos\xi = 0$ as initial guess $\xi_0 \to -\pi/2$. Logarithmic scale is used on both axes.

for solving the linear systems.

If grid values are mapped into vectors using the natural ordering the matrices arising from the discretization will be sparse and banded. The matrix $A \in \mathbb{R}^{2MN \times 2MN}$ has upper and lower bandwidths MN, and fewer than $16MN$ nonzero elements. If the system were to be solved using direct methods one would typically choose an different ordering of the elements in order to reduce the bandwidth. Even in this case the storage requirements of a direct method becomes prohibitive, and instead Krylov subspace projection methods have been considered [28].

This class of methods has several attractive properties, and can be given special interpretation in the context of Newton iteration. The methods have in common that approximate solutions $\mathbf{x}_m$ are found in subspaces $\mathbf{x}_0 + \mathcal{K}_m$ of dimension m, where $\mathbf{x}_0$ is some initial guess,

$$\mathcal{K}_m(A, \mathbf{r}_0) = \text{span}\{\mathbf{r}_0, A\mathbf{r}_0, A^2\mathbf{r}_0, \ldots, A^{m-1}\mathbf{r}_0\}$$

is the Krylov subspace, and $\mathbf{r}_0 := \mathbf{b} - A\mathbf{x}_0$ is the initial residual. The approximations satisfy a Petrov-Galerkin condition $\mathbf{b} - A\mathbf{x}_m \perp \mathcal{L}_m$, where $\mathcal{L}_m$ is a subspace of dimension m whose basis will depend on the method chosen. We note that the basis of the Krylov subspace is computed by repeated matrix-vector multiplications, and in fact most of the work is needed to compute and orthogonalize this basis.

The choice of Krylov method is to large extent determined by the matrix A. Since the matrices considered here are non-symmetric the Generalized Minimal Residual Method (GMRES) [28, 30] has been chosen. For GMRES one takes $\mathcal{L}_m = A\mathcal{K}_m$ resulting in approximations such that the residual norm is minimized over all vectors in $\mathbf{x}_0 + \mathcal{K}_m$ [28]. Other possible choices include the Conjugate Gradient Squared method (CGS), the Biconjugate Gradient Stabilized method (Bi-CGSTAB) and the Quasi-Minimal Residual method (QMR). QMR requires the matrix transpose to form $\mathcal{L}_m$ and has not been considered since some additional difficulties are involved for efficient parallel implementation [2]. CGS and Bi-CGSTAB are associated with smaller storage requirements than GMRES, but need twice as many matrix-vector products per iteration. A thorough exposition of Krylov methods and related topics can be found in [28].

Although there is an upper bound on the number of iterations needed to find the solution in exact arithmetics, Krylov methods are often used to find a solution to a given tolerance. Otherwise the work would most likely exceed that of a direct solver, and for GMRES the storage requirements increase linearly with the number of iterations since the entire Krylov basis has to be stored. To limit storage requirements restarting is used. An upper limit on the dimension of the Krylov subspace is determined and if convergence has not occurred when this limit is reached, iteration is restarted with the current approximation as a new initial guess. With restarting convergence of GMRES is no longer guaranteed but the approach works well for most cases.

The convergence of Krylov methods is strongly dependent on the spectra of the matrix A. Often more tractable problems can be obtained by preconditioning the system, i.e. by solving the system $M_L^{-1} A M_R^{-1} \tilde{\mathbf{x}} = M_L^{-1} \mathbf{f}$, where the matrices M_L and M_R are the left and right preconditioners. The solution to the original system is given by $\mathbf{x} = M_R^{-1} \tilde{\mathbf{x}}$. By appropriately choosing preconditioners which can be efficiently evaluated the convergence rate can be greatly increased at a low computational cost. In a typical application preconditioning is performed by solving linear systems rather than explicitly computing the inverses.

For all computations presented here left Jacobi preconditioning has been used. The preconditioning matrix M_L is taken to be the main diagonal of A. Thus the number of floating point operations needed to apply the preconditioner scales linearly with the dimension of the system. In a distributed parallel setting the Jacobi preconditioning scales perfectly.

A preconditioned version of GMRES is given in Algorithm 2. The minimization problem giving $\mathbf{y}_m$ is cheap to solve since the matrix $\bar{H}_m \in \mathbb{R}^{(m+1)\times m}$ with non-zero elements $h_{i,j}$ is upper Hessenberg. Thus the minimizer can be computed by performing m Givens-rotations followed by a backwards substitution on a triangular system of dimension m. If the algorithm should break down, i.e. any of the $h_{j+1,j} = 0$, the exact solution is contained in the current Krylov subspace.

In the context of Newton iteration there are a few additional points to be noted. The matrix appearing in the linear systems is the Jacobian $\partial G/\partial \xi$. Since Krylov methods require only the action of the matrix on vectors we can use finite

Choose initial guess $\mathbf{x}_0$
while (not convergence)
  Let $\mathbf{r} = M_L^{-1}(\mathbf{b} - A\mathbf{x}_0)$
  Let $\beta = \|\mathbf{r}\|_2$
  Let $\mathbf{v}_1 = \mathbf{r}/\beta$
  for $j = 1, 2, \ldots, m$
    Compute $\mathbf{w} = M_L^{-1} A M_R^{-1} \mathbf{v}_j$
    for $i = 1, 2, \ldots, j$
      $h_{i,j} = \mathbf{w}^T \mathbf{v}_i$
      $\mathbf{w} = \mathbf{w} - h_{i,j}\mathbf{v}_i$
    end
    Compute $h_{j+1,j} = \|\mathbf{w}\|_2$
    Compute $\mathbf{v}_{j+1} = \mathbf{w}/h_{j+1,j}$
  end
  Let $V_m = [\mathbf{v}_1, \ldots, \mathbf{v}_m]$
  Let $\bar{H}_m = [h_{i,j}]$ (zeros for $h_{i,j}$ not computed above)
  Compute $\mathbf{y}_m = \text{argmin}_y \|\beta \mathbf{e}_1 - \bar{H}_m \mathbf{y}\|_2$
  Let $\mathbf{x}_0 = \mathbf{x}_0 + M_R^{-1} V_m \mathbf{y}_m$
end

Algorithm 2: Preconditioned GMRES(m) iteration for solving $A\mathbf{x} = \mathbf{b}$.

difference approximations to avoid explicit representation of the Jacobian, e.g.

$$\left(\frac{\partial G}{\partial \xi}(\xi_i)\right) \cdot v \approx \frac{G(\xi_i + \delta v) - G(\xi_i)}{\delta}, \quad 0 < \delta \ll 1.$$

For the phase-field equations the evaluation of $G(\xi)$ is expensive compared to computing matrix-by-vector products. In most simulations the Jacobian is therefore explicitly formed, despite the increased storage requirements.

As previously noted in Section 2.3.1 the linear behavior of $G(\xi)$ assumed in the formulation of Newton's method may be a poor model far from the root. For this reason it is not always necessary to compute the exact Newton direction, and a "sufficiently accurate" approximation will do. It is possible to relate the accuracy needed in GMRES to the residual norm in the current Newton step, and formulate sufficient conditions for super-linear or quadratic convergence [8]. In [7] Brown and Saad use this to formulate non-linear Krylov algorithms with stronger coupling between the linear and non-linear solvers. Among the algorithms considered is a Newton-GMRES algorithm based on backtracking

line search techniques.

2.3.3 Orthogonalization scheme in GMRES

The inner loop of Algorithm 2 orthogonalizes the newly computed basis vector $\mathbf{w}$ of the Krylov subspace against all previously computed vectors $\{\mathbf{v_1}, \ldots, \mathbf{v_j}\}$. There are several mathematically equivalent formulations of which perhaps the two most famous are the Classical Gram-Schmidt procedure (CGS),

$$\begin{aligned} &\mathbf{w} = \mathbf{w} - [\mathbf{v_1}, \ldots, \mathbf{v_n}] \cdot \left([\mathbf{v_1}, \ldots, \mathbf{v_n}]^T \mathbf{w} \right) \\ &\text{Let } \mathbf{v}_{j+1} = \mathbf{w} / \|\mathbf{w}\|_2 \end{aligned}$$

and the Modified Gram-Schmidt procedure (MGS) given in Algorithm 2,

$$\begin{aligned} &\textbf{for } i = 1, 2, \ldots, j \\ &\quad \mathbf{w} = \mathbf{w} - (\mathbf{w}^T \mathbf{v}_i) \mathbf{v}_i \\ &\textbf{end} \\ &\text{Let } \mathbf{v}_{j+1} = \mathbf{w} / \|\mathbf{w}\|_2 \end{aligned}$$

From a numerical point of view MGS is to prefer since it is less sensitive to round-off errors [5]. For computational purposes CGS is expected to perform better since it is formulated using level 2 linear algebra operations instead of level 1 [16]. In a parallel setting the differences become even more important. MGS requires $m(m+1)/2$ global reductions for each outer iteration with Krylov dimension m. With CGS only $2m$ reductions are needed per outer iteration. Since $m \approx 5 - 20$ is common the CGS algorithm scales better than MGS for increasing number of processors [13].

The one-dimensional numerical experiments presented in Section 3.1 have been carried out using the GMRES implementation from SPARSKIT which utilizes MGS orthogonalization [27]. The two-dimensional timings presented in Section 3.3 are based on the GMRES routine from PSPARSLIB where CGS orthogonalization is employed [29].

One should note that there are several other algorithms for computing orthogonal subspaces, including QR-factorization and basis generation based on Newton polynomials. In computations not described here iterative versions of CGS and MGS guaranteeing that computed subspaces are numerically orthogonal have been investigated. The iterative versions had little or no impact on the convergence rate of GMRES, and have therefore not been used in this paper.

2.3.4 Numerical examples

Figures 2 and 3 show two examples of the non-linear solver applied to a typical system arising from dendritic solidification simulation in two dimensions. Data is extracted from computation after the fastest transients have vanished, and inserted in (2.12) resulting from a BDF(1)-scheme. The benefits of the backtracking line search modification are clear from Figure 2. The Euclidean

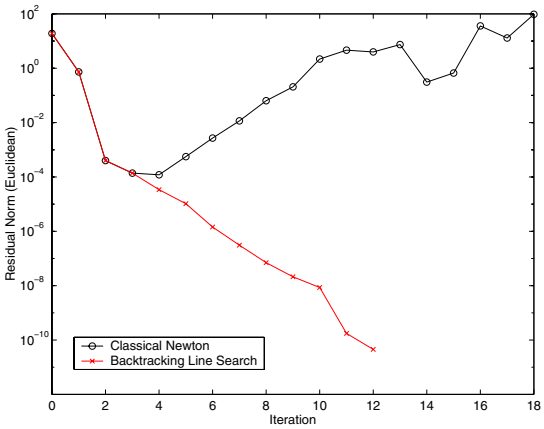


Figure 2: Convergence of Newton-GMRES algorithm for a typical system (2.12) arising from dendritic solidification simulation.

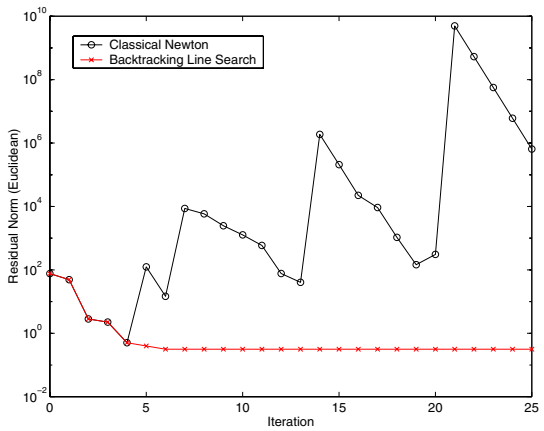


Figure 3: Convergence of Newton-GMRES algorithm applied to (2.12) with twice as large time step as in Figure 2.

norm of the residual exceeds $4 \cdot 10^{16}$ in the 19th iteration for the classical Newton algorithm, whereas Algorithm 1 converges to the desired tolerance in ten iterations.

Figure 3 shows convergence rates for the same data but with the time step twice as large as in Figure 2. The backtracking line search modification does not yield a solution to the given tolerance due to lack of convergence in the GMRES iteration. Such behavior can be observed for many computations. For purposes of simulation the upper bound on the time step is therefore often determined by the convergence of GMRES rather than the stability restrictions of the BDF-scheme. By increasing the dimension of the Krylov subspace the system can be solved also for the larger time step. Since this approach is associated with increased storage requirements, as discussed in Section 3.4, it is not practical for large grids.

To fully exploit the relaxed stability restrictions of the BDF-scheme it would be necessary to find a better preconditioner than the Jacobi preconditioning applied here. Some experiments have been performed using incomplete LU factorization [28, 29] and sparse approximate inverse [9, 10] techniques. Improved convergence rates could be observed but because of additional administration associated with the more elaborate preconditioners only minor reductions in simulation time were achieved. On parallel architectures the increased communication lead to little or no gain in simulation time.

2.4 A fast Poisson solver

For the scheme based on equation splitting only one linear system must be solved per time step. The GMRES method described in Section 2.3.2 can be used, but under certain simplifying assumptions more efficient algorithms can be found. We note for example that with natural ordering of the grid-points in one space-dimension it is possible to define a second order accurate difference operator D_u resulting in tridiagonal systems. Thus a banded solver from e.g. LAPACK can be used instead of iterative methods [4].

In two space dimensions a full banded solver should not be used directly. For the standard five-point stencil with natural ordering the matrix of dimension MN has upper and lower bandwidths M, and fewer than $5MN$ non-zero elements. The storage requirements of a banded solver becomes prohibitive unless a different mapping of grid-values to column vectors is used, or equivalently, a sparse banded solver is employed. Using fast Poisson techniques one can however reduce the bandwidth prior to solving the system thus allowing the use of efficient LAPACK routines [23].

To formulate an algorithm we note that any function $f_{i,j}$ defined on the uniform grid can be written

$$f_{i,j} = \frac{\widehat{f}_{i,j}(0)}{N} + \frac{2}{N} \sum_{k=1}^{N-1} \widehat{f}_{i,j}(k) \cos\big(\omega(2j+1)k\big), \quad \omega = \frac{\pi}{N-1}, \tag{2.19}$$

via a discrete cosine transformation (DCT) in the j-direction [32]. For known

functions f the coefficients can be explicitly computed,

$$\widehat{f}_{i,j}(k) = \sum_{j=0}^{N-1} f_{i,j} \cos\big(\omega(2j+1)k\big).$$

By equation (2.19) difference operators acting in the j-direction can be replaced by algebraic expansions. For example,

$$u_{i,j+1} - 2u_{i,j} + u_{i,j-1} = \frac{2}{N} \sum_{k=1}^{N-1} -4\sin^2(\omega k)\widehat{u}_{i,j}(k) \cos\big(\omega(2j+1)k\big).$$

Applying a DCT to (2.9) therefore results in a tridiagonal linear system for the $\widehat{u}_{i,j}$. The system can be solved by LAPACK routines, and the temperature $u^n_{i,j}$ determined from (2.19). In principle one could perform a DCT in the i-direction as well, which would result in a diagonal system. The amount of work needed to compute the DCT is larger than the amount of work needed to solve the tridiagonal system. Hence the second transform is not performed.

The use of the DCT guarantees that zero Neumann conditions are satisfied at the boundaries. It is possible to extend the algorithm such that inhomogeneous Neumann conditions are handled as well. Dirichlet conditions can be handled by the discrete sine transform rather than the DCT. We note however that the fast Poisson solver relies heavily on the uniform spatial discretization, and constant coefficients in the difference operator. It can be generalized for some special cases, such as adaptively refined grids where each refinement is uniform [19].

The DCT of a vector of length n is most efficiently computed using a complex Fast Fourier Transform (FFT) of length $n/2$, where the elements are obtained by permuting data in the original vector [23]. Optimized FFT routines are available in the public domain. Here we have chosen to work with the FFTW library [14]. Apart from a constant factor the DCT is it's own inverse, so the programming effort needed to implement a fast Poisson solver given an FFT-routine is small. The storage requirements can be kept low by using in-place transforms, where the input vector to the FFT-routine is overwritten by its Fourier transform.

A parallel version of the fast Poisson solver has been implemented but due to permutation of the data vectors a low parallel efficiency is obtained. It has not been used for any simulations presented here.

3 Computational Results

3.1 Order of accuracy

We investigate the order of accuracy in time by considering the semi-discrete system (2.7) in the spherically symmetric case. Dimensionless parameters are chosen to agree with pure nickel for a certain undercooling, $\alpha = 400$, $m = 0.05$ and $\Delta = 0.5$ [34]. It is necessary to have $\epsilon \lesssim 0.005$ to avoid unphysical oscillations in the phase field variable for this particular set of parameters [1, 15].

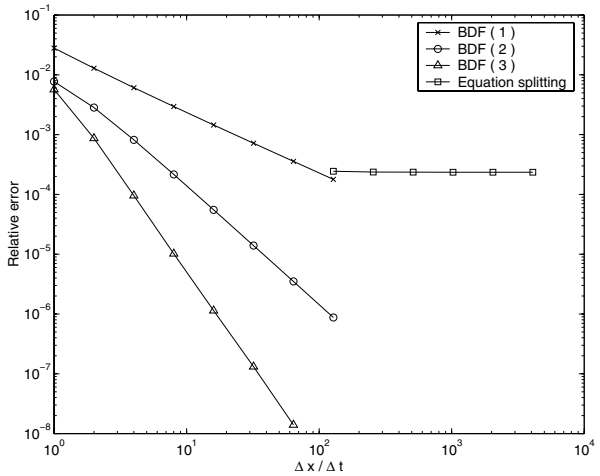


Figure 4: Approximate relative errors in computed solutions measured in Euclidean norm.

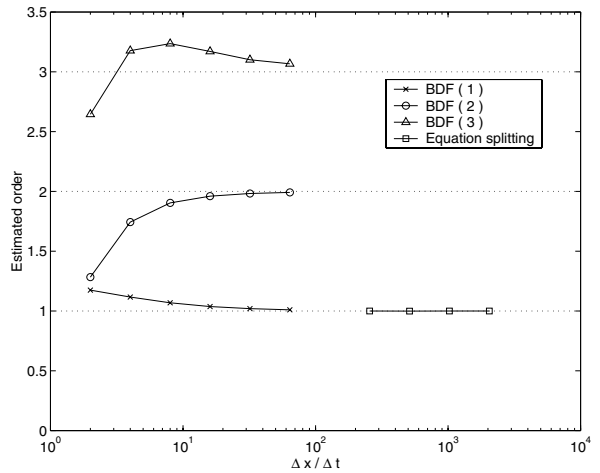


Figure 5: Order of convergence estimated by (3.2) using Euclidean norm.

We choose $\epsilon = 0.0025$, which is the smallest value considered in [34]. The results presented here have been obtained in the sharp interface limit of the asymptotics [3]. The errors and asymptotic convergence rates depicted in Figures 4 and 5 are almost identical when the thin interface correction is employed.

Computations are performed on the domain $[0, 2]$ using the second order accurate discretization discussed in Section 2.1, with $\Delta x = \epsilon/2$. The phase field equations are solved in time until $T = 0.13$ starting from a spherical solid of radius $r_0 = 0.5$ and uniform temperature $u = -1$. Initial data is needed on more than one time level for BDF-methods of order two or higher. We first solve the phase field equations for $0 \leq t \leq 0.03$ using a BDF(3)-method with half the smallest step size used in the convergence experiments. By saving the computed solutions at appropriate time levels, simulation can then be initiated from $t = 0.03$ with accurate initial data on all time levels. At this time the fastest initial transients have vanished.

When computations are initiated at $t = 0.03$ the solid-liquid interface is located at $r \approx 0.72$. At the final time $t = 0.13$ the location is $r \approx 1.02$, so the interface is moving with an average dimensionless speed 3. The grid size and integration time have been chosen such that the temperature at the outer boundary $r = 2$ remains close to -1 throughout the simulation.

Time stepping for the approach based on equation splitting uses the forward Euler method for the phase field as in (2.8), and the Crank-Nicholson scheme to update temperature. Due to stability smaller time steps have been chosen than for the implicit scheme. The one dimensional case is particularly simple since the linear systems are tridiagonal, and can be solved directly without using a fast Poisson solver. The number of floating point operations per time step is proportional to the number of grid points.

A number of additional parameters related to the non-linear solver must be set. These parameters have an impact on the convergence of the iterative methods, and hence the total execution time of the program. To ensure that errors from the solver are small compared to errors from discretization we set tolerances close to the limits imposed by IEEE floating point arithmetics. The non-linear systems are solved by the backtracking line search modification of Newton's method without enforcing the directional derivative condition. Iteration is interrupted when the discrete maximum norm of the residual (2.12) is smaller than 10^{-12}. We iterate over Krylov subspaces of dimension 20 using an implementation of GMRES taken from SPARSKIT [27] to determine Newton directions. Left Jacobi preconditioning is employed and convergence is said to have occurred when the Euclidean vector norm of the residual is smaller than 10^{-14}.

Newton's method typically converges in 3-5 iterations. With the possible exception of the first one or two iterations convergence is quadratic. The backtracking line search modification significantly reduces the number of iterations for the larger time steps considered.

Figure 4 shows approximate relative errors in the computed solutions. The values were obtained by comparison to the solution expected to be the most accurate, that is, the one computed with the smallest time step and highest order

of accuracy. Attempts to compare results from phase field simulations to sharp interface solutions computed using a numerical method from [31] have been made. As the asymptotic order of convergence of the sharp interface scheme was found to be 1/2, and a more accurate scheme would necessitate solving a large full system of non-linear equations this approach was not pursued further.

We expect the norm of the error in computed solutions to behave as

$$c\Delta t^p + \mathcal{O}(\Delta t^{p+1}) \tag{3.1}$$

for a fixed time as $\Delta t \downarrow 0$ since ϵ and the grid spacing are kept fixed. By determining the constants c and p, errors from temporal discretization of the semi-discrete system can be related to errors from spatial discretization.

The order of convergence is estimated by

$$p = \log_2 \left(\frac{\left\| [\Phi^N_{2\Delta t} \; ; \; U^N_{2\Delta t}] - [\Phi^N_{\Delta t} \; ; \; U^N_{\Delta t}] \right\|}{\left\| [\Phi^N_{\Delta t} \; ; \; U^N_{\Delta t}] - [\Phi^N_{\Delta t/2} \; ; \; U^N_{\Delta t/2}] \right\|} \right), \tag{3.2}$$

where the superscript denotes the time level at which the solutions are compared, and the subscript denotes the time step used (c.f. Richardson extrapolation). $[\Phi; U]$ should be interpreted as a column vector containing the grid values of ϕ and u. The estimate is relevant for sufficiently small time steps when the leading order error term dominates over higher order terms in (3.1). Equation (3.2) is attractive because it estimates p by comparing computed solutions, and does not require the actual error.

Estimated convergence rates are shown in Figure 5. The order of the fully implicit discretizations approach the order of the underlying BDF-methods as Δt is decreased. The scheme based on equation splitting is first order accurate but seems to converge to a slightly different solution, see Figure 4. These results were obtained for specific dimensionless parameters in the phase field model, but the asymptotic convergence rates have been observed for other choices as well. Almost identical curves are obtained when other norms, such as the maximum norm, are used in (3.2).

The error constant c in (3.1) can be approximated by

$$c = \frac{\left\| [\Phi^N_{2\Delta t} \; ; \; U^N_{2\Delta t}] - [\Phi^N_{\Delta t} \; ; \; U^N_{\Delta t}] \right\|}{(2^p - 1)\Delta t^p}, \tag{3.3}$$

where p is the limit of (3.2) as $\Delta t \downarrow 0$. From Figure 5 we see that p is close to its limit only for the smaller time steps, and hence relevant estimates of c can not be obtained for the larger Δt.

Approximated values of c and p for the finest time step are given in Table 1. The differences compared to the next finest time step are below 5% for all schemes. For the BDF-methods the error constant grows by a factor of approximately 100 as the order is increased. Since the largest time step considered is $1.25 \cdot 10^{-3}$ one benefits from using a more accurate BDF-method, which is also seen from Figure 4.

Scheme	c	p
BDF(1)	$3.4 \cdot 10^2$	1.01
BDF(2)	$8.4 \cdot 10^4$	1.99
BDF(3)	$9.8 \cdot 10^6$	3.07
Equation splitting	$1.2 \cdot 10^2$	1.00

Table 1: Order of convergence and error constant estimated by (3.2) and (3.3) for the finest time step. Errors are measured in Euclidean norm of the solution vector.

The number of floating point operations needed for the scheme based on equation splitting is inversely proportional to the time step. With all output from the program removed slightly more than a minute is needed for the largest time step. It is more difficult to predict the execution time for the implicit scheme since it depends on the convergence of iterative methods. Initially the run time decreases with the time step since the predictor yields a more accurate initial guess for Newton's method. More non-linear systems must be solved as the time step is decreased, and after a certain point the execution time increases. The second order accurate BDF-scheme requires approximately 90 minutes for the time step $\Delta x/\Delta t = 8$. The execution time can be reduced to 27 minutes with a relative increase of 1% in the error by changing the tolerances to 10^{-6} and 10^{-8} in the non-linear solver and GMRES respectively. Increasing the tolerances by a factor 1000 more results in a 38% relative increase of the error, and a total run time of 13 minutes. The run times can probably be reduced further by employing a more efficient preconditioner.

3.2 Errors in the propagation speed

The scheme based on equation splitting may fail to capture the correct propagation speed of the interface. Such behavior is more frequently observed as the propagation speed increases. Figure 6 shows results from a simulation performed with $\alpha = 400$, $m = 0.05$, $\Delta = 1.0$, and $\epsilon = \Delta x = 0.005$. Growth is initiated from a spherical seed of radius 0.5 and uniform undercooling. Time steps have been chosen smaller than or equal to $5 \cdot 10^{-5}$, which is within the stability domains of the considered schemes. The errors from the splitting approach are much larger for this case than for the one discussed above.

We note two things. There is a significant difference between the computed solutions for the two different time steps when the splitting approach is used, which can be explained by the first order accuracy. Secondly, the deviation in interface position increases with time which indicates that the interfacial velocity is not captured correctly.

The BDF(2)-discretization is less sensitive to the choice of time step, and reducing Δt by a factor 10 does not alter the plotted solution. To obtain visual

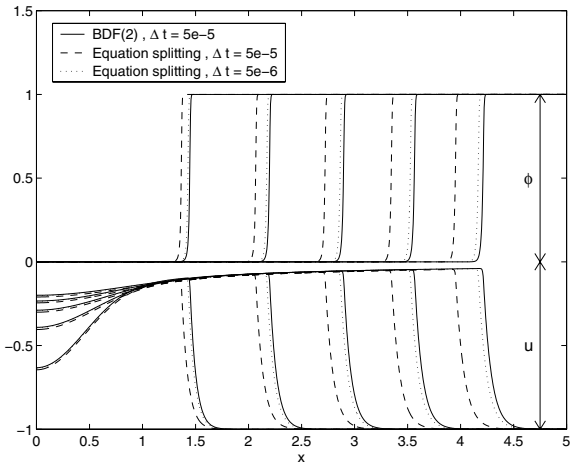


Figure 6: Phase field and temperature for a case where the scheme based on equation splitting fails to capture the correct propagation speed. The curves show the solution at times $0.05, 0.1, \ldots, 0.25$, moving from left to right.

agreement between the two schemes a Δt of the order 10^{-8} must be used in the scheme based on splitting. At this point the execution time is a few times larger than for the implicit approach.

3.3 Execution time in two space dimensions

The spherically symmetric case is sufficient for investigating the asymptotic order of convergence of the time stepping schemes, but we are interested in performing simulations in two and three space dimensions. The linear systems arising from the splitting approach will no longer be tridiagonal and can not be solved using direct methods. For purposes of simulation it is therefore more relevant to compare execution times to achieved accuracy in two space dimensions.

We choose the same physical parameters as above: $\alpha = 400$, $m = 0.05$ and $\Delta = 0.5$. In addition to these we set the anisotropy parameters in (2.3) to $\gamma = 0.015$, $k = 4$ and $\theta_0 = 0$. Simulations are performed on a 201×201-point discretization of the domain $[0,1] \times [0,1]$, with $\Delta x = \Delta y = \epsilon = 0.005$. The grid is coarser and ϵ larger than for the one dimensional simulations to allow computation on a single computer within a couple of hours.

To obtain initial data the phase field equations are solved for $0 \leq t \leq 0.015$ using a BDF(3)-method with time step $\Delta t = 7.8125 \cdot 10^{-5}$, starting from a phase field given by [3]

$$\phi(x,y,0) = \frac{1}{2}\left[\tanh\left(\frac{\sqrt{x^2+y^2}-0.05}{2\sqrt{2}\epsilon}\right)+1\right],$$

and $u(x,y) = -\phi(x,y)$.

In Figure 7 the achieved accuracy compared to a reference solution is plotted versus the execution time. The smallest time step used is given in the plot, and each successive data point corresponds to an increase in Δt by a factor two. It is possible to use larger time steps for the BDF(2)- and BDF(3)-schemes but large execution times are needed due to poor convergence in the GMRES iteration.

Figure 7 suggests that the fully implicit scheme is more efficient for accurate simulation. We note that the measured execution times exhibit the same dependence on tolerances in the non-linear solver observed in the one-dimensional simulations. By using the parallel implementation described in [2] the run times can be reduced further for the fully implicit scheme.

3.4 Storage requirements in two space dimensions

Extension to three space dimension implies significantly larger grids. It is therefore relevant to also compare the required storage for the two different time stepping schemes. Table 2 summarizes the amount of computer memory needed for the different algorithms.

The variations for the BDF-scheme depends on the exact algorithms used. Using a classical or damped Newton iteration instead of a the backtracking

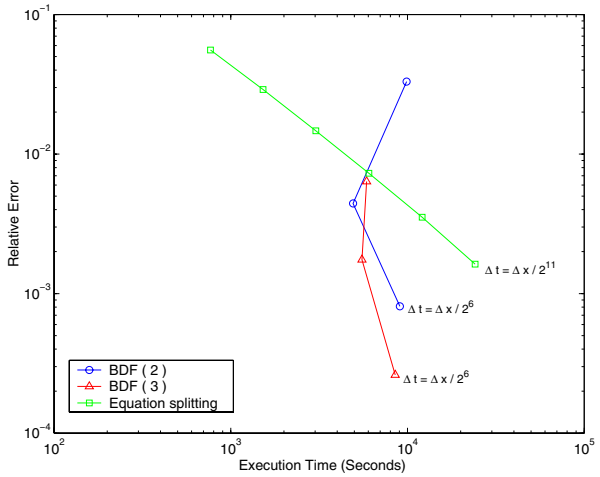


Figure 7: Relative error at time $t = 0.065$ compared to a reference solution computed using BDF(3) with $\Delta t = 7.8125 \cdot 10^{-5}$. The error is measured in Euclidean vector norm.

Purpose of storage	Size (`double` elements)	
	BDF(k)	Semi-explicit
Computed solution	$2(k+1)MN$	$3MN$
Explicit Jacobian	$(0-16)MN$	–
Newton solver	$(6-10)MN$	–
Linear solver	$(10+4m)MN$	$3(M+N)$
Preconditioner (Jacobi)	$(0-2)MN$	–
Total	$((18-40)+2k+4m)MN$	$3MN$

Table 2: Approximate storage requirements for the different time stepping schemes on a grid of $M \times N$ points. The storage requirements for GMRES(m) have been based on the PSPARSLIB implementation [29].

line search modification leads to smaller storage requirements. It is possible to use a finite-difference approximation of the Jacobian in the GMRES iteration, in which case the explicit Jacobian does not have to be stored at the expense of increased computational complexity. Precomputing the elements of M_L^{-1} reduces the execution time when the preconditioner is applied. It is however not necessary for the preconditioning to work. One should note that with typical values of m being between 5 and 20 the Krylov subspace accounts for most of the memory needed.

Since the semi-explicit scheme requires less storage it can be useful when doing computations on large grids with limited computational resources. It also renders smaller simulation in three dimensions possible without having to use adaptive grid refinement.

4 Conclusions

We have investigated two different approaches to time-stepping for numerical simulation of dendritic solidification. The first scheme is semi-explicit and computationally cheap per time step, both in terms of floating point operations and storage requirements. Since it is limited to first order accuracy in time it may be necessary to use small time steps to obtain accurate solutions. This is especially true when the solidification front propagates fast since the velocity may not be captured correctly.

A second scheme based on BDF-methods allows up to sixth order accuracy in time and relaxes the restrictions on the time step imposed by the semi-explicit scheme. It requires a non-linear system of equations to be solved each time step which for the experiments described here is done by a Newton-GMRES algorithm.

For sufficiently low error tolerances one benefits from using the fully implicit algorithm. On large grids the storage requirements may however become prohibitive. To fully exploit the relaxed stability restrictions one also needs to find a linear solver capable of handling larger time steps. Some attempts to use ILU and SPAI preconditioning techniques in the GMRES algorithm have been made but did not yield the desired results. Further investigation of preconditioners is therefore required to take full advantage of the implicit scheme. Alternatively, one could consider other linear solvers such as multi-level methods.

For efficient computation in realistic physical settings one should use the phase-field model in three space dimensions. Such an extension would require adaptive grid refinement to keep the storage requirements low. Some adaptivity should also be used in the time stepping.

We believe that the fully implicit scheme is more appropriate for simulation in three dimensions. The non-linear solver does not depend on any special properties of the grid or boundary conditions, and only the discretization would have to be changed. Furthermore, the relaxed stability restrictions are important for simulating solidification of alloys where additional time scales differing by several orders of magnitude are introduced in the equations.

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Paper 2

Computation of Dendrites on Parallel Distributed Memory Architectures

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Abstract. A code for simulating the solidification of a pure material from its undercooled melt based on a phase field approach has been written for parallel distributed memory architectures using MPI. The numerical scheme is based on finite differences and results in large sparse non-linear systems which are solved by a backtracking line search modification of Newton's method combined with GMRES. Experiments conducted on an IBM SP2 and networks of Sun Ultra 5 workstations show that the code scales well with the number of processors.

1 Introduction

We are interested in accurately simulating the solidification of a pure material from its undercooled melt taking into account effects of surface tension and kinetic undercooling. This is perhaps the simplest physical process in which one can observe the growth of dendritic crystals, characterized by complicated tree-like patterns. Some properties of solids such as electric conductivity and mechanical strength are determined by the micro structures produced upon solidification and one would like to better understand the pattern formation in order to obtain desired properties in manufacture.

The classical formulation of the solidification problem results in heat equations defined in the solid and melt, coupled with boundary conditions at the solid-melt interface. Solving this problem has proved difficult since it implies explicit tracking of a moving boundary. To avoid this difficulty the implementation has been based on a phase field approach.

By introducing a smooth approximation of a thermodynamic order function taking on distinct values in solid and melt, the so called phase field, a set of reaction-diffusion equations valid in the entire computational domain can be derived. The model used here is based on a derivation from basic thermodynamic principles [17] which ensures that the entropy of the system is monotonically increasing in time. In this formulation the moving boundary is replaced by a diffuse interface of width ϵ, in which the phase field changes value from zero (solid) to one (melt). One can prove by rigorous asymptotic analysis that the sharp interface model is recovered as the interface thickness

$\epsilon \to 0$ [3]. The governing equations can be formulated

$$\frac{\epsilon^2}{m}\frac{\partial \phi}{\partial t} = \phi(1-\phi)\big(\phi - \frac{1}{2} + 30\epsilon\alpha\Delta u\phi(1-\phi)\big) + \epsilon^2\nabla_a^2\phi, \tag{1}$$

$$\frac{\partial u}{\partial t} = -\frac{1}{\Delta}\frac{\partial}{\partial t}\Big(\phi^3(10 - 15\phi + 6\phi^2)\Big) + \nabla^2 u, \tag{2}$$

where $\phi(\mathbf{x},t)$ is the phase field and $u(\mathbf{x},t)$ is the dimensionless temperature scaled such that the melting temperature is 0 and the initial temperature is -1 on the boundaries. α, Δ and m are dimensionless positive physical constants. The Laplacian in (1) is non-linear in ϕ and has been modified to account for preferred growth directions in the material, see Sect. 2.1.

There is no need for explicit representation of the solid-melt interface since its location is implicitly given by level sets of ϕ, e.g. $\{\mathbf{x}(t) : \phi(\mathbf{x}(t),t) = 1/2\}$, which greatly simplifies the handling of merging interfaces and computation in two or three space dimensions. In the sharp interface formulation there is a discontinuity in the normal derivative of the temperature across the interface which due to the smoothness of the approximation is here replaced by a rapid transition. This makes it simpler to formulate numerical approximations.

However, in order to accurately resolve the interface the grid resolution must be of order ϵ which implies large computational grids since ϵ should be small in order to be close to the sharp interface formulation. Also, the introduction of the phase field variable increases the dimensionality of the problem. In three dimensions the solid-melt interface is given by a surface (2D), but to avoid explicit representation of this surface one introduces a variable defined in the entire computational domain (3D). These are the major drawbacks of the phase field approach and some motivations for writing a parallel implementation.

Several papers have been published in which both qualitative [11] and quantitative [1,9,19] agreement between phase field models and sharp interface formulations have been investigated. Physical problems such as solidification of binary alloys [18] and dendritic solidification under external flow [16] have also been addressed using a phase field approach. Other approaches include direct simulation of the sharp interface equations based on front tracking [15] or level set methods [4]. Some of the papers [1,9,15,16] describe methods based on adaptive meshing.

This implementation is part of an ongoing work aimed at finding computationally efficient numerical methods for simulation of dendritic growth. Since the phase field variable is not conserved an accurate numerical method must be used in order to obtain the correct propagation speed of the solid-melt interface. The main purpose here is to investigate whether an implicit second order scheme can be used to obtain reliable results within times comparable to semi-explicit first order schemes based on a splitting approach.

2 Numerical Method

In this section the numerical method used to solve (1) and (2) is introduced. We will first see how anisotropy can be included in (1) and complete the problem formulation by giving initial and boundary conditions. Via the method of lines the partial differential equations can be replaced by a large non-linear system of equations, which must be solved each time step. This is done using a modification of Newton's method, which basically reduces the numerical method to solving sparse linear systems. For this purpose the Generalized Minimal Residual Method (GMRES) [14] has been employed.

2.1 Mathematical Formulation

For many materials, including metals, the surface energy of the solid-melt interface will depend on orientation. To account for these effects in the phase field model one needs to modify the parameter ϵ for the gradient terms, or equivalently [19], use

$$\nabla_a^2 = -\frac{\partial}{\partial x}\Big(\eta(\theta)\eta'(\theta)\frac{\partial}{\partial y}\Big) + \frac{\partial}{\partial y}\Big(\eta(\theta)\eta'(\theta)\frac{\partial}{\partial x}\Big) + \nabla\big(\eta^2(\theta)\nabla\big), \qquad (3)$$

in (1), where η is the function used to model the anisotropy and θ is the angle between the normal to the interface and some reference direction,

$$\eta(\theta) = 1 + \gamma\cos(k\theta), \qquad \theta = \arctan\Big(\frac{\partial\phi/\partial y}{\partial\phi/\partial x}\Big) - \theta_0. \qquad (4)$$

γ determines the magnitude of anisotropy and k is the mode number. Since θ is determined by derivatives of ϕ the anisotropic Laplacian (3) is non-linear.

The phase field equations are solved on rectangular domains with zero Neumann conditions prescribed for both ϕ and u at the boundaries ($(\nabla\phi)\cdot\hat{n} = (\nabla u)\cdot\hat{n} = 0$). The implementation can handle a wide variety of initial conditions, the most common choice being

$$\phi(x,y,0) = \frac{1}{2}\Big[\tanh\Big(\frac{\sqrt{x^2+y^2} - r_0(\tan^{-1}(y/x))}{2\sqrt{2}\epsilon}\Big) + 1\Big], \qquad (5)$$

$$u(x,y,0) = -1, \qquad (6)$$

which corresponds to a solid whose shape is given (in polar coordinates) by the function r_0, and a uniform temperature distribution equal to the undercooling. The exact form of (5) is taken from the asymptotic approximation to the solution [3,17].

2.2 Discretization

Equations (1)–(4) have been discretized in space using second order finite differences on uniform Cartesian grids of $N_x \times N_y$ points,

$$\mathbf{x}_{i,j} = \big\{x_{min} + i\Delta x\,,\, y_{min} + j\Delta y\big\} \quad ,\ 0 \le i < N_x\ ,\ 0 \le j < N_y. \qquad (7)$$

The value of a gridfunction f defined on (7) in the gridpoint $\mathbf{x}_{i,j}$ will be denoted $f_{i,j}$. It is desirable to minimize the width of the stencil for several reasons, one being that the amount of communication needed to compute the action of the difference operator decreases. For this purpose approximations to certain gridfunctions will be needed between gridpoints, as denoted by non-integer indices in the formulas below. The standard five-point stencil can be used to approximate the ordinary Laplacian. For (3) we use,

$$\begin{aligned}\nabla_a^2\phi_{i,j} = &- D_{-x}\Big((\eta\eta')_{i+1/2,j} D_{0y}\frac{\phi_{i,j}+\phi_{i+1,j}}{2}\Big) \\ &+ D_{-y}\Big((\eta\eta')_{i,j+1/2} D_{0x}\frac{\phi_{i,j}+\phi_{i,j+1}}{2}\Big) \\ &+ D_{-x}\Big((\eta^2)_{i+1/2,j} D_{+x}\phi_{i,j}\Big) + D_{-y}\Big((\eta^2)_{i,j+1/2} D_{+y}\phi_{i,j}\Big),\end{aligned} \tag{8}$$

where the difference operators are defined as

$$D_{+x}f_{i,j} = \big(f_{i+1,j}-f_{i,j}\big)/\Delta x, \qquad D_{+y}f_{i,j} = \big(f_{i,j+1}-f_{i,j}\big)/\Delta y, \tag{9}$$

$$D_{-x}f_{i,j} = \big(f_{i,j}-f_{i-1,j}\big)/\Delta x, \qquad D_{-y}f_{i,j} = \big(f_{i,j}-f_{i,j-1}\big)/\Delta x, \tag{10}$$

$$D_{0x}f_{i,j} = \big(D_{+x}+D_{-x}\big)f_{i,j}/2, \qquad D_{0y}f_{i,j} = \big(D_{+y}+D_{-y}\big)f_{i,j}/2, \tag{11}$$

$\eta(\theta)$ is given by (4), and θ approximated by

$$\theta_{i+1/2,j} = \arctan\left(\frac{D_{0y}(\phi_{i,j}+\phi_{i+1,j})}{2D_{+x}\phi_{i,j}}\right) - \theta_0, \tag{12}$$

$$\theta_{i,j+1/2} = \arctan\left(\frac{2D_{+y}\phi_{i,j}}{D_{0x}(\phi_{i,j}+\phi_{i,j+1})}\right) - \theta_0. \tag{13}$$

between the gridpoints. This expression requires all eight neighboring points, which should be compared to just four neighboring points for the standard five-point stencil.

Inserting the spatial approximations in (1)–(4) one obtains a large system of ordinary differential equations (ODE) where the solution consists of approximations to ϕ and u in all grid points. The ODE can be solved using standard methods found in literature. Here we have chosen to work with the backward difference formulas (BDF) [8]. For a general ODE $y'(t) = f(t,y)$, where $y(t_0),\ldots,y(t_{n-1})$ are known, they take the form

$$\sum_{i=0}^{k}\alpha_i^{(k)} y(t_{n-i}) = \Delta t f\big((t_n, y(t_n)\big), \tag{14}$$

where $1 \le k \le 6$ is the order of accuracy and $\alpha_i^{(k)}$ are known constants. Thus one can obtain a high order of accuracy in the temporal discretization with very little work. The generalization to adaptive time stepping is also relatively straight-forward [8]. The BDF-methods have desirable stability properties. In the sense of theory of numerical methods for ODE, the first and second order methods are A-stable.

2.3 Non-linear Solver

With the discretization of space and time discussed in Sect. 2.2 the solution can be advanced in time by solving a non-linear system of equations,

$$\sum_{i=0}^{k} \alpha_i^{(k)} \begin{bmatrix} \boldsymbol{\Phi}_{n-i} \\ U_{n-i} + P(\boldsymbol{\Phi}_{n-i})/\Delta \end{bmatrix} - \Delta t \begin{bmatrix} m\widetilde{\nabla}_a^2 \boldsymbol{\Phi}_n + mF(\boldsymbol{\Phi}_n, U_n)/\epsilon^2 \\ \widetilde{\nabla}^2 U_n \end{bmatrix} = 0. \quad (15)$$

For brevity of notation, all non-linear terms in (1) except the anisotropic Laplacian have been collected in F, the non-linear terms in (2) collected in P, and approximations of spatial derivatives are denoted by tilde. The vectors $\boldsymbol{\Phi}_n$ and U_n contain grid values of ϕ and u at time t_n to be discussed in more detail in Sect. 3.1 below.

Equation (15) is solved by Newton iteration. The well-known advantages are quadratic convergence in the vicinity of a root and the fact that the solution can be determined by solving "a few" linear systems with the Jacobian as matrix.

Unfortunately Newton's method will only converge if the initial guess is sufficiently close to the root. A backtracking line search technique [2,5] is used to increase the radius of convergence. The underlying idea is to adaptively choose a damping factor such that there is a "sufficient" decrease in the residual, and the directional derivative for the next iterate is not "too small". This increases the radius of convergence while quadratic convergence is retained in the vicinity of the root. The extra work consists mostly of evaluating the non-linear function.

The major part of the computational work in Newton's method is needed for solving linear systems. For this application the matrices will be sparse and banded, see Sect. 3.1. The storage requirements of a direct solver are prohibitive in this case, even if the matrix is permuted to decrease the bandwidth.

Krylov subspace projection methods [12] have several attractive properties for solving sparse linear systems. The storage requirements are only a few vectors and they do not need the matrix itself but only its action on vectors[1]. In the context of Newton iteration the action of the Jacobian can be approximated by a difference quotient and the linear systems need not be solved exactly. In [2] it is shown how non-linear Krylov subspace correction methods can be formulated, including a Newton-GMRES algorithm based on backtracking line search techniques.

Since the matrices considered here are non-symmetric GMRES [12,14] has been chosen. Other possible choices include the Conjugate Gradient Squared Method (CGS) and the Biconjugate Gradient Stabilized Method (Bi-CGSTAB) [12]. GMRES requires more storage than both CGS and Bi-CGSTAB but only half the number of matrix-vector products need to be

[1] Some Krylov methods also need the action of the matrix transpose and have not been considered here.

computed. Restarting is used to limit the memory usage. The current implementation only has Jacobi preconditioning [12].

The numerical method employed here is second order accurate and different from the one chosen by many other scientists, see e.g. [9,19]. Instead of solving (15), Φ_n and U_n are updated separately according to

1. Given Φ_{n-1} and U_{n-1}, compute Φ_n from (1) using an explicit method (e.g. Forward Euler).
2. Solve (2) for U_n using an implicit method (e.g. Crank-Nicholson), approximating the time derivative of $P(\Phi)$ using Φ_n (as computed in 1) and Φ_{n-1}.

This approach is more efficient both when it comes to floating point operations and storage requirements since no non-linear equations have to be solved and the linear systems have dimension $N_x N_y$ instead of $2N_x N_y$. There are however step size restrictions due to stability and the order of accuracy is only one.

3 Parallelization

The numerical method described in Sect. 2 has been implemented in C for parallel distributed memory architectures based on a SPMD (Single Program Multiple Data) model. One of the critical issues for good performance is to choose a data distribution which minimizes communication, as discussed in Sect. 3.1. The more technical details of the implementation such as memory requirements and communication are then discussed in Sect. 3.2. The performance of the code is experimentally investigated in Sect. 4.1.

3.1 Data Distribution

Apart from evaluating (15) and the Jacobian of (15) a few key computational kernels can be identified from Newton's method and GMRES, namely computing matrix-vector products, vector updates, norms and scalar products. Since the last two are global operations and independent of data distribution, and the vector update is strictly local, it makes sense to choose a data distribution such that (15) can be efficiently evaluated. The communication pattern of this operation is identical to that of computing the matrix-vector products.

Assuming that the solution is represented on the grid (7), and that the vectors $\Phi_n, U_n \in \mathbb{R}^{N_x N_y}$ contain grid values sorted in increasing order of $jN_x + i$ ("natural ordering" [12]), the Jacobian of (15) will have the structure shown in Fig. 1. For a direct solver one would typically use another ordering of the elements to reduce the bandwidth. For GMRES this will not alter the storage requirements and the simple ordering has therefore been kept to simplify the implementation of the difference approximation.

The matrix in Fig. 1 can be seen as consisting of four block matrices showing dependencies of data defined on the grid. The structure of the upper left and lower right blocks are determined by the difference approximations of spatial derivatives. The remaining two blocks are diagonal and contain terms coming from the coupling between (1) and (2). Distributing the vectors Φ_n and U_n row wise across the processes will give a row-cyclic distribution of the Jacobian (as indicated by the shaded area in Fig. 1) and minimize communication.

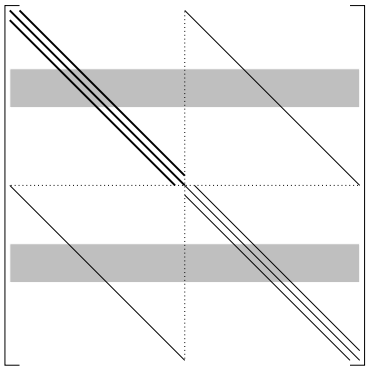


Fig. 1. Jacobian of (15). The shaded area shows the part of the matrix residing on the same process.

To evaluate (15) each process would have to send $2N_x + 1$ elements to its two neighbors. The communication can be overlapped by computation since only a relatively small fraction of the elements depend on external data (approximately $2p/N_y$ on p processes). This is accomplished by non-blocking MPI-routines [6].

3.2 Implementation

Since the numerical method used here is more expensive per time step than the common choice discussed in Sect. 2.3 the primary focus has been to write an efficient implementation to allow simulation within comparable time. To this end existing numerical libraries have been used to a large extent. The only numerical routine written specifically for this application is the backtracking line search modification of Newton's method. All calls to basic linear algebra routines are made to ESSL (IBM SP2) or BLAS (Sun workstations). The GMRES implementation is taken from the PSPARSLIB library [13].

Inter-process communication has been implemented using MPI [6]. Non-blocking routines are used wherever possible to overlap computation and communication. Many of the operations needed for GMRES and the Newton iteration are inherently global and will introduce synchronization points. These are unavoidable in the Newton iteration since all global reductions are used either to determine whether convergence has occurred or to update the approximate solution.

The GMRES implementation uses the classical Gram-Schmidt algorithm [12] to orthogonalize the newly computed vector against the previously computed vectors in each inner iteration. While this is advantageous for numerical stability concerns it may result in poor parallel performance since two global reductions are used for each orthogonalization. It is possible to orthogonalize the entire Krylov basis at once using e.g. the m-step GMRES(m) algorithm [10], thus reducing the number of synchronization points. Such algorithms are not supported in the current release of PSPARSLIB.

Table 1 shows the distribution of the total run time over the different routines for one particular simulation. Notably computing the analytic Jacobian of (15) accounts for almost half of the total time. Still using the analytic expression instead of difference approximations significantly reduces the run time for most simulations at the expense of larger storage requirements. A more detailed look at the routines for computing (15) and its Jacobian show that almost 75% of the time is spent inside the trigonometric functions `sin`, `cos` and `atan2` from the C math library. Thus there is a large part of the code which can not be optimized further. Temporary storage of computed values is used to avoid excessive calls to the trigonometric functions.

Table 1. Distribution of total run time over communication and computational routines for a simulation on a 400×400 grid running on 4 IBM SP2 processors. Results were obtained by `gprof`. The time used for call counts was excluded before fractions were computed.

Routine	Fraction of run time
Analytic Jacobian of (15)	47.7 %
Evaluating (15)	23.8 %
ESSL	10.9 %
Matrix-vector products	5.7 %
Preconditioning	1.4 %
Communication	1.1 %

Both GMRES and the Newton solver are based on reverse communication protocols. The internal work in these routines make up the major part of the time used by ESSL. The 9% not accounted for by Table 1 are overhead,

initialization and unidentified routines. This fraction agrees well with the measured speedup, see Table 3.

The approximate memory usage of the program is shown in Table 2. The reciprocals of the matrix diagonal are stored for the Jacobi preconditioner to avoid divisions. Simpler versions of Newton iteration will require less storage, as well as using difference quotients to approximate the action of the Jacobian. One could also decrease the needed storage by choosing another Krylov method. Typically the dimensions of the Krylov subspaces are small, say 5–20, and the gain in storage is not sufficient to justify the increase in computational complexity.

Table 2. Approximate storage requirements per process measured in `sizeof(double)`. Simulation on $N_x \times N_y$–grid using a BDF(k)-method and Krylov subspaces of dimension m running on p processes.

Purpose of allocated storage	Size
Solution at different time levels	$2(k+1)\,N_xN_y/p$
Action of difference operator	$12\,N_x$
Newton solver	$10\,N_xN_y/p$
Krylov solver (PSPARSLIB [13])	$(10+4m)\,N_xN_y/p$
Jacobian of (15)	$16\,N_xN_y/p$
Preconditioner (Jacobi)	$2N_xN_y/p$
Total	$\approx (40+2k+4m)\,N_xN_y/p$

An effort has been made to write modular code and the numerical methods (apart from the discretization) can be replaced with a relative small amount of work. The generalization to similar physical problems, such as solidification of binary alloys [18], is also straight-forward. Both numerical and physical parameters used for simulation are read from an input file for flexibility.

4 Computational Results

4.1 Performance

The scalability of the implementation has been investigated on two different architectures, the IBM SP2 *Strindberg* located at PDC, Stockholm, and networks of Sun Ultra 5 workstations. The (approximate) computational performance lies in the range of 80–150 Mflops on a single thin IBM SP2 node, depending on grid size and data. Peak performances of up to 230 Mflops have been obtained.

Tables 3 and 4 show speedup on the different architectures for a few different grid sizes. Speedup is formally defined as the execution time of the fastest serial algorithm on one processor divided by the execution time of the

parallel algorithm [7] and is a measure of the number of processors which can be efficiently used to solve a given problem. The results shown were obtained by comparison to a modified version of the parallel code, in which all the communication has been removed. The timings were measured over 100 time steps using `MPI_Wtime`. Realistic parameters were used and initial data was chosen such that the fastest transients had died out. The results should be representative for a typical simulation.

Table 3. Speedup on an IBM SP2 for different grid sizes

	Grid size			
Nodes	200×200	400×400	600×600	800×800
2	1.92	1.89	4.27	4.73
4	3.56	3.63	8.14	9.15
8	6.56	6.34	15.30	17.22
16	11.25	11.38	26.24	31.45
32	12.89	17.36	44.10	51.37

Table 4. Speedup on clusters of Sun workstations (Ultra 5) for different grid sizes

	Grid size		
Computers	200×200	400×400	600×600
2	1.91	2.61	4.09
4	3.83	5.17	31.03
8	6.14	10.00	60.60
16	7.34	16.49	93.78

The speedup for the larger grids is more than optimal. For these simulations the numerical parameters used for the one processor job are different from the ones used on several processors due to RAM memory limitations. In most cases the dimensions of the Krylov subspaces have been decreased (compare Table 2). For the 600×600-grid on one Sun workstation, finite difference approximations were used to evaluate the action of the Jacobian.

The results are encouraging even for the smaller grids. All simulations on eight processors show an efficiency, defined as the ratio of speedup to the number of processors, exceeding 75%. On the IBM SP2 the efficiency is more than 70% for all grids on 16 processors.

Another scalability metric is the scaled speedup [7], which is a measure of how long a given parallel program would have taken to run on a hypothetical serial processor with the same amount of RAM memory as the parallel machine. The motivation for this metric is that with a larger machine one is able to solve larger problems, which is not reflected in the fixed-size speedup. By increasing the problem size linearly with the number of processors memory and surface-to-volume effects remain constant and usually the efficiency is higher than for the fixed-size case which is limited by Amdahl's law [7].

Since the considered implementation is based on iterative methods it is not obvious how to increase the problem size. Parameters such as grid resolution, time step and tolerances all affect execution time and performance. Furthermore, the convergence is to large extent dependent on data. For these reasons two different limits have been investigated: increased size of physical domain with all numerical parameters kept fixed, and decreased grid spacing with the physical problem kept fixed. In both cases the problem size was increased such that approximately 400^2 and 200^2 gridpoints reside on each processor on the IBM SP2 and Sun workstations respectively. The results are shown in Table 5.

Table 5. Scaled speedup for increased size of the physical domain with numerical parameters kept fixed (case 1), and decreased grid spacing with all other parameters kept fixed (case 2).

	IBM SP2		Sun workstations	
Nodes	Case 1	Case 2	Case 1	Case 2
2	1.99	1.92	1.92	1.67
4	3.97	3.63	3.72	3.31
8	7.94	7.60	6.78	6.14
16	15.70	12.90	12.71	10.04

The scaled speedup is closer to the optimal than the fixed-size speedup as expected. When the size of the physical domain is increased there is only a slight decrease in efficiency due to the global reductions. The results are not as good when the grid is refined since a larger number of iterations are required for GMRES to converge and thus more global reductions are needed. It is a possible that a better efficiency can be obtained by changing the time step as well as the grid spacing.

4.2 Computed Dendrites

Results from two different computer simulations are given in this section. For both cases four-fold anisotropy has been used with the x-axis taken as reference direction, which corresponds to $k = 4$ and $\theta_0 = 0$ in (4).

The first example is a reproduction of results presented by Braun and Murray [1]. The physical parameters used for the simulation were $\alpha = 400$, $m = 0.035$, $\epsilon = 0.005$, $\gamma = 0.015$ and $\Delta = 0.25$. The initial shape of the solid was given by (5) with $r_0(\psi) = 0.05\,(1 + 0.5\cos(\psi))$, and the initial temperature was taken to be $u(x, y, 0) = -\phi(x, y, 0)$. Computations were carried out on a 601×601-point discretization of $[0, 3] \times [0, 3]$ with constant time step $\Delta t = 0.001$. The shape of the solid in Fig. 2 agrees with the results presented in [1, Fig. 1].

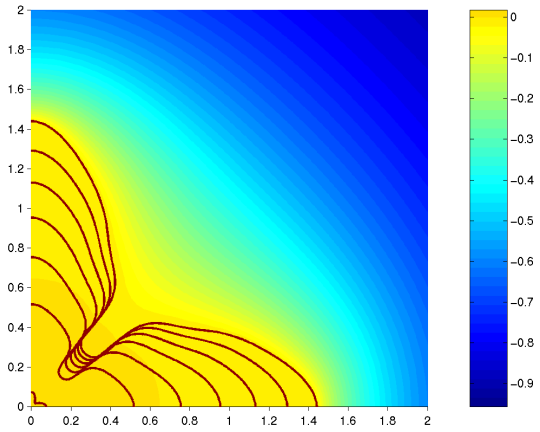


Fig. 2. Growth of dendrite with physical parameters and initial data as in [1]. The solid lines show the location of the solid-melt interface at the time levels $t = 0.0, 0.25, \ldots, 1.5$. The background color represents the heat distribution at $t = 1.5$.

The parameters for the second simulation were chosen to model pure nickel, $\alpha = 400$, $m = 0.05$, $\epsilon = 0.005$, $\gamma = 0.025$ and $\Delta = 0.5$. It should be noted that the exact value of γ is unknown, but the choice made here is well within the range of values chosen by other scientists, see e.g. [9,11,19]. The undercooling Δ is larger than in the simulation above causing the dendrite to grow faster. To further stimulate side-arm growth a random perturbation of 1% amplitude was added to the non-linear function $F(\phi, u)$ in (15), which can be interpreted as thermal noise at the interface [19].

The computation was initiated with a circular seed of radius 0.02, and a uniform temperature equal to -1. Results obtained from calculations on a 451×451-point discretization of $[0, 4.5] \times [0, 4.5]$ with $\Delta t = 0.0005$ are shown in Fig 3.

In both simulations presented here smaller time steps have been chosen than motivated by stability and convergence radius of the Newton iteration. Empirical choices were made in an attempt to decrease the wall clock time needed for the entire simulation. For larger time steps the initial guess for the Newton iteration (taken to be the solution at the old time level) will be further from the root, and convergence in the backtracking line search framework is only linear[2]. The time needed to solve the arising linear systems will also increase. Adaptive BDF-methods provide a more systematic way of selecting time steps.

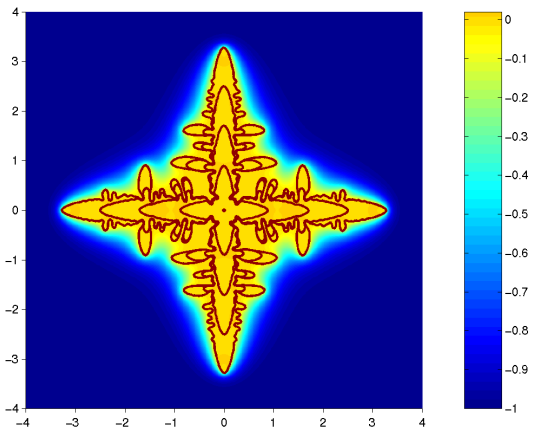


Fig. 3. Growth of nickel dendrite starting from a small circular initial seed. The solid lines show the location of the solid-melt interface at the time levels $t = 0.0, 0.075, \ldots, 0.3$ and the background color represents the heat distribution at $t = 0.3$. The solution has been computed in one quarter of the domain and reflected around the x- and y-axes.

Newton iteration was performed until the discrete Euclidean norm of the residual (as given by (15)) was smaller than 10^{-4}. Convergence was typically reached in 3–5 iterations, apart from the first time steps where there are still fast transients coming from initial data. The largest part of the residual is located in the vicinity of the solid-melt interface but the residual in each component of the solution remains much smaller than 10^{-4}. The error tolerance can be decreased at the expense of larger computational time so that only

[2] Without the backtracking line search modification Newton's method may diverge for these cases.

the discretization error remains. More sophisticated convergence criteria can also be implemented with a small amount of work.

Solutions obtained by the first order method described in Sect. 2.3 have been inserted in (15) and the norm of the residual for the second order discretization computed to get a feeling for how large the errors due to the splitting are. With parameters from the simulation in Fig. 3, and initial data taken from various time levels in the second order accurate simulations, the norm of (15) after one time step is approximately 0.03–0.04 for $\Delta t = 10^{-4}$, and approximately 0.001–0.002 for $\Delta t = 10^{-5}$. It is not clear at this point how this affects the global solution but it is a topic for further investigation.

The wall clock time required to compute Fig. 3 was approximately 24 hours on four Sun workstations. This particular computation has not been performed with the first order scheme but numerical experiments indicate that the wall clock time required would be about 36 hours running on a single Sun workstation with $\Delta t = 10^{-5}$. The increase in computational cost for using the second order method would be about 2.7 if the wall clock time is scaled by the number of computers, which seems to be a justifiable expense for the increased accuracy.

5 Conclusions

A second order accurate numerical method for simulating dendritic solidification has been introduced. This method is different from the first order method chosen by many other scientists and computationally more expensive per time step. It was shown how the scheme can be implemented on parallel architectures to allow computation on larger domains and/or with finer grid resolution.

The code scales well with the number of processors, both on parallel computers and networks of workstations. Performance is reasonable for sparse matrix computations. Numerical experiments indicate that the increased amount of RAM memory made available by the parallelization can be exploited to allow accurate simulation within an acceptable increase of run time. A more careful investigation of the errors in the two numerical methods is however needed to confirm this conclusion.

The uniform discretization of space and constant time stepping is expected to be replaced by adaptive methods in the future.

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Paper 3

Third order asymptotics of a phase-field model

Christer Andersson

Abstract

A third order asymptotic analysis of a phase-field model of solidification is presented in the limit of vanishing diffuse interface thickness ϵ. Agreement with the sharp interface formulation is investigated for general functions modeling the release of latent heat. It is shown that for the originally proposed fifth degree polynomial a thin interface correction [A. Karma and W.-J. Rappel, Phys. Rev. E **53** R3017 (1996)] can be employed to eliminate $\mathcal{O}(\epsilon)$-errors. From analysis and numerical computation it is concluded that if there are polynomials eliminating $\mathcal{O}(\epsilon^2)$-errors they must be of degree ten or higher. For computation one would prefer not to use polynomials of such high degree, and for practical purposes there is a bound on the accuracy of the phase-field model.

1 Introduction

Over the past decade phase-field models have become increasingly popular in numerical simulation of dendritic solidification [2, 8, 10, 12]. The need to explicitly track a geometrically complicated solid-liquid interface is avoided by deriving a set of reaction-diffusion equations valid in the entire domain. The equations are expressed in terms of the phase-field ϕ which takes on distinct values in solid and melt, and has a smooth transition between these values in a thin region corresponding to a diffuse interface.

There are several phase-field models. Here we have chosen to work with a thermodynamically consistent phase-field model derived by Wang et al. [11], which on dimensionless form can be written

$$\frac{\epsilon^2}{m}\partial_t\phi = \epsilon\alpha\Delta up'(\phi) - \frac{1}{4}g'(\phi) + \epsilon^2\nabla^2\phi, \tag{1.1}$$

$$\partial_t\Big(u + \frac{1}{\Delta}p(\phi)\Big) = \nabla^2 u, \tag{1.2}$$

where u is the dimensionless temperature, α the inverse capillary length, Δ the undercooling, m the interfacial mobility and ϵ an approximate diffuse interface thickness. The phase-field variable takes the values 0 and 1 in solid and liquid respectively. The function $g(\phi) = \phi^2(1-\phi)^2$ is a symmetric double well potential. The function $p(\phi)$ describing the release of latent heat satisfies the

conditions

$$\begin{cases} p(0) = 0 \\ p(1) = 1 \\ p'(0) = p'(1) = 0 \\ p''(0) = p''(1) = 0 \end{cases}, \tag{1.3}$$

guaranteeing that the free energy of the system has a local minima in the solid and liquid phases for all temperatures [11].

An asymptotic analysis is necessary in order to show that solutions of the phase-field equations converge to solutions of the sharp interface problem as $\epsilon \to 0$. The temperature is expected to satisfy the following dimensionless interface conditions [4, 11]

$$\begin{aligned} u_\Gamma &= -\frac{1}{6\sqrt{2}\alpha\Delta}\Big(\frac{v}{m} + \mathcal{K}\Big), \\ \partial_n u\Big|_{\text{solid}}^{\text{liquid}} &= -\frac{v}{\Delta}. \end{aligned} \tag{1.4}$$

The first equation is the interfacial temperature as prescribed by the Gibbs-Thomson condition. The second equation, or the jump condition, states that the interfacial velocity is proportional to the discontinuity in the normal derivative of the temperature across the interface.

In the early asymptotic analysis first-order approximate solutions to the phase-field equations were constructed [4, 5]. This is sufficient to show that the interface conditions (1.4) are satisfied with errors of size $\mathcal{O}(\epsilon)$, thus guaranteeing convergence as $\epsilon \to 0$. The resulting approximations can be physically interpreted as having a constant temperature in the diffuse interface.

Subsequently more accurate approximations have been constructed by allowing a variation of the temperature within the interface [1, 6, 7]. The asymptotics of Karma and Rappel proved to be especially useful due to the observation that a small *thin interface correction* to the parameter m eliminates a spurious $\mathcal{O}(\epsilon)$-term in the interfacial temperature [2, 7]. By letting

$$\frac{1}{m} = \frac{1}{m_{sharp}} + \epsilon\alpha C,$$

where m_{sharp} is the value of m from the early asymptotics, and C is a numerical constant, the interface conditions (1.4) are satisfied with errors $\mathcal{O}(\epsilon^2)$. Since the required grid resolution is determined by ϵ a small correction to m allows simulation on smaller grids without loss of accuracy.

The purpose of this paper is to investigate whether a similar correction can be determined such that the interfacial errors become of size $\mathcal{O}(\epsilon^3)$. As observed in [1] the thin interface correction is made possible because of properties of $p(\phi)$ and $g(\phi)$ in the phase-field model. In the ensuing asymptotics, which is carried one order higher than previously, additional constraints on $p(\phi)$ arise. These constraints are not satisfied by the polynomial $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ suggested by Wang et al. [11, Model I]. As will be seen from the asymptotics,

polynomials $p(\phi)$ must be of at least degree eight to uniquely determine the interfacial temperature to third order.

From a modeling point of view $p(\phi)$ can be a general function satisfying the pointwise constraints (1.3). In the ensuing discussion polynomials will often be used to clarify results, although this assumption is not required for the analysis. Due to difficulties associated with the evaluation of the third order interfacial temperature we shall especially consider polynomials $p(\phi)$ of degree nine. A thin interface correction will eliminate some error terms but not all. The error has a time dependent term which disappears if the interfacial curvature and velocity are constant to lowest order. It can be argued that this error is "nice" in the sense that it is transient for dendrite tips. There is also a time-independent error term which we have not been able to eliminate, thus implying that there is an upper bound on the accuracy of the phase-field equations for the polynomials considered here.

The remainder of the paper is outlined as follows. Section 2 describes the underlying assumptions of the formal asymptotic analysis. In particular, some preliminary observations on the structure of solutions are collected in Lemma 2 of Section 2.1. A third order asymptotic approximation to the solution of (1.1)–(1.2) is constructed in Section 3. Maple has been used to evaluate all expressions for particular choices of $p(\phi)$ except the $\mathcal{O}(\epsilon^2)$ contribution to the interfacial temperature. This last expression is numerically approximated using a C++ code for semi-symbolic computation described in Section 3.5. Section 4 compares the interface conditions arising from the phase-field model to those of the sharp interface formulation (1.4).

2 The solution approach

We shall consider the phase-field model (1.1)–(1.2) in one space dimension under assumption of radial symmetry. It is assumed that solidification propagates from left to right, and that all parameters except ϵ are $\mathcal{O}(1)$ as $\epsilon \to 0$ to allow representation of the solutions as asymptotic expansions.

It is convenient to work in a moving coordinate system $r(x,t;\epsilon) = x - \Gamma(t;\epsilon)$ attached to the interface, given implicitly by $\Gamma(t;\epsilon)$ defined such that $\phi(\Gamma(t;\epsilon),t;\epsilon) = 1/2$. Noting that r is a signed distance function we can identify negative values with solid, and positive values with liquid for $|r| \gg \mathcal{O}(\epsilon)$. In the inner region $r = \mathcal{O}(\epsilon)$ the stretched coordinate $\rho = r/\epsilon$ is introduced, and the derivatives in (1.1)–(1.2) are transformed according to

$$\begin{aligned} \nabla^2 &\to \epsilon^{-2}\partial_{\rho\rho} + \epsilon^{-1}\mathcal{K}(t;\epsilon)\partial_\rho, \\ \partial_t &\to \partial_t - \epsilon^{-1}v(t;\epsilon)\partial_\rho, \end{aligned}$$

where $v(t;\epsilon) = \partial_t\Gamma(t;\epsilon)$ is the propagation speed of the interface.

Assuming that solutions to the phase-field equations approach their asymptotic limits smoothly as $\epsilon \to 0$ we can represent the inner solutions by power

series expansions,

$$\begin{aligned} \phi(\rho,t;\epsilon) &= \sum_{j=0}^{\infty} \epsilon^j \phi_j(\rho,t;\epsilon), \\ u(\rho,t;\epsilon) &= \sum_{j=0}^{\infty} \epsilon^j u_j(\rho,t;\epsilon). \end{aligned} \tag{2.1}$$

A similar ansatz is used for the outer solutions with dependence on r instead of ρ, and using superscripts $-$ and $+$ to denote solid and liquid respectively. If the asymptotic ansatz holds, the interfacial properties satisfy

$$\begin{aligned} \Gamma(t;\epsilon) &= \sum_{j=0}^{\infty} \epsilon^j \Gamma_j(t;\epsilon), \\ \mathcal{K}(t;\epsilon) &= \sum_{j=0}^{\infty} \epsilon^j \mathcal{K}_j(t;\epsilon). \end{aligned}$$

In the common domain of validity $\epsilon \ll |r| \ll 1$ the inner and outer solutions must agree, leading to the matching conditions

$$\begin{aligned} \lim_{\rho \to \pm\infty} u_k(\rho,t;\epsilon) &= \sum_{j=0}^{k} \frac{1}{j!} \partial_r^j u_{k-j}^{\pm}(0,t;\epsilon)\rho^j, \\ \lim_{\rho \to \pm\infty} \phi_k(\rho,t;\epsilon) &= \begin{cases} (1 \pm 1)/2 & \text{if } k = 0, \\ 0 & \text{otherwise.} \end{cases} \end{aligned} \tag{2.2}$$

The first condition is obtained by Taylor expanding the outer solutions to the left and right of Γ and identifying powers of ϵ. It holds if the $u_j^{\pm}$ are sufficiently smooth at $r = 0^{\pm}$ as $\epsilon \to 0$. The second condition ensures that the phase-field is constant and equal to 0 or 1 away from the interface. In addition to (2.2) it follows from the definition of $\Gamma(t;\epsilon)$ that

$$\phi_k(0,t;\epsilon) = \begin{cases} 1/2 & \text{if } k = 0 \\ 0 & \text{otherwise} \end{cases}. \tag{2.3}$$

To solve the phase-field equations in the entire domain one must also prescribe boundary conditions on the outer solutions. The local relations at the interface should hold regardless of outer boundary conditions. Thus they are not needed to determine interfacial conditions resulting from the phase-field model.

Inserting the asymptotic ansatz (2.1) in (1.1), and applying the change to

the moving coordinate system we obtain

$$
\begin{aligned}
0 =&\epsilon^0\Big[\partial_{\rho\rho}\phi_0 - \frac{1}{4}g'(\phi_0)\Big] + \\
&\epsilon^1\Big[\partial_{\rho\rho}\phi_1 - \frac{1}{4}g''(\phi_0)\phi_1 + \alpha\Delta u_0 p'(\phi_0) + \Big(\frac{v_0}{m} + \mathcal{K}_0\Big)\partial_\rho\phi_0\Big] + \\
&\epsilon^2\Big[\partial_{\rho\rho}\phi_2 - \frac{1}{4}g''(\phi_0)\phi_2 - \frac{1}{8}g'''(\phi_0)\phi_1^2 + \alpha\Delta u_1 p'(\phi_0) + \alpha\Delta u_0 p''(\phi_0)\phi_1 + \\
&\quad\Big(\frac{v_0}{m} + \mathcal{K}_0\Big)\partial_\rho\phi_1 + \Big(\frac{v_1}{m} + \mathcal{K}_1\Big)\partial_\rho\phi_0 - \frac{1}{m}\partial_t\phi_0\Big] + \\
&\epsilon^3\Big[\partial_{\rho\rho}\phi_3 - \frac{1}{4}g''(\phi_0)\phi_3 - \frac{1}{24}g^{(iv)}(\phi_0)\phi_1^3 - \frac{1}{4}g'''(\phi_0)\phi_1\phi_2 + \\
&\quad\alpha\Delta u_2 p'(\phi_0) + \alpha\Delta u_1 p''(\phi_0)\phi_1 + \alpha\Delta u_0\Big(p''(\phi_0)\phi_2 + \frac{1}{2}p'''(\phi_0)\phi_1^2\Big) + \\
&\quad\Big(\frac{v_0}{m} + \mathcal{K}_0\Big)\partial_\rho\phi_2 + \Big(\frac{v_1}{m} + \mathcal{K}_1\Big)\partial_\rho\phi_1 + \Big(\frac{v_2}{m} + \mathcal{K}_2\Big)\partial_\rho\phi_2 - \frac{1}{m}\partial_t\phi_1\Big] + \mathcal{O}(\epsilon^4),
\end{aligned}
\tag{2.4}
$$

for the phase-field, and similarly by inserting (2.1) in (1.2),

$$
\begin{aligned}
0 =&\epsilon^0\Big[\partial_{\rho\rho}u_0\Big] + \\
&\epsilon^1\Big[\partial_{\rho\rho}u_1 + \Big(v_0 + \mathcal{K}_0\Big)\partial_\rho u_0 + \frac{v_0}{\Delta}p'(\phi_0)\partial_\rho\phi_0\Big] + \\
&\epsilon^2\Big[\partial_{\rho\rho}u_2 + \Big(v_0 + \mathcal{K}_0\Big)\partial_\rho u_1 + \Big(v_1 + \mathcal{K}_1\Big)\partial_\rho u_0 + \frac{v_0}{\Delta}p'(\phi_0)\partial_\rho\phi_1 + \\
&\quad\frac{v_1}{\Delta}p'(\phi_0)\partial_\rho\phi_0 + \frac{v_0}{\Delta}p''(\phi_0)\phi_1\partial_\rho\phi_0 - \partial_t u_0\Big] + \mathcal{O}(\epsilon^3).
\end{aligned}
\tag{2.5}
$$

Throughout the analysis we shall assume that $g(\phi) = \phi^2(1-\phi)^2$. In Sections 3.5 and 4, where the interface conditions are determined to third order, $p(\phi)$ will be considered to be a polynomial of degree nine. One should note that the asymptotics up to that point does not need this assumption.

2.1 Preliminary results

Before proceeding with the asymptotic analysis two lemmas with preliminary results are formulated allowing us to be more brief in the future. The first lemma states some observations on how expressions can be rewritten in terms of the zero-order approximation to the phase-field variable, ϕ_0. The second lemma can be used to express ϕ_j, $j \geq 1$ in terms of known functions by use of integral operators.

Lemma 1. *Let $\phi_0(\rho)$ be defined by*

$$
\phi_0(\rho) = \frac{1}{1 + \exp(-\rho/\sqrt{2})}. \tag{2.6}
$$

Then

(a) the first two derivatives of $\phi_0(\rho)$ *are given by*

$$\begin{aligned} \phi_0'(\rho) &= \phi_0(\rho)(1-\phi_0(\rho))/\sqrt{2}, \\ \phi_0''(\rho) &= \phi_0(\rho)(1-\phi_0(\rho))(1-2\phi_0(\rho))/2, \end{aligned} \tag{2.7}$$

(b) the derivative of a function $f(\phi_0(\rho))$ *can be written*

$$\frac{d}{d\rho} f(\phi_0(\rho)) = \frac{\phi_0(\rho)(1-\phi_0(\rho))}{\sqrt{2}} \frac{d}{d\phi_0} f(\phi_0), \tag{2.8}$$

(c) integrals over ρ *can be transformed to integrals over* $\phi_0(\rho)$ *according to*

$$\int_{\rho_0}^{\rho_1} F(\phi_0(\rho))\phi_0'(\rho)\,d\rho = \int_{\phi_0(\rho_0)}^{\phi_0(\rho_1)} F(\psi)\,d\psi, \tag{2.9}$$

$$\int_{\rho_0}^{\rho_1} F(\phi_0(\rho))\,d\rho = \int_{\phi_0(\rho_0)}^{\phi_0(\rho_1)} \frac{\sqrt{2}F(\psi)}{\psi(1-\psi)}\,d\psi. \tag{2.10}$$

Proof. Statement (a) follows from (2.6) and straight-forward calculations.

The chain rule of differentiation gives

$$\frac{d}{d\rho} f(\phi_0) = \frac{d\phi_0}{d\rho} \cdot \frac{d}{d\phi_0} f(\phi_0) = \frac{\phi_0(\rho)(1-\phi_0(\rho))}{\sqrt{2}} \frac{d}{d\phi_0} f(\phi_0),$$

after applying (2.7), thus proving (b).

There is a one-to-one correspondence between ϕ_0 and ρ, and the first integral in (2.9) follows directly from variable substitution. (2.10) is derived from (2.9) by introducing the help function $G(\phi_0) = F(\phi_0)/\phi_0'$ and noting that

$$\int_{\rho_0}^{\rho_1} F(\phi_0(\rho))\,d\rho = \int_{\rho_0}^{\rho_1} G(\phi_0(\rho))\phi_0'(\rho)\,d\rho = \int_{\phi_0(\rho_0)}^{\phi_0(\rho_1)} G(\psi)\,d\psi.$$

From the definition of $G(\phi_0)$ and (2.7) it follows that also (2.10) holds, and the proof is complete. □

Lemma 2. *Let* $\phi_0(\rho)$ *be defined by* (2.6). *The solution to the ordinary differential equation*

$$\partial_{\rho\rho}\phi(\rho) - \frac{1}{4} g''(\phi_0(\rho))\phi(\rho) = F(\phi_0(\rho)), \tag{2.11}$$

can be written as

$$\phi(\rho) = \frac{\phi_0(\rho)(1-\phi_0(\rho))}{\sqrt{2}} \left(\Phi_0 + \int_{1/2}^{\phi_0(\rho)} \frac{2\sqrt{2}}{\psi^3(1-\psi)^3} \Big(\Phi_1 + \int_0^{\psi} F(\zeta)\,d\zeta \Big)\,d\psi \right),$$

subject to the solvability condition

$$\int_0^1 F(\psi)\,d\psi = 0. \tag{2.12}$$

Here Φ_0 *and* Φ_1 *are constants of integration.*

If $F(0) = F(1) = 0$, *the additional conditions* $\phi(0) = 0$ *and* $\lim_{\rho\to\pm\infty}\phi(\rho) = 0$, *are satisfied for* $\Phi_0 = \Phi_1 = 0$.

Proof. Note that $\phi_0'(\rho)$ is a null function to the differential operator in the left hand side of (2.11),

$$\mathcal{P} = \frac{d^2}{d\rho^2} - \frac{1}{4}g''(\phi_0(\rho)).$$

We have

$$\mathcal{P}\phi_0'(\rho) = \frac{d}{d\rho}\Big(\phi_0''(\rho) - \frac{1}{4}g'(\phi_0(\rho))\Big),$$

where the parenthesis is always equal to zero by Lemma 1*a* and the definition of $g(\phi_0)$. For a solution $\phi(\rho) = \mathcal{P}^{-1}F(\phi_0(\rho))$ to exist the right-hand side must be orthogonal to the null function, i.e.

$$0 = \int_{-\infty}^{\infty} F(\phi_0(\rho))\phi_0'(\rho)d\rho = \int_0^1 F(\psi)d\psi,$$

after applying (2.9).

To find $\phi(\rho)$ we introduce the function $\varphi = \sqrt{2}\phi/(\phi_0(1-\phi_0))$. The differential operator $\mathcal{P}$ can be written in terms of ϕ_0 by (2.8),

$$\mathcal{P} = \frac{\phi_0(1-\phi_0)}{2}\frac{d}{d\phi_0}\Big(\phi_0(1-\phi_0)\frac{d}{d\phi_0}\Big) - \frac{1}{4}g''(\phi_0),$$

and after some calculus we obtain

$$\mathcal{P}\phi = \mathcal{P}\frac{\phi_0(1-\phi_0)\varphi}{\sqrt{2}} = \frac{d}{d\phi_0}\Big(\frac{\phi_0^3(1-\phi_0)^3}{2\sqrt{2}}\frac{d\varphi}{d\phi_0}\Big).$$

Hence

$$\frac{d\varphi}{d\phi_0} = \frac{2\sqrt{2}}{\phi_0^3(1-\phi_0)^3}\Big(\Phi_1 + \int_0^{\phi_0} F(\zeta)d\zeta\Big),$$

where Φ_1 is a constant of integration. Integrating once more and using the definition of φ we get the solution

$$\phi(\rho) = \frac{\phi_0(1-\phi_0)}{\sqrt{2}}\left(\Phi_0 + \int_{1/2}^{\phi_0}\frac{2\sqrt{2}}{\psi^3(1-\psi)^3}\Big(\Phi_1 + \int_0^{\psi} F(\zeta)d\zeta\Big)d\psi\right), \qquad (2.13)$$

where Φ_0 is a second constant of integration.

Finally, the constants Φ_0 and Φ_1 are determined for a special case useful in the ensuing asymptotics. Note that

$$\phi(\rho = 0) = \phi(\phi_0 = 1/2) = \frac{\Phi_0}{4\sqrt{2}},$$

so $\Phi_0 = 0$ must hold for $\phi(\rho = 0) = 0$, corresponding to the definition of interface location by (2.3). By investigating two mutually exclusive cases in the limits $\rho \to \pm\infty$ the constant Φ_1 can be determined.

Assume that the outer integral in (2.13),

$$\int_{1/2}^{\phi_0}\frac{2\sqrt{2}}{\psi^3(1-\psi)^3}\Big(\Phi_1 + \int_0^{\psi} F(\zeta)d\zeta\Big)d\psi, \qquad (2.14)$$

is unbounded as $\rho \to \pm\infty$. By l'Hospital's rule the limiting values of $\phi(\rho)$ are given by

$$\lim_{\rho\to\pm\infty} \phi(\rho) = \lim_{\phi_0\to 0,1} \frac{-2}{\phi_0(1-\phi_0)(1-2\phi_0)}\Big(\Phi_1 + \int_0^{\phi_0} F(\zeta)d\zeta\Big).$$

The integral approaches zero as $\phi_0 \to 0$ and $\phi_0 \to 1$, where the latter limit is a consequence of the solvability condition. Hence Φ_1 must be equal to zero for $\phi(\rho)$ to remain bounded. Applying l'Hospital's rule once more gives the explicit limits,

$$\lim_{\phi_0\to 0,1} \phi = \lim_{\phi_0\to 0,1} \frac{-2F(\phi_0)}{1-6\phi_0(1-\phi_0)} = \lim_{\phi_0\to 0,1} -2F(\phi_0).$$

Since $F(0) = F(1) = 0$ by assumption ϕ satisfies the matching condition (2.2).

If the integral (2.14) is bounded as $\rho \to \pm\infty$ l'Hospital's rule is no longer applicable. We consider the derivative in order to determine Φ_1,

$$\frac{d\phi}{d\rho} = \frac{\sqrt{2}}{\phi_0(1-\phi_0)}\Big(\Phi_1 + \int_0^{\phi_0} F(\zeta)d\zeta\Big) + \frac{1-2\phi_0}{\sqrt{2}}\phi$$

By the same argument used above, the integral approaches zero as $\phi_0 \to 0, 1$. The term proportional to ϕ disappears since $\lim_{\rho\to\pm\infty} \phi(\rho) = 0$ follows trivially from the assumption of a bounded outer integral in (2.13). Thus $\Phi_1 = 0$ unless $\phi'(\rho)$ becomes unbounded as $\rho \to \pm\infty$, contradicting the observation that ϕ approaches zero.

Another application of l'Hospital's rule shows that for $\Phi_1 = 0$ the derivative $\phi'(\rho)$ converges to $\sqrt{2}F(0)$ and $-\sqrt{2}F(1)$ as $\phi_0 \to 0$ and $\phi_0 \to 1$, respectively. By assumption $F(0) = F(1) = 0$ so the solution satisfies the matching condition (2.2) also when (2.14) is bounded.

We conclude the proof by noting that the integral (2.14) can be bounded for $\rho \to \infty$, and unbounded for $\rho \to -\infty$, or the other way around. The two cases discussed above cover this behavior as well since the two limits are independent of each other.

□

3 Asymptotics

In Sections 3.1–3.3 approximations to $u_j(\rho, t; \epsilon)$ and $\phi_j(\rho, t; \epsilon)$ are computed for $j = 0, 1, 2$ according to the following procedure. By integrating the $\mathcal{O}(\epsilon^j)$-term of (2.5) twice $u_j(\rho, t; \epsilon)$ is determined. Two constants of integration c_j and l_j appear, corresponding to constant temperature and linear growth in ρ, respectively.

The constant c_j can not be determined at this point in the analysis. It is however possible to identify certain constraints on $p(\phi)$ which must be satisfied if the outer temperature is to be continuous across the interface. This is required to uniquely identify the temperature for comparison to (1.4).

The jump condition to order $\mathcal{O}(\epsilon^j)$ is verified by comparing the polynomial behavior of $u_j(\rho,t;\epsilon)$ to the matching conditions (2.2) as $\rho \to \pm\infty$. The constant l_j depends on the outer solutions and will for relevant functions $p(\phi)$ disappear from the interface conditions. It is therefore not determined explicitly.

After $u_j(\rho,t;\epsilon)$ is known the solvability condition (2.12) for $\phi_j(\rho,t;\epsilon)$ is investigated. It is used to find the value of c_{j-1}, which uniquely determines the contribution $u^{\pm}_{j-1}(0,t;\epsilon)$ to the interfacial temperature. The contribution ϕ_j to the phase-field variable is then computed directly from Lemma 2.

It is natural to work in the scaled inner coordinate ρ when determining u_j, whereas the lowest order approximation to the phase-field ϕ_0 is more convenient for computing ϕ_j. For brevity of notation the explicit dependence on ρ and ϕ_0 is often dropped, as well as time dependence. We adopt the convention that integrals over Latin letters are computed with respect to ρ, and integrals over Greek letters are computed with respect to ϕ_0.

3.1 Zero-order inner solution

The zero-order equation for the temperature can be integrated twice to yield

$$u_0(\rho) = c_0 + l_0\rho.$$

It is clear from the matching condition (2.2) that $l_0 = 0$, unless the temperature becomes unbounded in the lowest order.

Using the chain rule of differentiation we note that

$$\partial_{\rho\rho}\phi_0 = \frac{1}{2}\frac{d}{d\phi_0}\big(\partial_\rho\phi_0\big)^2.$$

Inserting this expression in the zero-order equation from (2.4) and integrating with respect to ϕ_0 yields

$$\big(\partial_\rho\phi_0\big)^2 = \frac{1}{2}g(\phi_0) + const. \tag{3.1}$$

According to (2.2) the constant of integration must be zero, and the equation is separable with solution

$$\phi_0(\rho) = \frac{const_1}{1+e^{-\rho/\sqrt{2}}} + \frac{const_2}{1+e^{\rho/\sqrt{2}}},$$

where the two different terms are the result of taking the square root of (3.1). Applying the matching conditions again and enforcing $\phi_0(0,t;\epsilon) = 1/2$ the zero-order inner solution can be written

$$\phi_0(\rho) = \frac{1}{1+e^{-\rho/\sqrt{2}}} = \frac{1}{2}\Big(\tanh\frac{\rho}{2\sqrt{2}} + 1\Big).$$

3.2 First order inner solution

Integrating the $\mathcal{O}(\epsilon)$-term of (2.5) twice with respect to ρ gives the temperature

$$u_1(\rho) = -(v_0 + \mathcal{K}_0)u_0\rho - \frac{v_0}{\Delta}\int_0^\rho p(\phi_0(y))dy + c_1 + l_1\rho. \tag{3.2}$$

Rewriting the integral so that it remains bounded as $\rho \to \pm\infty$ we get

$$u_1(\rho) = \begin{cases} c_1 + \frac{v_0}{\Delta}\int_\rho^0 p(\phi_0(y))dy + \big(l_1 - (v_0 + \mathcal{K}_0)u_0\big)\rho & \rho < 0 \\ c_1 - \frac{v_0}{\Delta}\int_0^\rho \big(p(\phi_0(y)) - 1\big)dy + \big(l_1 - (v_0 + \mathcal{K}_0)u_0 - \frac{v_0}{\Delta}\big)\rho & \rho \geq 0 \end{cases}. \tag{3.3}$$

By comparison to the matching conditions (2.2),

$$\partial_r u_0\Big|_-^+ = \partial_r u_0^+(0) - \partial_r u_0^-(0) = -\frac{v_0}{\Delta},$$

so the jump condition is satisfied to lowest order in the outer solution. The outer temperature satisfies

$$\begin{aligned} u_1\Big|_-^+ = u_1^+(0) - u_1^-(0) &= -\frac{v_0}{\Delta}\left(\int_0^\infty \big(p(\phi_0(y)) - 1\big)dy + \int_{-\infty}^0 p(\phi_0(y))dy\right) \\ &= -\frac{v_0}{\Delta}\int_0^\infty \big(p(\phi_0(y)) + p(1-\phi_0(y)) - 1\big)dy, \end{aligned}$$

where a variable substitution has been performed in the second integral, and the observation $\phi_0(-\rho) = 1 - \phi_0(\rho)$ used.

Thus the outer temperature is continuous across the interface if and only if

$$\int_0^\infty \big(p(\phi_0(y)) + p(1-\phi_0(y)) - 1\big) = 0. \tag{3.4}$$

The polynomial $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ suggested by Wang et al. [11, Model I] satisfies the stronger condition $p(\phi) + p(1-\phi) = 1$. All other polynomials satisfying (3.4) and the thermodynamic conditions (1.3) are of degree 7 or higher.

According to Lemma 2 the solvability condition

$$0 = -\int_0^1 \left(\alpha\Delta u_0 p'(\psi) + \Big(\frac{v_0}{m} + \mathcal{K}_0\Big)\frac{\psi(1-\psi)}{\sqrt{2}}\right)d\psi = -\alpha\Delta u_0 - \frac{1}{6\sqrt{2}}\Big(\frac{v_0}{m} + \mathcal{K}_0\Big)$$

must hold, which uniquely identifies the hitherto unknown constant

$$c_0 = -\frac{1}{6\sqrt{2}\alpha\Delta}\Big(\frac{v_0}{m} + \mathcal{K}_0\Big). \tag{3.5}$$

We see that the interfacial temperature is correct to lowest order.

The equation for ϕ_1 given by (2.4), together with the matching conditions (2.2) and (2.3) has the solution

$$\phi_1 = \frac{\phi_0(1-\phi_0)}{\sqrt{2}} \int_{1/2}^{\phi_0} \frac{2\sqrt{2}}{\psi^3(1-\psi)^3} \int_0^{\psi} \left(-\alpha\Delta u_0 p'(\zeta) - \Big(\frac{v_0}{m} + \mathcal{K}_0\Big) \frac{\zeta(1-\zeta)}{\sqrt{2}} \right) d\zeta d\psi$$

as given by Lemma 2. Since u_0 is known the expression can be written on the more compact form

$$\phi_1 = -\Big(\frac{v_0}{m} + \mathcal{K}_0\Big) \frac{\phi_0(1-\phi_0)}{\sqrt{2}} \int_{1/2}^{\phi_0} \frac{(3-2\psi)\psi^2 - p(\psi)}{3\psi^3(1-\psi)^3} d\psi. \tag{3.6}$$

3.3 Second order inner solution

The second order temperature term u_2 is determined by integrating the appropriate equation from (2.5) twice,

$$\begin{aligned} u_2(\rho) = c_2 + l_2\rho - \frac{v_0}{\Delta} \int_0^{\rho} p'(\phi_0(y))\phi_1(y)\,dy - \frac{v_1}{\Delta} \int_0^{\rho} p(\phi_0(y))dy \\ - \big(v_0 + \mathcal{K}_0\big) \int_0^{\rho} u_1(y)\,dy + \frac{(\partial_t u_0)\rho^2}{2}. \end{aligned} \tag{3.7}$$

By l'Hospital's rule we can identify the behavior of u_2 as $\rho \to \pm\infty$,

$$\begin{aligned} \lim_{\rho\to-\infty} u_2(\rho) \sim \rho^2 \left[\frac{(\partial_t u_0)}{2} - \frac{v_0 + \mathcal{K}_0}{2} \left(l_1 - (v_0 + \mathcal{K}_0)u_0 \right) \right] + \\ \rho \left[l_2 - (v_0 + \mathcal{K}_0)c_1 - \frac{v_0(v_0 + \mathcal{K}_0)}{\Delta} \int_{-\infty}^{0} p(\phi_0(y))dy \right] + \\ \left[c_2 + \frac{v_0(v_0 + \mathcal{K}_0)}{\Delta} \int_{-\infty}^{0} \int_{-\infty}^{y} p(\phi_0)dzdy + \frac{v_0}{\Delta} \int_{-\infty}^{0} p'(\phi_0)\phi_1 dy + \right. \\ \left. \frac{v_1}{\Delta} \int_{-\infty}^{0} p(\phi_0)dy \right] \\ \lim_{\rho\to\infty} u_2(\rho) \sim \rho^2 \left[\frac{(\partial_t u_0)}{2} - \frac{v_0 + \mathcal{K}_0}{2} \left(l_1 - (v_0 + \mathcal{K}_0)u_0 - \frac{v_0}{\Delta} \right) \right] + \\ \rho \left[l_2 - (v_0 + \mathcal{K}_0)c_1 + \frac{v_0(v_0 + \mathcal{K}_0)}{\Delta} \int_0^{\infty} ((p(\phi_0(y)) - 1)dy - \frac{v_1}{\Delta} \right] + \\ \left[c_2 - \frac{v_0(v_0 + \mathcal{K}_0)}{\Delta} \int_0^{\infty} \int_y^{\infty} (p(\phi_0) - 1)dzdy - \frac{v_0}{\Delta} \int_0^{\infty} p'(\phi_0)\phi_1 dy - \right. \\ \left. \frac{v_1}{\Delta} \int_0^{\infty} (p(\phi_0(y)) - 1)dy \right] \end{aligned}$$

Let us assume that $p(\phi)$ satisfies (3.4) so that $u_1^{\pm}$ is continuous across the interface. By comparison to the matching conditions (2.2) we see that $\partial_r u_1^{\pm}$, corresponding to linear growth in $u_2(\rho)$, satisfies the jump condition,

$$\partial_r u_1 \Big|_-^+ = -\frac{v_1}{\Delta}.$$

From the constant terms the jump in outer temperature across the interface can be identified,

$$u_2 \Big|_-^+ = -2(v_0 + \mathcal{K}_0)\frac{v_0}{\Delta}\int_0^\infty \int_y^\infty (p(\phi_0) - 1) dz dy - \frac{v_0}{\Delta}\int_{-\infty}^\infty p'(\phi_0)\phi_1 dy$$

Inserting ϕ_1 and transforming the integrals using Lemma 1 we find that the temperature across the interface is continuous for general v_0 and $\mathcal{K}_0$ if and only if

$$\begin{aligned} 0 &= \int_0^1 p'(\psi) \int_{1/2}^{\psi} \frac{(3-2\zeta)\zeta^2 - p(\zeta)}{\zeta^3(1-\zeta)^3} d\zeta d\psi \\ 0 &= \int_{1/2}^1 \frac{2}{\psi(1-\psi)} \int_\psi^1 \frac{p(\zeta)-1}{\zeta(1-\zeta)} d\zeta d\psi \end{aligned} . \tag{3.8}$$

The standard fifth degree polynomial suggested by Wang et al. [11, Model I] does not meet these two constraints and can therefore never be expected to yield errors smaller than $\mathcal{O}(\epsilon^2)$. Polynomials $p(\phi)$ must be of at least eighth degree to satisfy the integral constraints imposed by (3.4) and (3.8), together with the conditions (1.3) given by the derivation.

The constant c_1 needed to determine $u_1^{\pm}(0)$ is given by the solvability condition from Lemma 2,

$$\begin{aligned} 0 =& \frac{1}{8}\int_0^1 g'''(\phi_0)\phi_1^2 d\phi_0 - \alpha\Delta u_0 \int_0^1 p''(\phi_0)\phi_1 d\phi_0 - \alpha\Delta \int_0^1 u_1 p'(\phi_0) d\phi_0 - \\ & \Big(\frac{v_0}{m} + \mathcal{K}_0\Big)\int_0^1 \partial_\rho \phi_1 d\phi_0 - \Big(\frac{v_1}{m} + \mathcal{K}_1\Big)\int_0^1 \partial_\rho \phi_0 d\phi_0 + \int_0^1 \frac{\partial_t \phi_0}{m} d\phi_0. \end{aligned}$$

Since c_1 appears only in the integral of $u_1 p'(\phi_0)$ simple manipulation gives

$$\begin{aligned} c_1 =& \frac{1}{8}\int_0^1 \frac{g'''(\phi_0)\phi_1^2}{\alpha\Delta} d\phi_0 - u_0 \int_0^1 p''(\phi_0)\phi_1 d\phi_0 - \int_0^1 (u_1 - c_1) p'(\phi_0) d\phi_0 - \\ & \Big(\frac{v_0}{m} + \mathcal{K}_0\Big)\int_0^1 \frac{\partial_\rho \phi_1}{\alpha\Delta} d\phi_0 - \Big(\frac{v_1}{m} + \mathcal{K}_1\Big)\int_0^1 \frac{\partial_\rho \phi_0}{\alpha\Delta} d\phi_0 + \int_0^1 \frac{\partial_t \phi_0}{\alpha\Delta m} d\phi_0. \end{aligned} \tag{3.9}$$

The outer temperature is now known to second order at the interface, and equal to $u_0^{\pm}(0) + \epsilon u_1^{\pm}(0)$ assuming that (3.4) holds. Consider for example the polynomial $p(\phi) = \sum_{i=0}^7 a_i \phi^i$ of degree 7. All coefficients can be expressed in

terms of the leading coefficient a_7 by solving (1.3) and (3.4). By use of (3.3), (3.5), (3.9) and the matching conditions we obtain

$$\begin{aligned} u_0^{\pm}(0) + \epsilon u_1^{\pm}(0) &= c_0 + \epsilon\left[c_1 + \frac{v_0}{\Delta}\int_{-\infty}^{0} p(\phi_0(y))\,dy\right] \\ &= -\frac{1}{6\sqrt{2}\alpha\Delta}\Big(\frac{v_0+\epsilon v_1}{m} + \mathcal{K}_0 + \epsilon\mathcal{K}_1\Big) - \frac{\epsilon v_0}{6\sqrt{2}\Delta}\Big(\frac{209}{35} + \frac{520-a_7}{12012}a_7\Big) \end{aligned}$$

The first term corresponds to the Gibbs-Thomson condition (1.4), and the second term is an error of size $\mathcal{O}(\epsilon)$. Following Karma and Rappel [7] we can determine a thin interface correction to m which pushes the errors to $\mathcal{O}(\epsilon^2)$,

$$\frac{1}{m} = \frac{1}{m_{sharp}} + \epsilon\alpha\Big(\frac{209}{35} + \frac{520-a_7}{12012}a_7\Big). \tag{3.10}$$

From (3.10) we observe that by choosing $a_7 = 260 \pm \frac{2}{5}\sqrt{870805}$ second order accuracy is achieved for $m = m_{sharp}$, thus eliminating the need for a thin interface correction. From a computational point of view it is preferable to choose $a_7 = 0$ and employ a thin interface correction since it is cheaper in terms of floating point operations, and less sensitive to round-off errors.

The second order contribution to the phase-field is given by Lemma 2. After some simplification it can be written as

$$\begin{aligned} \phi_2(\rho) =& \frac{\phi_0(1-\phi_0)}{\sqrt{2}}\int_0^{\phi 0(\rho)} \frac{2\sqrt{2}}{\psi^3(1-\psi)^3}\int_0^{\psi}\Big[\frac{1}{8}g'''(\zeta)\phi_1^2 - \alpha\Delta u_0 p''(\zeta)\phi_1 - \\ & \alpha\Delta u_1 p'(\zeta) - \Big(\frac{v_0}{m} + \mathcal{K}_0\Big)\frac{d\phi_1(\zeta)}{d\zeta}\frac{\zeta(1-\zeta)}{\sqrt{2}} - \Big(\frac{v_1}{m} + \mathcal{K}_1\Big)\frac{\zeta(1-\zeta)}{\sqrt{2}}\Big]d\zeta d\psi. \end{aligned}$$

3.4 Third order solvability

To push the errors to $\mathcal{O}(\epsilon^3)$ one must also consider ϕ_3 and u_3. They can be constructed by the same procedure used in Sections 3.2 and 3.3. Since our main concern is the interface conditions explicit expressions for ϕ_3 and u_3 will be omitted.

The jump condition can be verified by formulating the integral representation of u_3 and identifying linear terms in ρ, c.f. (3.7) of Section 3.3. An investigation shows that

$$\partial_r u_2\Big|_-^+ = -\frac{v_2}{\Delta}.$$

The constant c_2 uniquely identifying $u_2^{\pm}(0)$ is given by the third order solvability

condition, which according to (2.4) and Lemma 2 becomes

$$0 = \int_0^1 \Bigg\{ -\frac{\partial_t \phi_1}{m} + \frac{1}{24} g^{(iv)}(\phi_0)\phi_1^3 - \frac{1}{4} g'''(\phi_0)\phi_1\phi_2 + \alpha\Delta u_2 p'(\phi_0) + \\ \alpha\Delta u_1 p''(\phi_0)\phi_1 + \alpha\Delta u_0 \Big(p''(\phi_0)\phi_2 + \frac{1}{2} p'''(\phi_0)\phi_1^2 \Big) + \\ \Big(\frac{v_0}{m} + \mathcal{K}_0 \Big) \partial_\rho \phi_2 + \Big(\frac{v_1}{m} + \mathcal{K}_1 \Big) \partial_\rho \phi_1 + \Big(\frac{v_2}{m} + \mathcal{K}_2 \Big) \partial_\rho \phi_0 \Bigg\} d\phi_0 . \tag{3.11}$$

The same technique used to obtain (3.9) can be employed to express c_2 explicitly in terms of integrals from (3.11). We have however not been able to compute c_2 symbolically in Maple or Mathematica for given polynomials $p(\phi)$. Instead a program for semi-symbolic computation has been implemented in C++ to allow investigation of $\mathcal{O}(\epsilon^2)$-terms in the interfacial temperature for polynomials of degree 8 and 9.

3.5 Evaluation of third order solvability

From a programming point of view there are some difficulties associated with the evaluation of (3.11) since there are parameters and constants which are not a priori known. The implementation must therefore handle certain quantities symbolically, namely

- parameters in the phase-field model: α, Δ and m
- propagation velocity and interfacial curvature as well as some time derivatives: v_0, v_1, v_2, $\mathcal{K}_0$, $\mathcal{K}_1$, $\mathcal{K}_2$, $\partial_t v_0$ and $\partial_t \mathcal{K}_0$
- constants from lower order approximations: l_1 and l_2

Depending on $p(\phi)$ there may also be additional symbolic constants not determined uniquely by (1.3), (3.4) and (3.8).

As seen from the asymptotics polynomials $p(\phi)$ must be of degree eight or higher to allow continuity of the outer temperature to errors of size $\mathcal{O}(\epsilon^3)$. There are two unique polynomials of degree eight satisfying all necessary constraints, as plotted in Figure 1. It is possible to find two families of ninth degree polynomials with one free coefficient. The two ninth degree polynomials plotted in Figure 2 satisfy the stronger form of (3.4), $p(\phi) + p(1-\phi) = 1$, and seem to be of special importance.

A program has been written in C++ based on the GiNaC framework for symbolic computation [3]. This allows convenient representation of symbolic constants and a syntax close to mathematics. In addition, GiNaC is based on the CLN library thus enabling computation in arbitrary precision.

Due to the structure of the asymptotic analysis, where more accurate approximations are constructed through integrals involving lower order approximations, computation of integrals form an important part of the program.

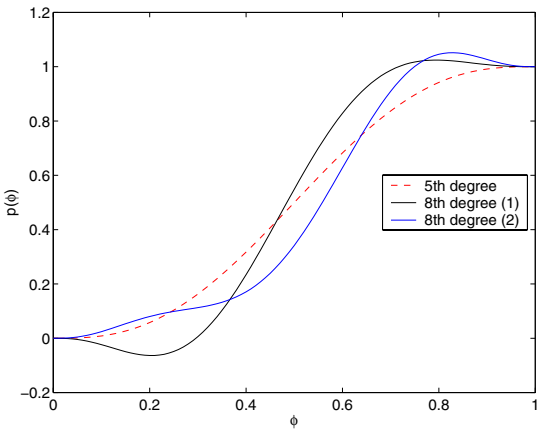


Figure 1: Polynomials $p(\phi)$ of degree eight satisfying (1.3), (3.4) and (3.8).

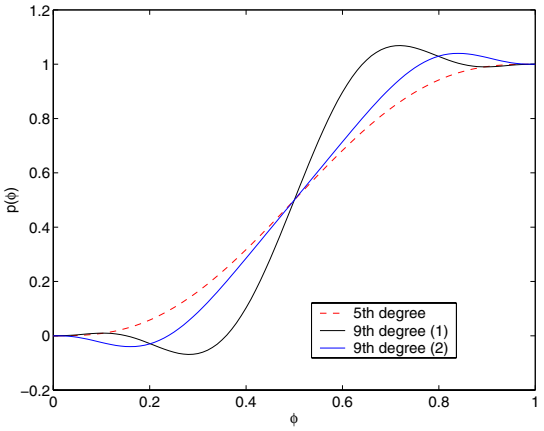


Figure 2: Polynomials $p(\phi)$ of degree nine satisfying (1.3), (3.8), and the stronger form of (3.4), $p(\phi) + p(1-\phi) = 1$.

GiNaC has been extended with a class `integral`. The class is semi-symbolic in the sense that integrals are manipulated symbolically, but evaluated numerically using Clenshaw-Curtis quadrature. The integration routine accepts a user-supplied tolerance, and attempts to compute $\int_a^b f(x)dx$ such that

$$\frac{\text{absolute error}}{\int_a^b |f(x)|dx} < \text{tol},$$

given an upper bound on the number of function evaluations. No special algorithms are employed for multiple integrals, which are just considered to be integrals of integrals.

The solvability condition (3.11) can be written as a sum of numerical expressions multiplied by one or more symbolic constants. A numerical expression in this context is a standard mathematical expression or function which does not contain any of the symbolic constants mentioned above. When evaluated it returns a pure numerical value. `integral` objects are automatically transformed by

$$\text{Expression} \to \sum_k \text{NumExpression}_k \cdot \prod_{j \in J_k} \text{Symbol}_j.$$

The transformation guarantees that the estimate of the absolute error in the integration routine can be computed independent of symbolic constants.

There are some numerical difficulties not handled directly by the integration algorithm. Consider for example the integral

$$\int_0^{1/2} x \ln x dx = -\frac{1 + 2\ln 2}{16}.$$

The integrand is bounded and equal to zero as $x \downarrow 0$. When evaluated in C++ however `0.0*log(0.0)` results in the "not a number" value (`NaN`) as prescribed by the IEEE floating point standard.

Certain integrands in the solvability condition exhibit this behavior for $\phi_0 = 0$ and $\phi_0 = 1$. They are evaluated with a special function call which returns 0 when $\phi_0 = 0$ and/or $\phi_0 = 1$ in the floating point representation. Since CLN allows user-defined precision only one or two points in the integration interval need special treatment.

To improve accuracy and efficiency expressions for ϕ_1 and u_1 computed by Maple for ninth-degree $p(\phi)$ are used instead of the integral representations (3.3) and (3.6). The expressions ϕ_2 and u_2 are defined via integrals since the expressions returned by Maple are long and contain complex values in the intermediate stages of evaluation.

The program described above has been verified by comparison to Maple and Mathematica for a large number of numeric and symbolic expressions. Figure 3 shows an example of convergence compared to a symbolically computed solution from Mathematica.

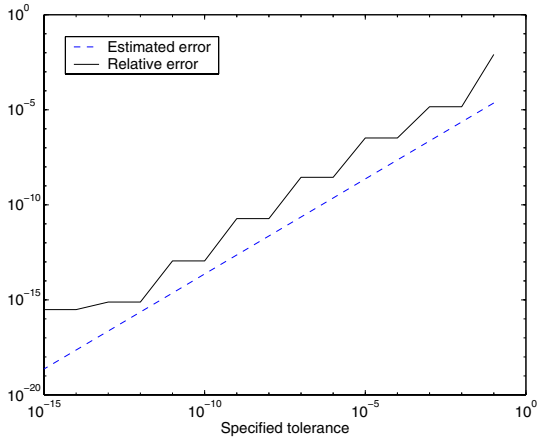


Figure 3: Actual and estimated error in integral $\int_0^1 \phi_1(\phi_0)^3 d\phi_0$ with $v_0/m+\mathcal{K}_0 = 1$ for computation in IEEE double precision.

4 Interface conditions

The asymptotic approximation constructed in Sections 3.1– 3.4 uniquely identify the interface conditions resulting from the phase-field model up to terms of size $\mathcal{O}(\epsilon^3)$. The jump condition has already been shown to agree with the sharp interface formulation (1.4) but it remains to investigate the interfacial temperature.

The Gibbs-Thomson condition is formulated in terms of the outer solution. The polynomial $p(\phi)$ must therefore satisfy (1.3), (3.4) and (3.8) so that a well-defined interfacial temperature exists. The temperature is computed from (3.2), (3.5), (3.7) and the matching conditions (2.2),

$$\begin{aligned} u_0^{\pm}(0) + \epsilon u_1^{\pm}(0) + \epsilon^2 u_2^{\pm}(0) = c_0 + \epsilon \left[c_1 + \frac{v_0}{\Delta} \int_{-\infty}^{0} p(\phi_0(y)) dy \right] + \\ \epsilon^2 \left[c_2 + \frac{v_0(v_0+\mathcal{K}_0)}{\Delta} \int_{-\infty}^{0} \int_{-\infty}^{y} p(\phi_0(z)) dz dy + \right. \\ \left. \frac{v_0}{\Delta} \int_{-\infty}^{0} p'(\phi_0(y))\phi_1(y) dy + \frac{v_1}{\Delta} \int_{-\infty}^{0} p(\phi_0(y)) dy \right], \end{aligned} \tag{4.1}$$

or the equivalent expression obtained in the limit $\rho \to \infty$. The constant c_0 is known, c_1 can be symbolically computed from (3.9) by Maple, as can the integrals. The C++ implementation is used to approximate c_2.

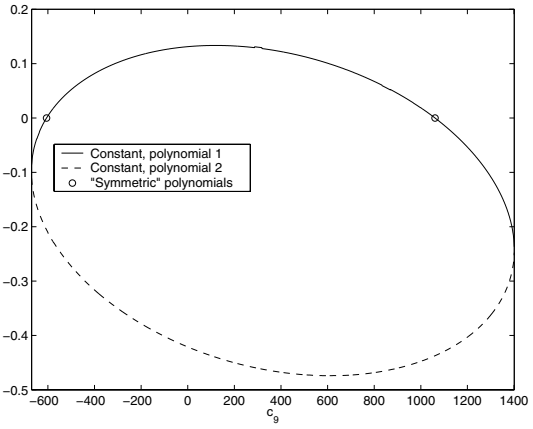


Figure 4: The numerical constant multiplying the anomalous $\epsilon^2 v_0 l_1$-term of the interfacial temperature as function of leading coefficient for polynomials $p(\phi)$ of degree nine.

The evaluated third order interfacial temperature (4.1) can be written as the sum of the value predicted by the Gibbs-Thomson condition and a number of anomalous terms. Each anomalous term has the form of a numerical constant multiplied by a product of symbolic constants. We shall especially consider the term $constant \cdot \epsilon^2 v_0 l_1$. Unless the constant in this term is zero the value of l_1 must be known for comparison to the sharp interface conditions (1.4). Such a dependence is clearly not satisfactory for practical purposes since l_1 depends on the boundary conditions prescribed on $u(r,t)$. An approximate analytic solution would be required in the entire domain before simulation is possible.

Figure 4 shows how the numerical constant multiplying $\epsilon^2 v_0 l_1$ depends on the choice of leading coefficient a_9 in $p(\phi)$. The plot has been generated by evaluating (4.1) for all possible integer values resulting in real $p(\phi)$. The polynomials of degree eight plotted in Figure 1 correspond to the special cases with $a_9 = 0$. The polynomials of degree nine plotted in Figure 2 are marked with circles.

The $\epsilon^2 v_0 l_1$-term disappears only for polynomials satisfying the stronger form of (3.4). The same is true for many of the other error terms and in this context the interfacial temperature can be written

$$u^{\pm}(0) \approx -\frac{1}{6\sqrt{2}\alpha\Delta}\Big(\frac{v_0+\epsilon v_1+\epsilon^2 v_2}{m} + (\mathcal{K}_0+\epsilon\mathcal{K}_1+\epsilon^2\mathcal{K}_2)\Big) + E_0\frac{\epsilon v_0+\epsilon^2 v_1}{\Delta}$$
$$+\epsilon^2\frac{E_1}{\alpha\Delta}\partial_t\Big(\frac{v_0}{m}+\mathcal{K}_0\Big) + \epsilon^2\frac{E_2}{\alpha\Delta}\Big(\frac{v_0}{m}+\mathcal{K}_0\Big)^3 + \mathcal{O}(\epsilon^3),$$

where E_0, E_1 and E_2 are numerically evaluated constants of size 0.1. Terms smaller than 10^{-4} have been discarded and are considered to arise from numerical integration.

Errors proportional to E_0 can be pushed to $\mathcal{O}(\epsilon^3)$ by a thin interface correction to m,

$$\frac{1}{m} = \frac{1}{m_{sharp}} + 6\sqrt{2}\epsilon\alpha E_0.$$

We note that the analytic expression $E_0 = \int_{-\infty}^{0} p(\phi(y))dy$ can be identified from (4.1). Hence the correction to m is identical to the one determined by second order asymptotics. The constant E_1 multiplies a time-derivative and disappears when the dendrite tip reaches steady state to lowest order in the moving coordinate system. The last error term, proportional to E_2, does not disappear and implies that polynomials of degree 9 can not result in a third order accurate phase-field model.

5 Conclusions

We have investigated the asymptotic behavior of a phase-field model of solidification (1.1)–(1.2) in the limit of vanishing diffuse interface thickness ϵ. During the analysis two constraints (3.4) and (3.8) on the choice of $p(\phi)$ emerged. The constraints must be satisfied if the outer temperature is to be continuous to third order. From numerical computation we conclude that there are additional constraints on $p(\phi)$ needed for solutions of the phase-field model to agree with the Gibbs-Thomson condition to third order.

Since the final step in the asymptotic analysis has been performed numerically no closed expression can be given for these constraints. In theory one could identify them from the integral representation of the second order constant contribution to the temperature, c_2. Due to the length of the expression such a process would be tedious, and reduction to significant terms error-prone. A less general approach would be to evaluate c_2 symbolically for given $p(\phi)$ and thereby identifying parameters in p leading to third order accuracy, c.f. (3.10) where conditions for second order accuracy were determined for seventh degree polynomials.

Here c_2 has been evaluated numerically for polynomials of degree nine. We have not been able to find a polynomial satisfying all constraints so if one does exist it must be of degree ten or higher. For computational purposes it is questionable whether the increased degree can be motivated by the more accurate interface conditions since the evaluation of $p(\phi)$ and its derivatives would be more sensitive to round-off errors.

We conclude by noting that it may be possible to obtain third order agreement with the sharp interface conditions by considering non-polynomial $p(\phi)$, or by using a generalized phase-field model [9].

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Paper 4

Analysis of a phase-field model for solidification in the limit of vanishing undercooling

Christer Andersson

Abstract

Phase-field models are attractive for numerical simulation of solidification. The derivation ensures that moving boundaries in the sharp interface formulation are replaced by thin regions of width ϵ given implicitly by the solution. The need to explicitly track the location of geometrically complex solid-liquid interfaces is eliminated.

The sharp interface problem is formally recovered as the diffuse interface thickness $\epsilon \to 0$. Karma and Rappel have presented an asymptotic analysis which allows certain restrictions on ϵ to be relaxed by adding a small correction to a parameter in the phase-field equations [A. Karma and W.-J. Rappel, Phys. Rev. E **53** R3017 (1996)]. As a consequence computation can be performed on smaller grids without loss of accuracy in the interface conditions. This is of especial importance at low undercoolings.

Here an alternative asymptotic analysis is presented where the undercooling tends to zero much faster than the diffuse interface thickness. The limit itself is not of interest in computation. It does however show that the correction predicted by Karma and Rappel eliminates the leading error in the interface conditions. The predicted temperature at the interface becomes independent of ϵ. Results of analysis are experimentally corroborated by investigating agreement between the phase-field and sharp interface models in one and two space dimensions.

1 Introduction

We are interested in solidification of a pure material from its undercooled melt at low undercoolings, taking into account effects of surface tension, kinetic undercooling and anisotropy. The classical mathematical formulation has the form of a Stefan problem. Two equations govern the diffusion of heat in the solid and liquid phases, coupled by boundary conditions prescribed at the moving solid-liquid interface. Under certain physical settings the crystals will exhibit dendritic shapes. The Stefan problem is unsuitable for numerical simulation in these cases since explicit tracking of geometrically complex interfaces is required.

Over the past decade phase-field models have emerged as a popular alternative for computation [5, 11, 14, 16]. By introducing a smooth approximation of an indicator function, the so-called *phase-field*, a set of equations valid in the

entire computational domain can be derived from thermodynamics. Thus there is no need to explicitly track solid-liquid interfaces. Here we shall consider a model where the phase-field variable satisfies

$$\phi(\mathbf{x},t) = \begin{cases} 0 & \text{if } \mathbf{x} \text{ in solid at time } t \\ 1 & \text{if } \mathbf{x} \text{ in liquid at time } t \end{cases}.$$

It is assumed that $\phi(\mathbf{x},t)$ varies smoothly between zero and one in thin regions of width $\mathcal{O}(\epsilon)$ corresponding to diffuse solid-liquid interfaces. Based on thermodynamically consistent formalism Wang et al. derive a phase-field model which on dimensionless form can be written as

$$\frac{\epsilon^2}{m}\partial_t\phi(\mathbf{x},t) = \epsilon\alpha\Delta u p'(\phi(\mathbf{x},t)) - \frac{1}{4}g'(\phi(\mathbf{x},t)) + \epsilon^2\nabla^2\phi(\mathbf{x},t), \tag{1.1}$$

$$\partial_t\Big(u(\mathbf{x},t) + \frac{1}{\Delta}p(\phi(\mathbf{x},t))\Big) = \nabla^2 u(\mathbf{x},t), \tag{1.2}$$

assuming that the heat capacities and thermal conductivities are the same in solid and liquid [13, 15]. Here $u(\mathbf{x},t)$ is the dimensionless temperature. The physical parameters Δ and m correspond to the undercooling and interfacial mobility, respectively. The product $\alpha\Delta$ can be interpreted as a scaled inverse capillary length. The function $g(\phi) = \phi^2(1-\phi)^2$ is a symmetric double well potential introduced in the derivation of the model. The function $p(\phi)$ describing the release of latent heat satisfies the pointwise constraints

$$\begin{cases} p(0) = 0 \\ p(1) = 1 \\ p'(0) = p'(1) = 0 \\ p''(0) = p''(1) = 0 \end{cases}, \tag{1.3}$$

guaranteeing that the free energy of the system has local minima in the solid and liquid phases. In this paper we shall mainly use the polynomial $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ as suggested in [15, Model I] although there are other possible choices.

To verify that solutions of the phase-field equations (1.1) and (1.2) agree with solutions of the Stefan problem it is necessary to perform an asymptotic analysis in the limit of vanishing diffuse interface thickness ϵ. It can be shown that $u(\mathbf{x},t)$ satisfies $\partial_t u = \nabla^2 u$ in the bulk solid and liquid phases as as $\epsilon \to 0$. The temperature at the interface is prescribed by the Gibbs-Thomson condition,

$$u_{interface} = -\frac{1}{6\sqrt{2}\alpha\Delta}\Big(\frac{v}{m} + \mathcal{K}\Big), \tag{1.4}$$

and the normal derivative has a discontinuity satisfying the jump condition

$$\partial_n u\Big|_{\text{solid}}^{\text{liquid}} = -\frac{v}{\Delta}, \tag{1.5}$$

as $\epsilon \to 0$.

For computational purposes it is impossible to let $\epsilon \to 0$ since the transition in the phase-field variable must be resolved across interfaces. The early asymptotic analysis showed that (1.4) and (1.5) are satisfied with errors of size $\mathcal{O}(\epsilon)$ which is sufficient to establish convergence [6, 7]. In recent years much interest has been devoted to constructing more accurate approximations since that allows simulation with larger grid resolution without loss of accuracy [2, 3, 8, 10]. In particular, Karma and Rappel have shown how second order accuracy can be obtained by an $\mathcal{O}(\epsilon)$-correction to a single parameter in the phase-field model [3, 10]. The correction determined by Karma and Rappel also allows larger ϵ to be used before unphysical oscillatory intermediate states appear in the transition region of the phase-field variable [1, 9].

The accuracy in the interface conditions is especially important for numerical simulation in the low undercooling regime. As Δ decreases the smallest geometric scale in the crystals increase and the solidification proceeds slower. Larger domains must be discretized to capture dendritic structures, and long-time integration is needed for them to arise. Karma and Rappel motivate the usefulness of their asymptotics for low undercolings by scaling and modeling arguments [10, 11]. Here an alternative motivation is presented by considering (1.1) and (1.2) in the limit $\Delta, \epsilon \to 0$, where $\epsilon \to 0$ "much slower" than Δ. While the limit itself is not of practical interest asymptotics is a useful technique for investigating the behaviour of the phase-field model.

The approximation scheme outlined in Section 2 uses the same techniques as for the analysis in the limit $\epsilon \to 0$ [3] for constant Δ. The main differences are related to the matching conditions discussed in Section 2.1. We also need to make stronger assumptions on the velocity and curvature terms as $\Delta \to 0$. First order asymptotic approximations are constructed formally in Section 3. Section 3.3 compares the interface conditions arising from the phase-field model to the sharp interface conditions (1.4)–(1.5). It is shown that an $\mathcal{O}(\epsilon)$-correction to the interfacial mobility increases the order of accuracy from "almost zero" to one, where the meaning of almost zero depends on how much faster than ϵ the undercooling Δ approaches zero. Some numerical experiments corroborating the conclusions of the analysis are presented in Section 4.

2 The solution approach

From the phase-field model we expect the solution to behave differently in different parts of the $(\mathbf{x}, t)$-domain. Away from diffuse interfaces $\phi(\mathbf{x}, t)$ should take on a constant value 0 or 1 depending on the phase, and the non-linear forcing terms in (1.1)–(1.2) disappear. To construct an approximate solution we shall therefore think of the domain as being divided into an "inner region" corresponding to the diffuse interface, and two "outer regions" corresponding to the bulk solid and liquid phases. For simplicity the analysis is carried out in one space dimension under assumption of radial symmetry.

Let the interface position $\Gamma(t)$ at time t be given implicitly by the expression $\phi(\Gamma(t), t) = 1/2$. Define the moving coordinate system $r(x, t) = x - \Gamma(t)$

attached to the interface. Assuming that the solidification propagates from left to right we can now identify negative values of r with solid, and positive values with liquid for $|r| \gg \mathcal{O}(\epsilon)$. In the inner region $r = \mathcal{O}(\epsilon)$ we introduce the stretched coordinate $\rho = r/\epsilon$. Derivatives over $(\mathbf{x}, t)$ can be converted to derivatives over (ρ, t) by

$$\begin{aligned} \nabla^2 &\to \epsilon^{-2}\partial_{\rho\rho} + \epsilon^{-1}\mathcal{K}(t)\partial_\rho, \\ \partial_t &\to \partial_t - \epsilon^{-1}v(t)\partial_\rho, \end{aligned}$$

where $\mathcal{K}(t)$ is the curvature and $v(t) = \partial_t\Gamma(t)$ the propagation speed of the interface.

Asymptotic approximations are constructed in the limit of vanishing undercooling with α and m kept fixed. For technical reasons we must also let $\epsilon \to 0$ but, as discussed in Section 2.1, it is assumed that ϵ approaches zero "much slower" than Δ. Assuming that solutions to the phase-field equations approach their asymptotic limits smoothly as $\Delta \to 0$ the inner solutions can be represented by power series expansions,

$$\begin{aligned} \phi(\rho, t; \Delta) &= \sum_{j=0}^{\infty} \Delta^j \phi_j(\rho, t; \Delta), \\ u(\rho, t; \Delta) &= \sum_{j=0}^{\infty} \Delta^j u_j(\rho, t; \Delta). \end{aligned} \tag{2.1}$$

A similar ansatz is used for the outer solutions with dependence on the unscaled coordinate r instead of ρ. Superscripts $-$ and $+$ are used to denote outer solutions in the bulk solid and liquid phases respectively. If the asymptotic expansions of ϕ and u hold the interfacial properties can also be represented by power series. We shall assume that

$$\begin{aligned} \mathcal{K}(t; \Delta) &= \sum_{j=1}^{\infty} \Delta^j \mathcal{K}_j(t; \Delta), \\ \Gamma(t; \Delta) &= \Gamma_0 + \sum_{j=1}^{\infty} \Delta^j \Gamma_j(t; \Delta). \end{aligned} \tag{2.2}$$

Note that the curvature and time-dependent terms in the interface position are of order Δ. This does not follow from the regularity assumptions on (2.1) but is required to avoid differential-algebraic systems in the approximation scheme. From physics one expects the propagation speed and curvature of dendrite tips to decrease with the undercooling. We emphasize that the asymptotics is used merely as a tool for investigating the behavior at low undercoolings. The limit $\Delta = 0$ corresponds to a stationary planar solidification front and is not of interest in numerical computation.

By inserting the asymptotic ansatz (2.1)–(2.2) in the phase-field model (1.1)–

(1.2) and applying a change of coordinates to the moving (ρ,t)-system we obtain

$$\begin{aligned}0 =&\Delta^0\Big[\partial_{\rho\rho}\phi_0-\frac{1}{4}g'(\phi_0)\Big]+\Delta^1\Big[\partial_{\rho\rho}\phi_1-\frac{1}{4}g''(\phi_0)\phi_1+\epsilon\alpha u_0p'(\phi_0)+\epsilon\Big(\frac{v_1}{m}+\mathcal{K}_1\Big)\phi_0'\Big]+\\&\Delta^2\Big[\partial_{\rho\rho}\phi_2-\frac{1}{4}g''(\phi_0)\phi_2+\epsilon\alpha u_1p'(\phi_0)+\epsilon\alpha u_0p''(\phi_0)\phi_1-\frac{1}{8}g'''(\phi_0)\phi_1^2+\\&\quad\epsilon\Big(\frac{v_1}{m}+\mathcal{K}_1\Big)\phi_1'+\epsilon\Big(\frac{v_2}{m}+\mathcal{K}_2\Big)\phi_0'\Big]+\mathcal{O}(\Delta^3),\end{aligned}\tag{2.3}$$

and

$$\begin{aligned}0 =&\Delta^0\Big[\partial_\rho u_0+\epsilon v_1p(\phi_0)-d_1\Big]+\\&\Delta^1\Big[\partial_\rho u_1+\epsilon v_2p(\phi_0)+\epsilon v_1p'(\phi_0)\phi_1+\epsilon\Big(\frac{v_1}{m}+\mathcal{K}_1\Big)u_0-d_2\Big]+\mathcal{O}(\Delta^2).\end{aligned}\tag{2.4}$$

The temperature equation has been integrated once with respect to ρ and constants of integration have been d_j introduced.

Approximations to $\phi(\rho,t)$ and $u(\rho,t)$ can now be determined by solving the $\mathcal{O}\left(\Delta^k\right)$-equations of (2.3)–(2.4) to obtain $\phi_k(\rho,t)$ and $u_k(\rho,t)$ for $k=0,1,2,\ldots$. The same procedure can be used to approximate $\phi^\pm(r,t)$ and $u^\pm(r,t)$ resulting in approximations valid in the entire computational domain, subject to the matching conditions discussed in Section 2.1.

The outer solutions are not explicitly determined since that requires boundary conditions for (1.1)–(1.2). The matching conditions below allow us to find the free boundary problem satisfied by the outer solutions $u^\pm(r,t)$ and $\phi^\pm(r,t)$ by examining the inner solutions only.

2.1 Matching conditions

We assume that the inner expansion is valid in some region $-R\lesssim\rho\lesssim R$ around $\rho=0$, where R is the matching point. Similarly the outer expansions are assumed to be valid in the domains $|\rho|\gtrsim R$. Exact inequalities are not used since there should be a common domain of validity where the inner and outer expansions represent the same solution.

From the definition of the phase-field variable we expect

$$\lim_{\rho\to\pm R}\phi_k(\rho,t)\approx\begin{cases}(1\pm1)/2 & \text{if } k=0\\ 0 & \text{otherwise}\end{cases}\tag{2.5}$$

to hold. By the definition of $\Gamma(t)$ it follows that ϕ_k must also satisfy

$$\phi_k(0,t)=\begin{cases}1/2 & \text{if } k=0\\ 0 & \text{otherwise}\end{cases}$$

so that the interface position is uniquely determined. Matching relations for the temperature are determined by Taylor expanding the outer solutions around

$r = 0$ and identifying powers of Δ,

$$\lim_{\rho \to \pm R} u_k(\rho, t) = \sum_{j=0}^{N} \frac{\epsilon^j}{j!} \partial_r^j u_k^{\pm}(0, t) \rho^j + \mathcal{O}\left((\epsilon\rho)^{N+1}\right). \tag{2.6}$$

The matching conditions (2.5) and (2.6) are formulated using the same techniques as for the asymptotics in ϵ. There are however important differences related to the definition of the inner and outer coordinate systems. In the limit of vanishing diffuse interface thickness one can assume that for sufficiently small ϵ there are constants D and $0 < \mu < 1$ such that the inner and outer approximations describe the same solution in the vicinity of

$$R(\epsilon) = D\epsilon^{\mu - 1}. \tag{2.7}$$

As $\epsilon \to 0$, ϵR approaches zero and R approaches infinity. Hence the truncation error in the equivalent of (2.6) disappears, and (2.5) is satisfied exactly [3].

These properties do not hold as $\Delta \to 0$ for fixed ϵ, when the phase-field model is no longer a singularly perturbed reaction-diffusion equation. It is therefore necessary to let ϵ approach zero with Δ. Let us assume that there are constants $C > 0$ and $0 < \nu \ll 1$ such that

$$\lim_{\Delta \to 0} \epsilon = C\Delta^{\nu}. \tag{2.8}$$

ϵ tends to zero much slower than Δ since $\nu \ll 1$ so equations (2.3) and (2.4) do not change to significant orders in Δ.

With the same assumptions on $R(\epsilon)$ as in (2.7) we have

$$\lim_{\Delta \to 0} \epsilon R(\epsilon) = \lim_{\Delta \to 0} DC^{\mu}\Delta^{\nu\mu} = 0,$$
$$\lim_{\Delta \to 0} R(\epsilon) = \lim_{\Delta \to 0} DC^{\mu-1}\Delta^{\nu(\mu-1)} = \infty.$$

As $\Delta \to 0$ the interfacial region becomes infinitesimally thin (in outer coordinates) guaranteeing that the truncation errors in (2.6) disappear. Furthermore, R can be replaced by infinity in (2.5), which allows us to uniquely determine $\phi_0(\rho, t)$ in Section 3.1.

Equations (2.7) and (2.8) are necessary to formally determine the rate of convergence to the sharp interface conditions. The values of the constants are of not needed in simulation.

3 Asymptotics

3.1 Zero-order inner solution

To determine ϕ_0 we note that the chain rule of differentiation gives

$$\partial_{\rho\rho}\phi_0 = \frac{1}{2}\frac{d}{d\phi_0}\left(\partial_{\rho}\phi_0\right)^2.$$

Inserting this expression in the zero-order equation from (2.3) and integrating with respect to ϕ_0 yields

$$\big(\partial_\rho \phi_0\big)^2 = \frac{1}{2} g(\phi_0) + const. \tag{3.1}$$

ϕ_0 is expected to approach a constant value as $|\rho|$ increases, and it follows from (2.5) and the definition of $g(\phi)$ that the constant in (3.1) must be zero. The equation is separable with solution

$$\phi_0(\rho) = \frac{const.}{1 + e^{-\rho/\sqrt{2}}} + \frac{const.}{1 + e^{\rho/\sqrt{2}}}$$

where the two different terms are the result of taking the square root of (3.1). The constants are determined from the matching condition (2.5) resulting in

$$\phi_0(\rho) = \frac{1}{1 + e^{-\rho/\sqrt{2}}} = \frac{1}{2}\Big(\tanh \frac{\rho}{2\sqrt{2}} + 1 \Big). \tag{3.2}$$

This expression satisfies the required normalization $\phi_0(0,t) = 1/2$.

Integrating the zero-order equation of (2.4) once with respect to ρ gives the zero-order inner approximation of the temperature,

$$u_0(\rho) = c_0 - \epsilon v_1 \int_0^\rho p\big(\phi_0(y)\big) dy + d_0 \rho, \tag{3.3}$$

where c_0 is a constant of integration. By (1.3) the expression can be rewritten so that the integral remains bounded as $|\rho|$ increases,

$$u_0(\rho) = \begin{cases} c_0 + \epsilon v_1 \int_\rho^0 p(\phi_0(y)) dy + d_0 \rho & \rho < 0 \\ c_0 - \epsilon v_1 \int_0^\rho \big(p(\phi_0(y)) - 1\big) dy + \big(d_0 - \epsilon v_1\big)\rho & \rho \geq 0 \end{cases} .$$

Comparing linear terms in ρ to the matching condition (2.6) shows that the outer temperature satisfies the jump condition

$$\partial_r u_0 \Big|_-^+ \equiv \partial_r u_0^+(0) - \partial_r u_0^-(0) = -v_1. \tag{3.4}$$

It follows from the assumption (2.2) that the outer temperature satisfies (1.5) to lowest order. A comparison of the constant terms gives

$$u_0 \Big|_-^+ = -\epsilon v_1 \int_0^R \big(p(\phi_0(y)) + p(1 - \phi_0(y)) - 1\big) dy.$$

If the polynomial $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ is used the integrand is always zero. Thus there is no jump in outer temperature to lowest order. Other choices of $p(\phi)$ may result in an outer temperature which is discontinuous across the interface, contrary to what one would expect from the Gibbs-Thomson condition.

3.2 Interfacial temperature to lowest order

The constant c_0 in (3.3) must be known to uniquely identify the interfacial temperature $u^{\pm}(0,t)$ resulting from the phase-field model. It is determined from the solvability condition for ϕ_1. The first order equation of (2.3) can be written as

$$\mathcal{L}\phi_1 \equiv \Big(\partial_{\rho\rho} - \frac{1}{4}g''(\phi_0)\Big)\phi_1 = -\epsilon\alpha u_0 p'(\phi_0) - \epsilon\Big(\frac{v_1}{m} + \mathcal{K}_1\Big)\phi_0'. \tag{3.5}$$

$\phi_0'(\rho,t)$ is a null function to the linear differential operator $\mathcal{L}$ since

$$\mathcal{L}\phi_0' = \partial_\rho\Big(\partial_{\rho\rho}\phi_0 - \frac{1}{4}g'(\phi_0)\Big),$$

and the equation in paranthesis was solved to obtain ϕ_0. For a solution ϕ_1 to exist ϕ_0' must be orthogonal to the right-hand side of (3.5), i.e.

$$0 = \epsilon\alpha\int_{-\infty}^{\infty} u_0(y)p'\big(\phi_0(y)\big)\phi_0'(y)\,dy + \epsilon\Big(\frac{v_1}{m} + \mathcal{K}_1\Big)\int_{-\infty}^{\infty}\big(\phi_0'(y)\big)^2 dy$$

The solvability condition can be evaluated for general $p(\phi)$ satisfying (1.3). It reduces to

$$c_0 = -\frac{1}{6\sqrt{2}\alpha}\Big(\frac{v_1}{m}+\mathcal{K}_1\Big) - \epsilon v_1\Big[\int_{-\infty}^{0} p^2\big(\phi_0(y)\big)\,dy + \int_0^{\infty} p(\phi_0(y))(p(\phi_0(y))-1)dy\Big], \tag{3.6}$$

which identifies the hitherto unknown constant c_0. For the fifth degree polynomial $p(\phi) = \phi^3(6\phi^2 - 15\phi + 10)$ we have

$$c_0 = -\frac{1}{6\sqrt{2}\alpha}\Big(\frac{v_1}{m} + \mathcal{K}_1\Big) + \Big(\frac{2827}{3360} - \ln 2\Big)\sqrt{2}\epsilon v_1\,.$$

At this stage we can proceed to determine $\phi_1(\rho,t)$ and $u_1(\rho,t)$ from the first order equations in (2.3)–(2.4). Since only ϕ_0 and u_0 are needed to determine the sharp interface problem satisfied by the outer solutions to first order this step is omitted here.

3.3 Interface conditions

The outer temperature satisfies the jump condition (1.5) with errors of size $\mathcal{O}(\Delta)$ according to (3.4). The outer temperature at the interface is given by (2.6) and (3.3),

$$u^{\pm}(0) = u_0^{\pm}(0) + \mathcal{O}(\Delta) = c_0 + \epsilon v_1\int_{-\infty}^{0} p(\phi_0(y))dy + \mathcal{O}(\Delta)\,. \tag{3.7}$$

Inserting (3.6) in the first order approximation $u_0^{\pm}(0)$ yields

$$u_0^{\pm}(0) = -\frac{1}{6\sqrt{2}\alpha}\Big(\frac{v_1}{m} + \mathcal{K}_1\Big) - \epsilon v_1\int_{-\infty}^{\infty} p(\phi_0(y))(p(\phi_0(y)) - 1)dy.$$

Hence the interfacial temperature can be expressed in terms of the function $p(\phi)$. By choosing $p(\phi_0) = \phi^3(6\phi^2 - 15\phi + 10)$ we find

$$\begin{aligned} u_0^{\pm}(0) &= -\frac{1}{6\sqrt{2}\alpha}\Big(\frac{v_1}{m} + \mathcal{K}_1\Big) + \epsilon\frac{209\sqrt{2}}{420}v_1 \\ &= -\frac{1}{6\sqrt{2}\alpha\Delta}\Big(\Big(\frac{1}{m} - \frac{209}{35}\epsilon\alpha\Big)v(t) + \mathcal{K}(t)\Big), \end{aligned} \tag{3.8}$$

which is the same expression obtained in the ϵ-asymptotics [3]. The temperature has an anomalous term not present in (1.4), and satisfies a Gibbs-Thomson condition for a different mobility. The interfacial temperature approaches the correct limit as $\Delta \to 0$, with errors decreasing as $\epsilon \sim \Delta^\nu$ to leading order.

If we for a given mobility m_0 instead use m given by

$$\frac{1}{m} = \frac{1}{m_0} + \frac{209}{35}\epsilon\alpha \tag{3.9}$$

in computation the leading error term of (3.8) is eliminated. The error in the Gibbs-Thomson condition becomes independent of the diffuse interface thickness. Better agreement between the phase-field and sharp interface models can be expected for small but non-zero values of Δ and ϵ when the thin interface correction (3.9) is applied.

We observe that (3.8) was obtained for a specific polynomial $p(\phi)$ of degree five. It is possible to find two polynomials $p(\phi)$ of degree seven eliminating the first anomalous term of (3.8) while satisfying (1.3) and $p(\phi) + p(1-\phi) = 1$ [3]. Thus the correct Gibbs-Thomson condition is satisfied for $m = m_0$, and there is no need to apply a thin interface correction.

4 Numerical experiments

4.1 One-dimensional computations

Numerical computations have been performed to corroborate the asymptotics. We shall consider the computed interfacial temperature compared to the values predicted by (1.4) and (3.8). By performing simulation without the thin interface correction (3.9) the accuracy of the different Gibbs-Thomson conditions can be compared. The primary aim is to investigate how well (3.8) predicted by analysis agrees with computation.

Simulations are performed in one space-dimension under assumptions of spherical symmetry for three different values of the undercooling Δ. We use parameters $\alpha = 400$, $m = 0.05$ and $\epsilon = 0.005$. Initial values for the phase-field are set by (3.2) such that $\mathcal{K} = 0.2\Delta$. The initial temperature is taken to be $-\phi(x, 0)$.

The phase-field equations (1.1)–(1.2) are solved using second order finite differences and a third order BDF-scheme for time integration. Discretization is uniform in space and time with $\Delta x = 0.0025$ and $\Delta t = 0.001$. Due to the

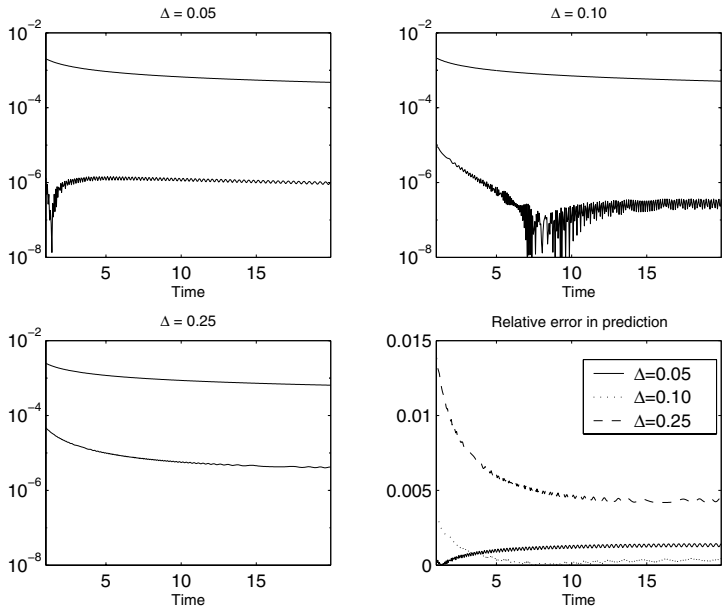


Figure 1: The first three plots show errors in computed interfacial temperature compared to (1.4) (upper curve) and (3.8) (lower curve). The lower right plot shows the relative difference between computed interfacial temperature and the value predicted by asymptotics (3.8). Absolute values are used for all plots.

implicit time stepping scheme a non-linear system must be solved each time step. A Newton-GMRES solver is employed to ensure that the residual is smaller than 10^{-8} in each grid point.

As the undercooling decreases the temperature will drop from the interfacial value to -1 over a larger interval. To ensure that the boundary conditions do not affect the extent of the temperature field the size of the computational grid is increased during the course of the simulation. When the temperature at the right boundary differs from -1 with more than 10^{-8} an additional 100 grid points are added. This ad-hoc adaptivity reduces execution time and does not introduce any artifacts visible in the computational results.

It is necessary to extract curvature, interfacial temperature and velocity from the computed solutions for comparison with (1.4). For this purpose a cubic spline interpolant is fitted to the phase-field each time step, and $\Gamma(t)$ such that $\phi(\Gamma(t),t) = 1/2$ is determined by Newton iteration. The curvature is given directly by $\Gamma(t)$ and the spherical symmetry assumption. Cubic spline interpolation gives the temperature $u(\Gamma(t),t)$, corresponding to c_0 given by (3.6). Before comparison the evaluated integral from (3.7) is added to obtain the outer temperature at the interface. The velocity is computed by fitting a third degree polynomial to the most recent values of $\Gamma(t)$ and evaluating its derivative.

Figure 1 shows the difference between computed interfacial temperature and the value predicted by the sharp interface condition (1.4), as well as the difference between computed interfacial temperature and the value predicted by asymptotics (3.8). For $\Delta = 0.05$ and $\Delta = 0.1$ the error changes sign in the interval due to transients from initial data which explains the local drops in magnitude. It is clear from the figure that the temperature predicted by analysis agrees better with computational results. The temperature predicted by asymptotics lies within 1.5% of the actual value for all cases considered.

Thus we can expect the thin interface correction (3.9) to result in better agreement between computed solutions and the sharp interface problem. This has been verified by performing computations using $m = m_0$, and m given by (3.9). Comparing computed interfacial temperature to the Gibbs-Thomson condition (1.4) for the two cases results in a plot almost identical to Figure 1, where the lower curves correspond to cases where the thin interface correction is applied. The magnitudes are the same but the curves are slightly different from Figure 1 since the two values of m correspond to different sharp interface problems.

In computations not presented here the same conclusions have been reached for constant curvature $\mathcal{K}(t) = 0.2\Delta$, and cases where the initial curvature does not depend on Δ.

4.2 Two-dimensional computations

Simulation in two space dimensions is more costly both in terms of floating point operations and storage requirements. We shall therefore only consider the undercooling $\Delta = 0.1$. Unlike the experiments in Section 4.1 the thin interface correction is applied here, which guarantees - in addition to the increased ac-

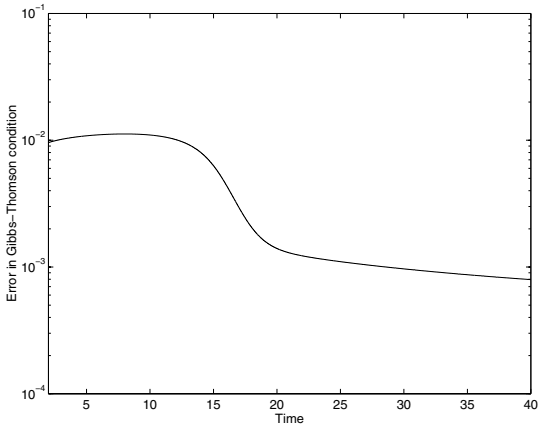


Figure 2: Absolute error in Gibbs-Thomson condition measured where the solid-liquid interface intersects the x-axis.

curacy - that larger ϵ can be used without introducing unphysical intermediate states in the phase-field variable [1, 9].

To account for effects of anisotropy in two space dimensions the Laplacian in (1.1) is replaced by

$$-\frac{\partial}{\partial x}\Big(\eta(\theta)\eta'(\theta)\frac{\partial}{\partial y}\Big)+\frac{\partial}{\partial y}\Big(\eta(\theta)\eta'(\theta)\frac{\partial}{\partial x}\Big)+\nabla\big(\eta^2(\theta)\nabla\big),$$

where $\eta(\theta)=1+\gamma\cos(k\theta)$, and θ is the angle between the solid-liquid interface and some reference direction. Here γ is the relative strength of the anisotropy and the integer k a mode number determining the number of symmetry axes [12]. With the inclusion of anisotropy the outer temperature is expected to satisfy a slightly different Gibbs-Thomson condition [16],

$$u_{interface}=-\frac{1}{6\sqrt{2}\alpha\Delta}\left(\frac{v}{m\eta(\theta)}+(\eta(\theta)+\eta''(\theta))\mathcal{K}\right). \tag{4.1}$$

When investigating agreement with (4.1) we shall consider the angle $\theta=0$ only, corresponding to the interfacial temperature where the solid-liquid interface intersects the x-axis. This simplifies approximation of the curvature which is given implicitly by derivatives of the phase-field variable. The interfacial temperature and velocity are determined as for the one-dimensional simulations described in Section 4.1.

Spatial derivatives in the phase-field model are approximated by second order

finite differences on a uniform Cartesian grid. Each time step the solution is updated in two stages according to the algorithm

1. Compute $\phi(t + \Delta t)$ using forward Euler scheme.

2. Compute $u(t+\Delta t)$ using backward Euler scheme with $\phi(t+\Delta t)$ prescribed by stage 1.

This algorithm is only first order accurate in Δt, but requires less computer memory than a higher order fully implicit scheme [4]. Furthermore, by prescribing zero Neumann conditions on $u(x, y, t)$ at all boundaries the linear system in stage 2 can be efficiently solved by a fast Poisson solver.

We use parameters for pure nickel in the phase-field model, $\alpha = 400$, $m_0 = 0.05$ and $\Delta = 0.1$ [16]. Fourfold anisotropy is employed with $\gamma = 0.02$ and preferred growth directions along the x- and y-axes. The diffuse interface thickness is set to $\epsilon = 0.04$ and the thin interface correction (3.9) is applied prior to computation. The computational grid has 3001×3001 points with $\Delta x = \Delta y = 0.02$, and the constant time step $\Delta t = 0.0025$ is used for time integration. The initial seed is circular with radius 5.0 and the temperature distribution is given by $u(x, y, 0) = -\phi(x, y, 0)$.

The absolute error in the Gibbs-Thomson condition (1.4) is plotted in Figure 2. There are long-lived transients from initial conditions and the decreasing error around $t = 15$ corresponds to a change of sign in the extracted temperature. At the final time the relative error in the interfacial temperature compared to (1.4) is about 20%, which is larger than what would be expected from Figure 1. In the corresponding simulation from Section 4.1 the error is less than 1% for $t = 20$. The two errors are not directly comparable since the curvature in the one-dimensional case is ten times smaller. One should also note that without the thin interface correction a smaller value of ϵ would have to used, thus leading to a larger grid.

Dendritic structures do not arise during the course of the simulation and the effects of anisotropy remain small even for much larger times. A second experiment has been performed to exemplify results expected from a typical computation. To reduce execution time a significantly smaller grid of 512×512 points is used. The smaller domain leads to the thermal field being affected by outer boundary conditions. Therefore the preferred growth directions has been chosen at a 45° angle to the x-axis so that the dendrite tip is as far from the edges of the domain as possible. The radius of the initial seed is given by the function $r(\theta) = 0.5(1 + 0.075\cos(\theta - \pi/4))$. All other parameters are the same as for the first simulation. Figure 3 shows the location of the solid-liquid interface at a few equidistant time levels.

5 Conclusions

The behavior of a phase-field model of solidification has been investigated in the limit of vanishing undercooling Δ and diffuse interface thickness ϵ, where $\epsilon \to 0$

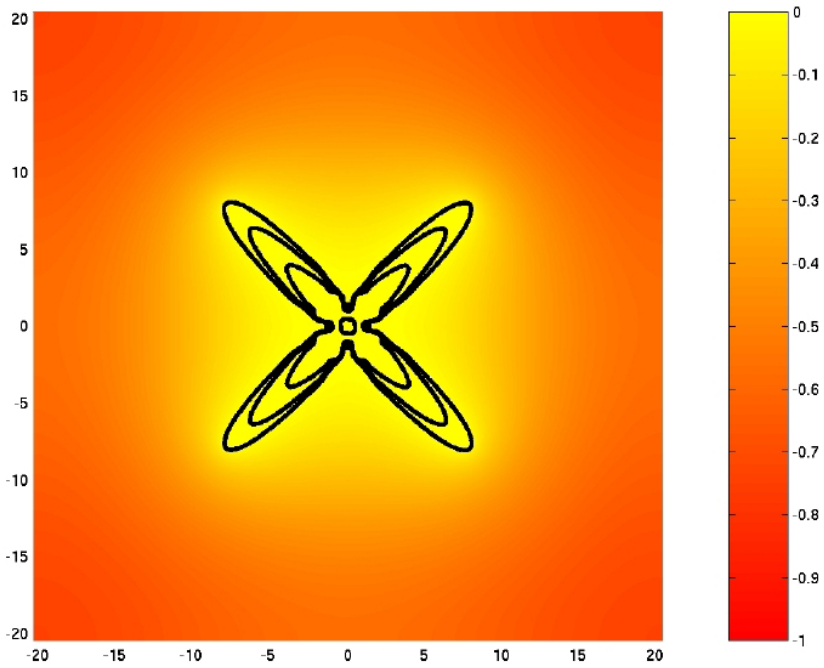


Figure 3: Growth of nickel dendrite for $\Delta = 0.1$. The solid lines show the location of the solid-liquid interface at times $t = 0, 50, 100, 150$, and the background color represents the heat distribution at $t = 150$. The solution has been computed in the upper right quadrant and reflected around the x- and y-axes.

much slower than Δ. The techniques used are the same as for the asymptotics in ϵ for constant Δ but stronger assumptions are needed on the interfacial velocity and curvature. By formal asymptotic analysis it is shown that a thin interface correction (3.9) can be employed to eliminate the leading $\mathcal{O}(\epsilon)$-error in the Gibbs-Thomson condition. The accuracy is increased from "almost zero" to one in Δ, where the meaning of almost zero depends on the constant $0 < \nu \ll 1$ determining how much slower ϵ approaches zero. The correction derived here is the same as one obtains from the ϵ-asymptotics [3] based on the observations of Karma and Rappel [10, 11].

The actual limit $\Delta \to 0$ is not interesting from a computational point of view. The analysis presented here does however provide an alternative motivation for the importance of the thin interface correction at low undercoolings which does not directly rely on scaling and modeling arguments [10, 11].

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